Regret and Information Avoidance*

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Abstract

Empirical evidence suggests that individuals selectively avoid information based on past decisions. We address these findings by studying an agent who behaves as if she trades off two conflicting effects of information. The first is a psychological cost from the regret about past decisions that are revealed to be suboptimal, whereas the second is the instrumental value in guiding future decisions. We take preferences over consumption-information pairs as primitive and capture information avoidance as a revealed preference for an information structure that is less informative in the Blackwell sense. Our main results are axiomatic representation theorems that account for both the instrumental value and the regret cost of information. We introduce new axioms linking the agent's consumption choices to her information choices. We also characterize the uniqueness of the preference parameters and derive comparative statics for the agent's information aversion.

KEYWORDS: Regret, information avoidance, value of information, preference for information, menu preference

JEL Code: D81, D83

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1 Introduction

Information avoidance describes an agent's unwillingness to obtain information that is both free and useful. Even though standard economic theories do not allow information to cause disutility in individual decision-making problems, substantial evidence indicates that information avoidance is a pervasive phenomenon in real-world settings.[1]

More puzzlingly, whether an individual seeks or avoids information can depend on what she previously chose. The pattern of interest has two features: the individual chooses whether to access an information source before knowing exactly what it will reveal, and this choice depends on an earlier decision. A closely related finding is documented by Sicherman et al. (2016), who study investors' decisions to log in to their retirement accounts. Logging in provides information about the performance of their current holdings that can guide subsequent investment decisions; not logging in avoids this information. Investors log in less often after stock market declines, and this tendency becomes stronger as the share of equity in their portfolios increases. Accounts with zero equity holdings do not show this pattern of avoidance. Thus, information avoidance is associated with what the investor previously chose to hold.

Related experiments document avoidance of feedback about previous allocation decisions (Schmidt, Schwieren, and Vollmann, 2020), avoidance of feedback about forgone outcomes in repeated choices (Reb and Connolly, 2009), and reduced information acquisition after a preliminary choice (Momsen and Schneider, 2024). If information is interpreted more broadly to include messages from a source, studies of post-decisional selective exposure document a preference for material supporting an earlier decision over material challenging it (Ehrlich et al., 1957; Frey, 1981, 1986; Frey and Wicklund, 1978; Jonas et al., 2001). Such choices are consistent with avoiding information that could make a forgone alternative appear better.

Why would an agent refuse to be better informed for future decisions due to a choice made in the past? One natural reason is regret. Information could reveal an agent's past decision to be suboptimal and cause her to experience a sense of loss from not having chosen a different alternative. We refer to this (psychological) effect as the *regret cost* of information.

While it can cause regret about past decisions, information can still guide future choices and hence retains instrumental value. For example, a physical examination that helps to screen for potential health problems might lead to the discovery that an old lifestyle was unhealthy. Inquiries about different positions for a new job might reveal to a worker that better early-career choices could have been made. Stock analyses for a new investment

[1] Lerman et al. (1999), Oster, Shoulson, and Dorsey (2013) and Persoskie et al. (2014) are part of a large literature documenting individuals avoiding medical information when at risk of certain diseases or health conditions. A rapidly growing literature studies investors' aversion to financial information. Examples include Karlsson, Loewenstein, and Seppi (2009), Sicherman et al. (2016) and Hilbert et al. (2022).

project might unveil that a better portfolio could have been constructed in the past. Because information often arrives between decisions made at different times, an agent must trade off its regret cost against its instrumental value. The central question is how this conflict shapes preference over information.

In this paper, we propose and axiomatize a model of information avoidance based exactly on the tradeoff between the regret cost and the instrumental value of any information source. In our model, an agent avoids a more informative source when (and only when) the increase in regret cost outweighs the increase in instrumental value. A previous decision (or the absence thereof) exposes the agent to (or protects the agent from) the possibility of regret, resulting in the dependence of information avoidance on past decisions.

Specifically, we build a model of decision making under uncertainty that involves choices in three periods. The uncertainty is captured by a set of payoff-relevant states and information is represented by Blackwell experiments over these states. In period 3 (the final period), the agent chooses an (Anscombe-Aumann) act from a menu of acts F after the arrival of information. That is, the act choice can be customized according to what the information reveals. In contrast, the menu of acts F itself has been selected from a menu of menus $\mathbb{F}$ in period 2 *before* the information arrives. Thus, the information could cause regret if it reveals that some unchosen menu G contains an act g that is better than all acts in F. In period 1, the agent chooses a *consumption-information pair* consisting of a menu of menus of acts and a Blackwell experiment, having in mind the subsequent choices of a menu and an act.

The menu-of-menus framework is inspired by a common decision procedure in which an agent gradually narrows down her set of options before making a final choice. It also provides a simple way to integrate both the regret cost and the instrumental value of information in one model. Including information sources as choice objects then allows us to study information avoidance directly through the agent's revealed preferences over these sources.

In our model, information avoidance is defined as an agent preferring a (Blackwell) less informative experiment to a more informative one. This general notion nests the common notion of information avoidance where an agent prefers no information to full information (fully revealing experiments) or partial information (general experiments).

Taking preferences over consumption-information pairs as primitive, our key result is a representation theorem that features an *informational tradeoff (IT) representation*, where the agent behaves *as if* she takes into account both the instrumental value and the regret cost when evaluating an information source. As a result, our model can generate both information-avoiding and information-seeking behavior depending on whether the regret cost or the instrumental value is dominant.

The information component of the choice domain is essential for interpreting the model as a theory of information avoidance. Preferences over menus of menus alone may reveal the information structure the agent expects to face, but they cannot reveal whether she would

prefer a less informative experiment. By treating Blackwell experiments as choice objects, the richer domain allows information avoidance to be defined directly as a revealed preference for less information.

In the axiomatization process of the IT representation, we obtain a subjective informational tradeoff (SIT) representation in which the agent's information is an unobserved preference parameter instead of a choice variable. We show that such a representation can be characterized from axioms on the agent's preference over menus of menus alone. Moreover, the unobserved information can be elicited from this preference.

The SIT representation is of interest on its own. Conceptually, the agent might still recognize both the regret cost and the instrumental value of the information structure she expects to face, even if it is not exogenously given to the modeler and is not her choice. Pragmatically, it is valuable to be able to elicit an agent's information because a modeler may not be able to observe this parameter of interest.

We have two key axioms for the SIT representation. The first key axiom is called "Ex-Ante Regret". It reflects the agent's desire to limit her options before the arrival of information. Formally, let F and G be two menus, and $\mathbb{F}$ a set of menus. The axiom states that if $\{F\} \succsim \{G\}$ and $F \in \mathbb{F}$, then $\mathbb{F} \succsim \mathbb{F} \cup \{G\}$.[2] Due to the existence of a better menu F, having the extra option of G makes $\mathbb{F}$ weakly less preferred. The intuition is that adding a menu that does not improve the optimal ex-ante material value can nevertheless increase anticipated regret.

The second key axiom is called "Interim Preference for Flexibility". It reflects the agent's preference to have more options after the arrival of information. Formally, the axiom states that $\mathbb{F} \cup \{F \cup G\} \succsim \mathbb{F} \cup \{F, G\}$ for any menu of menus $\mathbb{F}$ and any menus F and G. That is, when the sets of acts that can be ultimately chosen are the same, the agent prefers to keep more options open so she has more flexibility after the arrival of information.

Our axiomatization of the IT representation builds on the axiomatic characterization of the SIT representation. In addition to axioms for the SIT representation, we impose axioms to discipline the interaction between the agent's consumption choice and information choice.

We now describe the functional forms identified from our representation theorems.

Let Ω and X be finite sets of states and outcomes, respectively.

An information structure is a mapping $\sigma : \Omega \to \Delta(S)$ where S is a finite set of signals. An act is a mapping $f : \Omega \to \Delta(X)$. Let F be a menu of acts and $\mathbb{F}$ a menu of menus. The informational tradeoff (IT) representation for a pair $(\mathbb{F}, \sigma)$ is

$$W(\mathbb{F}, \sigma) := \max_{F \in \mathbb{F}} \sum_{s \in S} \sigma_s^{\pi} \Big[U(F, \mu_s^{\sigma}) - R(F, \mathbb{F}, \mu_s^{\sigma}) \Big] \tag{1}$$

[2] Despite its similarity with the main axiom (dominance) of Sarver (2008), our axiom is adapted to be imposed on the preference over menus of menus of acts instead of menus of lotteries.

where σ_s^π is the ex-ante probability that signal s is generated under prior π while μ_s^σ is the Bayesian posterior if s is observed,[3]

$$U(F, \mu_s^\sigma) := \max_{f \in F} \sum_{\omega \in \Omega} \mu_s^\sigma(\omega) u\big(f(\omega)\big)$$

captures the material value of a menu F under posterior μ_s^σ, where $u : \Delta(X) \to \mathbb{R}$ is an affine function over lotteries that captures the agent's taste over outcomes, and

$$R(F, \mathbb{F}, \mu_s^\sigma) := K \left[\max_{G \in \mathbb{F}} U(G, \mu_s^\sigma) - U(F, \mu_s^\sigma) \right]$$

captures the agent's regret for having chosen menu F from a set of menus $\mathbb{F}$ at posterior μ_s^σ. That is, the agent's regret is proportional to the difference between the highest material value she could have obtained from a menu in $\mathbb{F}$ and the material value of the menu she has chosen. The intensity of regret is captured by the scalar $K \geq 0$.

One special case of the IT representation is when σ is uninformative. Let $o : \Omega \to \Delta(\{0\})$ denote an uninformative information structure. Then $W(\mathbb{F}, o) = \max_{F \in \mathbb{F}} U(F, \pi)$ for any $\mathbb{F}$. The agent's posterior equals her prior, and anticipated regret is zero at an optimal menu.

In a SIT representation, a menu of menus of acts $\mathbb{F}$ is evaluated by

$$V(\mathbb{F}) = \max_{F \in \mathbb{F}} \sum_{s \in S} \sigma_s^\pi \Big[U(F, \mu_s^\sigma) - R(F, \mathbb{F}, \mu_s^\sigma) \Big] \tag{2}$$

where σ_s^π, μ_s^σ, $U(F, \mu_s^\sigma)$ and $R(F, \mathbb{F}, \mu_s^\sigma)$ are the same as in the IT representation. Although $W(\cdot, \cdot)$ and $V(\cdot)$ correspond to the same functional form, they describe functions over different domains. In particular, the information structure σ is a choice variable in an IT representation while it is an unobserved parameter in a SIT representation. Thus, the standalone SIT representation is not suitable to rationalize information avoidance.

Note that the agent experiences no regret when $\mathbb{F}$ contains only one menu. That is, if $\mathbb{F} = \{F\}$, then regret about choosing F from $\mathbb{F}$, $R(F, \mathbb{F}, \mu) \equiv 0$. This property of the representation may give the impression that we make the overly strong assumption that the agent cannot regret her choice of menu of menus in our model. However, as we discuss in Section 5.2, our model can be thought of as capturing the *additional* regret (on top of the regret associated with the choice of menu of menus) that the agent experiences because of her choice from a menu of menus.[4]

We proceed by illustrating our representations with the following numerical example.

[3]Formally, $\sigma_s^\pi = \sum_{\omega \in \Omega} \pi(\omega) \sigma(s \mid \omega)$ where $\pi \in \Delta(\Omega)$ is a prior and $\sigma(s \mid \omega)$ is the probability for observing signal s when the state is ω. For any signal s with $\sigma_s^\pi > 0$, the Bayesian posterior $\mu_s^\sigma \in \Delta(\Omega)$ is $\mu_s^\sigma(\omega) = \frac{\pi(\omega)\sigma(s|\omega)}{\sigma_s^\pi}$. The posterior μ_s^σ also depends on π; we suppress this dependence in the notation when the prior is clear from context. Without loss of generality, we set the posterior μ_s^σ to be uniform if $\sigma_s^\pi = 0$.

[4]Section 4.3 of Sarver (2008) provides a more detailed discussion on a closely related issue.

A numerical example. A college student aims to get a job with the best career prospect upon graduation. Each job is modeled as an act depending on the general labor market condition. Each major is a menu of jobs: the economics major could be Econ $= \{b, c\}$ whereas the computer science major could be CS $= \{e\}$ where b, c, e stand for banking (b), consulting (c), and engineering (e), respectively. Colleges are modeled as menus of majors with $\mathbb{F}_1 = \{\text{Econ},\ \text{CS}\} = \{\{b, c\}, \{e\}\}$, and $\mathbb{F}_2 = \{\text{Econ}\} = \{\{b, c\}\}$.

Uncertainty can be captured by a binary state space $\Omega = \{\omega_1, \omega_2\}$. The student's prior belief is $\pi(\omega_1) = \pi(\omega_2) = 0.5$. The state-dependent utilities are given in the left table.

u	b	c	e
ω_1	110	100	130
ω_2	60	90	50

σ	s_1	s_2
ω_1	1	0
ω_2	0	1

Suppose $K \geq 0$ and that the student learns the true state upon graduation, which corresponds to a fully revealing Blackwell experiment as in the right table.

We first illustrate the SIT representation. If the student has chosen the economics major from $\mathbb{F}_1 = \{\text{Econ},\ \text{CS}\}$, her ex-post regret will be $R(\text{Econ},\ \mathbb{F}_1, \mu^{\sigma}_{s_1}) = 20K$ if signal s_1 is observed and 0 if signal s_2 is observed.[5] The material utility (U) and ex-post regret (R) for choosing each major from $\mathbb{F}_1$ can be summarized by the following tables.

U	Econ	CS
s_1	110	130
s_2	90	50

R	Econ	CS
s_1	$20K$	0
s_2	0	$40K$

The ex-ante probability for each signal is $\sigma^{\pi}_{s_1} = \sigma^{\pi}_{s_2} = 0.5$. Therefore, the expected net value is $100 - 10K$ for the economics major in $\mathbb{F}_1$, and $90 - 20K$ for the computer science major. Since $K \geq 0$, the student would choose Econ from $\mathbb{F}_1$. Therefore, $V(\mathbb{F}_1) = 100 - 10K$.

The value for $\mathbb{F}_2$ is $V(\mathbb{F}_2) = 100$. With $K > 0$, the student would strictly prefer College 2 over College 1 because College 1 only exposes her to regret by offering the CS major.

To illustrate the IT representation, consider the parametrized information structures

σ_θ	s_1	s_2
ω_1	θ	$1-\theta$
ω_2	$1-\theta$	θ

where $\theta \in [0.5, 1]$ is the precision.

Suppose a student from College 1 can control the precision θ. Applying the IT representation, we have $W(\mathbb{F}_1, \sigma_1) = 100 - 10K$ and $W(\mathbb{F}_1, \sigma_{0.5}) = 95$. Therefore, as long as $K > 0.5$, the student would prefer no information ($\theta = 0.5$) to full information ($\theta = 1$).

[5] The signal s_1 reveals the state to be ω_1. The student's best option is b with $u(b(\omega_1)) = 110$ while career e with $u(e(\omega_1)) = 130$ is no longer available. The calculation of $R(\text{Econ},\ \mathbb{F}_1, \mu^{\sigma}_{s_2})$ is similar.

In Figure 1, we vary θ from 0.5 to 1 and draw the value of information as a function of θ. The upper curve corresponds to a regret intensity $K = 0.4$ while the lower curve is drawn under $K = 0.8$. Both curves are non-monotonic, thus allowing room for both information-avoiding and information-seeking behavior.

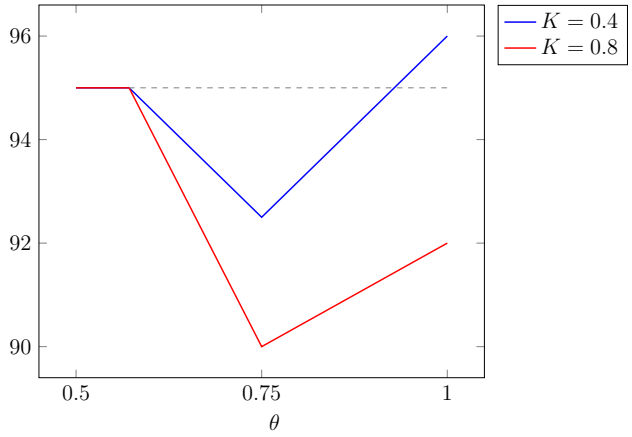


Figure 1: Value of Information σ_θ, $W(\mathbb{F}_1, \sigma_\theta)$, as a function of θ

The remainder of the paper is organized as follows. We set up the model in Section 2. We provide an axiomatization for the SIT representation in Section 3 to build towards the axiomatic theorem for the IT representation in Section 4. Section 5 concludes with a discussion of the related literature and extensions.

2 The Model

Let Ω be a finite set of states with $|\Omega| \geq 2$ and X be a finite set of prizes with $|X| \geq 2$. An *act* is a mapping $f : \Omega \to \Delta(X)$.[6] Let $\mathcal{F}$ be the set of all acts, endowed with the Euclidean metric d. A *menu* is a nonempty compact subset of $\mathcal{F}$, typically denoted by F, G. Let $\mathcal{M}$ be the set of all menus. Endow $\mathcal{M}$ with the Hausdorff metric d_h.[7] A *menu of menus* is a nonempty compact subset of $\mathcal{M}$, typically denoted by $\mathbb{F}, \mathbb{G}$. Let $\mathcal{D}$ be the set of all menus of menus. Endow $\mathcal{D}$ with the Hausdorff metric d_H.[8] Both $\mathcal{M}$ and $\mathcal{D}$ are compact.[9]

The set $\mathcal{F}$ is equipped with the standard mixture operation. If $f, g \in \mathcal{F}$ and $\alpha \in [0, 1]$, then $\alpha f + (1-\alpha)g$ is an act defined by $\big(\alpha f + (1-\alpha)g\big)(\omega) := \alpha f(\omega) + (1-\alpha)g(\omega)$. For any $F, G \in \mathcal{M}$ and $\alpha \in [0, 1]$, $\alpha F + (1-\alpha)G := \{\alpha f + (1-\alpha)g \mid f \in F \text{ and } g \in G\}$. For any $\mathbb{F}, \mathbb{G} \in \mathcal{D}$ and $\alpha \in [0, 1]$, $\alpha\mathbb{F} + (1-\alpha)\mathbb{G} := \{\alpha F + (1-\alpha)G \mid F \in \mathbb{F} \text{ and } G \in \mathbb{G}\}$.

[6] For finite Y, $\Delta(Y)$ is the probability simplex over Y, equipped with the usual mixture operation.

[7] This is the metric defined by $d_h(F, G) := \max\{\max_{f\in F} \min_{g\in G} d(f, g), \max_{f\in G} \min_{g\in F} d(f, g)\}$.

[8] d_H is defined by $d_H(\mathbb{F}, \mathbb{G}) := \max\{\max_{F\in\mathbb{F}} \min_{G\in\mathbb{G}} d_h(F, G), \max_{F\in\mathbb{G}} \min_{G\in\mathbb{F}} d_h(F, G)\}$.

[9] See, for example, Aliprantis and Border (2006, Theorem 3.85).

An *information structure* is a Blackwell experiment $\sigma : \Omega \to \Delta(S)$ where $S \subseteq [0,1]$ is a finite set of signals.[10] We write $\sigma(s \mid \omega)$ to denote the probability of observing signal s when the state is ω. Let $\mathcal{I}$ denote the set of all information structures.

The primitive of our model is a binary relation on $\mathcal{D} \times \mathcal{I}$, representing the agent's preference over pairs consisting of a menu of menus and an information structure. We have in mind an agent facing a three-period decision problem. In period 1, the agent jointly chooses a menu of menus $\mathbb{F}$ (the "consumption choice") and an information structure σ (the "information choice"). In period 2, the agent chooses a menu F from $\mathbb{F}$, anticipating the information to arrive after this choice. A signal $s \in S$ is then generated according to σ and observed by the agent. In period 3, the agent updates her belief and chooses an act f from F.[11] We do not explicitly model the agent's choices in periods 2 and 3, leaving them as part of the interpretations of the model. The timeline for events is summarized in Figure 2.

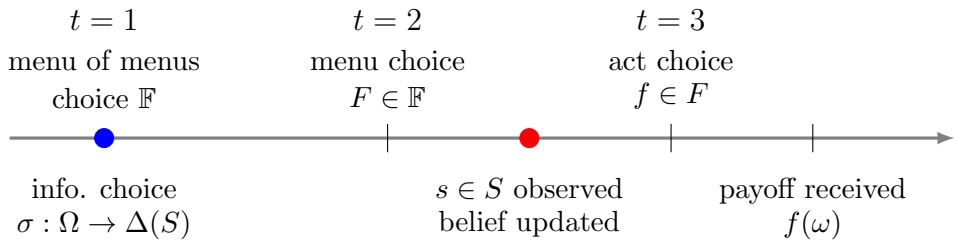


Figure 2: Timeline of Events

The baseline representation associates regret with the period-2 menu choice. After observing the signal, the agent compares the chosen menu with the other menus that were available in $\mathbb{F}$. It is still useful if we allow regrets associated with period-1 or period-3 consumption choices. Section 5.2 studies regret about the initial menu-of-menus choice and shows that the extension preserves consumption choices for a fixed information structure. Section 5.3 considers regret about the final choice of act triggered by knowing the state.

3 Subjective IT Representation

To build toward the axiomatization of the IT representation in Section 4, we first study a binary relation $\succsim$ on $\mathcal{D}$, representing the agent's preference over menus of menus of acts. In this restricted domain, the timeline is the same as in Figure 2 except that the agent chooses only $\mathbb{F}$ in period 1. The information structure is an unobserved preference parameter rather

[10] The restriction that the signal space S is a subset of $[0,1]$ is without loss of generality and to avoid set-theoretic bookkeeping over a "set of all sets".

[11] In some applications, it is more realistic for the information choice to be made after the menu choice. Our model can easily accommodate this change since our agent is both forward-looking and Bayesian.

than a choice variable. As mentioned in the introduction, this domain restriction makes it hard for us to interpret the representation as a theory of information avoidance. That said, the resulting subjective informational tradeoff (SIT) representation is useful as an intermediate step and also has an interpretation of its own. Section 3.1 explains how the anticipated information can be elicited from this preference.

DEFINITION 1: A *subjective informational tradeoff (SIT) representation* is a tuple (π, u, K, σ) that consists of a probability measure π on Ω, a non-constant affine function $u : \Delta(X) \to \mathbb{R}$, a constant $K \geq 0$, and an information structure $\sigma : \Omega \to \Delta(S)$ such that $\succsim$ can be represented by the function $V : \mathcal{D} \to \mathbb{R}$ defined by

$$V(\mathbb{F}) = \max_{F \in \mathbb{F}} \sum_{s \in S} \sigma_s^\pi \Big[U(F, \mu_s^\sigma) - R(F, \mathbb{F}, \mu_s^\sigma) \Big] \tag{3}$$

where $\sigma_s^\pi = \sum_{\omega \in \Omega} \pi(\omega)\sigma(s \mid \omega)$ is the ex-ante probability of signal s under prior π, μ_s^σ is the agent's Bayesian posterior after observing s,[12] and

$$U(F, \mu_s^\sigma) = \max_{f \in F} \sum_{\omega \in \Omega} \mu_s^\sigma(\omega) u\big(f(\omega)\big) \tag{4}$$

is the highest expected value that can be obtained by an act in F under posterior μ_s^σ. Moreover,

$$R(F, \mathbb{F}, \mu_s^\sigma) = K \left[\max_{G \in \mathbb{F}} U(G, \mu_s^\sigma) - U(F, \mu_s^\sigma) \right] \tag{5}$$

is the regret the agent experiences upon observing signal s, having chosen menu F from $\mathbb{F}$ before the signal arrived. Lastly, $K \geq 0$ represents the agent's regret intensity.

We impose eight axioms on $\succsim$. The first three axioms are standard in the setting of preferences over menus of lotteries and are simply adapted to our setting.

AXIOM 1—Weak Order: $\succsim$ *is complete and transitive.*

AXIOM 2—Continuity: *For any* $\mathbb{F} \in \mathcal{D}$*, the sets* $\{\mathbb{G} : \mathbb{F} \succsim \mathbb{G}\}$, $\{\mathbb{G} : \mathbb{G} \succsim \mathbb{F}\}$ *are closed.*

AXIOM 3—Independence: *For any* $\mathbb{F}, \mathbb{G}, \mathbb{H} \in \mathcal{D}$ *and any* $\alpha \in (0, 1)$,

$$\mathbb{F} \succsim \mathbb{G} \iff \alpha\mathbb{F} + (1-\alpha)\mathbb{H} \succsim \alpha\mathbb{G} + (1-\alpha)\mathbb{H}.$$

We refer readers to Dekel, Lipman, and Rustichini (2001), Dekel, Lipman, Rustichini, and Sarver (2007) and Kopylov (2009) for a discussion of these axioms.

To state the next axiom, we introduce the notion of a "critical" subset.

DEFINITION 2: We say that: (2.1) $\mathbb{G}$ is *critical* for $\mathbb{F}$ if $\mathbb{G} \subseteq \mathbb{F}$ and $\mathbb{H} \sim \mathbb{F}$ for any $\mathbb{H} \in \mathcal{D}$ satisfying $\mathbb{G} \subseteq \mathbb{H} \subseteq \mathbb{F}$. (2.2) For a finite $\mathbb{F} \in \mathcal{D}$, we say G is *critical for* F *in* $\mathbb{F}$ if $G \subseteq F \in \mathbb{F}$ and $(\mathbb{F} \setminus \{F\}) \cup \{H\} \sim \mathbb{F}$ for any menu $H \in \mathcal{M}$ satisfying $G \subseteq H \subseteq F$.

[12] If $\sigma_s^\pi > 0$, $\mu_s^\sigma(\omega) = \pi(\omega)\sigma(s \mid \omega)/\sigma_s^\pi$. The dependence of μ_s^σ on π is suppressed whenever the prior is clear from context. Without loss, we set μ_s^σ to be uniform whenever $\sigma_s^\pi = 0$.

In words, if $\mathbb{G}$ is critical for $\mathbb{F}$, then menus in $\mathbb{F}$ but outside $\mathbb{G}$ are irrelevant for the evaluation of $\mathbb{F}$. The intuition is similar for a menu G being critical for F in $\mathbb{F}$. It suffices to define this inner criticality for finite menus of menus and it makes sure that the replacement operation $(\mathbb{F} \setminus \{F\}) \cup \{H\}$ generates a compact set of menus.

Axiom 4—Finiteness: *There exists a natural number N such that: (4.1) For every $\mathbb{F}$, there exists $\mathbb{G}$ with $|\mathbb{G}| < N$ such that $\mathbb{G}$ is critical for $\mathbb{F}$; and (4.2) For every finite $\mathbb{F}$ and every $F \in \mathbb{F}$, there exists G with $|G| < N$ such that G is critical for F in $\mathbb{F}$.*

In essence, Axiom 4 states that, given a menu of menus, only finitely many menus within it matter for its evaluation; when the menu of menus is finite, only finitely many acts within any one of its menus matter for its evaluation. This restricts the agent to entertaining only finitely many beliefs over Ω as possible posteriors.

Axiom 4 is adapted from the Finiteness axiom from Stovall (2018), who studies preferences over menus of menus of lotteries. We refer the reader to Stovall (2018) for a more detailed discussion of this axiom and its connection with the finiteness axioms stated in Dekel, Lipman, and Rustichini (2009) and Kopylov (2009).

The following axiom is the key to capturing regret:

Axiom 5—Ex-Ante Regret: *If $\{F\} \succsim \{G\}$ and $F \in \mathbb{F}$, then $\mathbb{F} \succsim \mathbb{F} \cup \{G\}$.*

Axiom 5 is adapted from the Dominance axiom from Sarver (2008). Contrary to standard models, Axiom 5 allows for the possibility that $\mathbb{F} \succ \mathbb{F} \cup \{G\}$, that is, the agent may strictly prefer not to add a menu G to a menu of menus $\mathbb{F}$. The axiom concerns additions that do not improve the optimal ex-ante material value available from the menu of menus. Such an addition can nevertheless increase anticipated regret.

The next axiom reflects the fact that information can still benefit the agent because she values flexibility for her final act choice.

Axiom 6—Interim Preference for Flexibility: $\mathbb{F} \cup \{F \cup G\} \succsim \mathbb{F} \cup \{F, G\}$ *for any menu of menus $\mathbb{F}$ and menus F, G.*

Note that the two menus of menus in the comparison correspond to the same set of acts that can be ultimately chosen. They only differ in how much flexibility can be retained after the arrival of information. The agent can choose $F \cup G$ from the menu of menus on the left hand side and choose acts from both F and G after observing any signal. But she has to decide between F and G from the right-hand-side menu of menus before the information arrives. Selecting F means she has to let go of acts that are in G but not in F.

To state the next axiom, we introduce the notion of domination.

Definition 3: An act f *dominates* act g if $\{\{f(\omega)\}\} \succsim \{\{g(\omega)\}\}$ for all $\omega \in \Omega$. A menu F *dominates* menu G if for any $g \in G$, there exists $f \in F$ such that f dominates g. A menu of menus $\mathbb{F}$ *dominates* $\mathbb{G}$ if for any $G \in \mathbb{G}$, there exists $F \in \mathbb{F}$ such that F dominates G.

This definition is built on state-by-state comparisons between acts. We follow a common practice that uses ℓ to also denote the constant act that yields lottery ℓ in every state. For simplicity, we identify an act f with $\{\{f\}\}$ when writing preferences.

Axiom 7—Domination: *(7.1) If f dominates g, then $f \succsim g$ and $\{\{f,g\}\} \sim \{\{f\}\}$; (7.2) If $\mathbb{F}$ dominates $\mathbb{G}$, then $\mathbb{F} \sim \mathbb{F} \cup \mathbb{G}$.*

The first half of (7.1) imposes monotonicity on the agent's preference over acts. The second half of (7.1) states that adding a dominated act to a menu does not change an agent's attitude toward the menu. The intuition is that a dominated act contributes to neither the material value nor the regret. The intuition for part (7.2) is similar.

Axiom 8—Nontriviality: *There exist lotteries $\ell, \ell' \in \Delta(X)$ with $\ell \succ \ell'$.*

Theorem 1: *A binary relation $\succsim$ over $\mathcal{D}$ has a subjective informational tradeoff representation if and only if it satisfies Axioms 1-8.*

The proof of Theorem 1 is contained in Appendix B.

The necessity of the axioms is relatively easy to check.

There are two steps to prove the sufficiency of Axioms 1-8. First, we axiomatize a representation (through Theorem 6) tightly connected with the SIT representation. This so-called *partial regret* (PR) representation is for preferences over menus of menus of lotteries. The PR representation is the counterpart of the SIT representation within a framework using lotteries.[13] The setup for this framework and preliminary results are contained in Appendix A. In the second step, we follow the technique of Dillenberger, Lleras, Sadowski, and Takeoka (2014, henceforth DLST) and use a sequence of geometric arguments to establish a connection between preferences over menus of menus of acts (for the SIT representation) and preferences over menus of menus of lotteries (for the PR representation).

3.1 Uniqueness of the SIT Representation

A *distribution over posteriors,* denoted by ν, is a finitely-supported probability measure on $\Delta(\Omega)$. We say that ν is *induced by* a prior π and an information structure $\sigma : \Omega \to \Delta(S)$ if $\nu(\mu) = \sum_{s \in S} \mathbf{1}[\mu = \mu_s^\sigma]\sigma_s^\pi$ for any $\mu \in \Delta(\Omega)$, where $\mathbf{1}[\cdot]$ is the indicator function, and σ_s^π and μ_s^σ are defined as before.

Note from equation (3) that if two information structures σ and σ' induce the same distribution over posteriors given a prior π, then the SIT representations with parameters

[13] The agent chooses a menu of menus of lotteries in period 1 and then selects a menu from it in period 2. The uncertainty about her tastes over lotteries is resolved after the menu choice but before her lottery choice in period 3. Different from Sarver (2008), the agent still has some flexibility after the resolution of her taste uncertainty since she can choose the best lottery according to the realized taste. However, she is still exposed to the regret from letting go of a menu containing better lotteries.

(π, u, K, σ) and (π, u, K, σ') represent the same preference over $\mathcal{D}$. Therefore, we can only identify the information up to the distribution over posteriors it induces.

We say an information structure σ induces a *degenerate* distribution over posteriors if the induced distribution puts weight one on the prior.

THEOREM 2: *Suppose both $(\pi_0, u_0, K_0, \sigma_0)$ and (π, u, K, σ) represent $\succsim$. Then $\pi_0 = \pi$. $u_0 = au + b$ for some $a > 0$ and $b \in \mathbb{R}$. σ_0 and σ induce the same distribution over posteriors. $K_0 = K$ if σ induces a non-degenerate distribution over posteriors. If σ induces a degenerate distribution over posteriors, replacing K by any nonnegative scalar leaves the represented preference unchanged.*

The proof of Theorem 2 is relegated to Appendix C.1.

The identification of the prior π and the taste u follows from the Anscombe–Aumann act domain. Separate identification of the regret intensity K and the information σ is achieved through applying the SIT representation to two special types of menus of menus.

For any menu F, let $D(F) := \{\{f\} \mid f \in F\}$. We say a menu of menus $\mathbb{F}$ corresponds to an *early commitment* if $\mathbb{F} = D(F)$ for some F.

If $\mathbb{F}$ corresponds to an early commitment, $\mathbb{F}$ contains only singleton menus. That is, the agent has to commit to an act before she receives any information. Thus, information carries zero instrumental value and only contributes to regret. Formally,

$$V\big(D(F)\big) = \max_{f \in F} \left[(1 + K) \sum_{\omega \in \Omega} \pi(\omega) u(f(\omega)) \right] - K \sum_{s \in S} \max_{g \in F} \sum_{\omega} \pi(\omega) \sigma(s \mid \omega) u(g(\omega)). \tag{6}$$

This is the counterpart of the regret representation in Sarver (2008) in an act domain, with posterior beliefs taking the place of subjective tastes. The same functional form appears as the Subjective Late Information representation in Pennesi (2026). The identification obstacle remains on this restricted domain: a high regret intensity with little anticipated belief updating can generate the same choices as a lower regret intensity with more anticipated updating. For example, for $K > 0$, the specifications (K, σ) and $(2K, \frac{1}{2}\sigma + \frac{1}{2}o)$ give the same evaluation of every $D(F)$. Here the mixture means that the agent receives a signal from σ or a null signal with equal probability and observes which occurs. The mixture halves expected regret, while doubling K offsets this change.

The larger consumption domain resolves this problem. We say that a menu of menus $\mathbb{F}$ corresponds to *full flexibility* if $\mathbb{F} = \{F\}$ for some menu F. That is, the agent can make all her relevant decisions after the information arrives. Therefore, information causes zero regret and only carries instrumental value. With full flexibility, the SIT representation reduces to

$$V\big(\{F\}\big) = \sum_{s \in S} \max_{f \in F} \sum_{\omega \in \Omega} \pi(\omega) \sigma(s \mid \omega) u(f(\omega)). \tag{7}$$

Equation (7) corresponds to a finite version of the subjective learning representation in DLST. DLST show that the information structure can be uniquely identified up to the

induced distribution over posteriors. The identification of K follows once the distribution of posterior beliefs is identified, provided that this distribution is nondegenerate. Thus, the additional identification comes from observing the value of flexibility separately from anticipated regret, rather than from moving from menus of lotteries to menus of acts alone.

4 Informational Tradeoff Representation

With the axioms for the SIT representation in hand, we switch back to the larger choice domain $\mathcal{D} \times \mathcal{I}$. To minimize notation, we use $\succsim$ to denote a binary relation over $\mathcal{D} \times \mathcal{I}$ from this point on.[14]

The IT representation is the central representation of the paper because our question concerns preferences over information. Its domain allows the agent to compare information structures while holding the menu of menus fixed. Information avoidance can therefore be expressed directly as a preference for a less informative experiment. Preferences over menus of menus alone reveal the information an agent expects to receive, but do not directly reveal whether she would prefer to seek or avoid that information.

The SIT representation developed in Section 3 provides a building block for this analysis. For each fixed information structure σ, the conditional preference over menus of menus has a SIT representation. The additional axioms below connect these conditional preferences into a single preference over consumption-information pairs. Although the SIT representation is independently interesting, its role here is to help establish the behavioral foundations of IT. Comparative statics provide further implications of the IT representation; its connection to preferences over information is the reason for studying it.

Definition 4: A binary relation $\succsim$ over $\mathcal{D} \times \mathcal{I}$ has an *informational tradeoff (IT) representation* if there exists a triple (π, u, K) that consists of a probability measure π on Ω, a non-constant affine function $u : \Delta(X) \to \mathbb{R}$, and a constant $K \geq 0$ such that $\succsim$ can be represented by the function $W : \mathcal{D} \times \mathcal{I} \to \mathbb{R}$ defined by

$$W(\mathbb{F}, \sigma) := \max_{F \in \mathbb{F}} \sum_{s \in S} \sigma_s^\pi \left[U(F, \mu_s^\sigma) - R(F, \mathbb{F}, \mu_s^\sigma) \right] \tag{8}$$

where σ_s^π, μ_s^σ, material utility for a menu $U(F, \mu_s^\sigma)$, and the regret $R(F, \mathbb{F}, \mu_s^\sigma)$ are all defined in the same way as before.

As illustrated in the numerical example in the Introduction, given a menu of menus $\mathbb{F}$, it is possible that $W(\mathbb{F}, \sigma) > W(\mathbb{F}, \sigma')$ when σ' is Blackwell more informative than σ. Formally, we say an agent *avoids information* facing $\mathbb{F}$ if $(\mathbb{F}, \sigma) \succ (\mathbb{F}, \sigma')$ with σ' being Blackwell

[14] Jakobsen (2021) uses a similar primitive to axiomatize Bayesian persuasion models. In his paper, the modeler observes a sender's preference over information indexed by menus of acts and a receiver's choice correspondence from menus indexed by signals.

more informative than σ, denoted by $\sigma' \trianglerighteq \sigma$. Blackwell's informativeness theorem dictates that being Blackwell more informative is equivalent to generating a weakly higher expected utility in every standard decision problem, for every prior. Therefore, using such preference reversals is an intuitive (and in a sense canonical) way to define information avoidance.

Given a preference $\succsim$ over $\mathcal{D} \times \mathcal{I}$, each information structure $\sigma \in \mathcal{I}$ induces a preference over menus of menus of acts. Formally, a *conditional preference* $\succsim^\sigma$ is a binary relation over $\mathcal{D}$ where we say $\mathbb{F} \succsim^\sigma \mathbb{G}$ if $(\mathbb{F}, \sigma) \succsim (\mathbb{G}, \sigma)$.

In addition to previous axioms for the SIT representation (which will be imposed within each $\succsim^\sigma$), we impose new axioms to discipline the agent's choices between menus of menus across *different* information structures.

The first one is the standard rationality requirement.

Axiom 9—Weak Order: $\succsim$ *(as a preference over $\mathcal{D} \times \mathcal{I}$) is complete and transitive.*

Note that if a binary relation $\succsim$ over $\mathcal{D} \times \mathcal{I}$ is complete and transitive, then the induced conditional preferences $\succsim^\sigma$ must also be complete and transitive.

The next axiom imposes an independence requirement on mixing menus of menus with acts. For convenience, we write $\alpha\mathbb{F} + (1-\alpha)h$ for $\alpha\mathbb{F} + (1-\alpha)\{\{h\}\}$.

Axiom 10—Act Independence: *For any $(\mathbb{F}, \sigma)$, $(\mathbb{G}, \sigma')$, any act h and any $\alpha \in (0,1)$,*

$$(\mathbb{F}, \sigma) \succsim (\mathbb{G}, \sigma') \iff (\alpha\mathbb{F} + (1-\alpha)h, \sigma) \succsim (\alpha\mathbb{G} + (1-\alpha)h, \sigma').$$

Axiom 10 reflects the intuitive assumption that information should have no impact on the agent's preference over acts. This is because the information causes no regret and generates no instrumental value when the agent is constrained to some $\{\{h\}\}$. Therefore, the pair $(\alpha\mathbb{F} + (1-\alpha)h, \sigma)$ can be interpreted as if the agent has chosen σ as her information structure but faces uncertainty about whether she will get $\mathbb{F}$ or $\{\{h\}\}$. In this sense, the motivation for Axiom 10 is the same as any standard independence axiom.

An information structure $\sigma \in \mathcal{I}$ is *null* (or *uninformative*) if it always induces a degenerate distribution over posteriors under any prior.

Let $o : \Omega \to \Delta(\{0\})$ denote a specific null information structure. It has only one signal 0 which is obtained with probability one in every state. Our next axiom disciplines the agent's behavior when the observed information is exactly o.

Axiom 11—Strategic Rationality under Null Information (SRNI): *For any menus F, G, if $\{F\} \succsim^o \{G\}$, then $\{F \cup G\} \sim^o \{F\}$.*

Under null information, a menu that maximizes material value under the prior also maximizes the agent's objective and incurs no regret. In particular, the value of $\{F\}$ under o is $U(F, \pi)$. Axiom 11 is the standard strategic rationality requirement in this situation.

If $\succsim^o$ has a SIT representation, Theorem 2 guarantees the identification of some information structure i^o. We show that if $\succsim^o$ also satisfies Axiom 11 (SRNI), then the identified i^o and o both induce a degenerate distribution over posteriors.

LEMMA 1: *If $\succsim^o$ has a SIT representation (π^o, u^o, K^o, i^o), then $\succsim^o$ satisfies Axiom 11 if and only if i^o induces a degenerate distribution over posteriors.*

Proof. *"If."* If i^o induces a degenerate distribution over posteriors, then $\succsim^o$ is represented by $V^o(\mathbb{F}) = \max_{F\in\mathbb{F}} \max_{f\in F} \sum_{\omega\in\Omega} \pi^o(\omega)u^o(f(\omega))$. If $V^o(\{F\}) \geq V^o(\{G\})$, then $V^o(\{F\cup G\}) = V^o(\{F\})$. SRNI is thus satisfied.

"Only if." Suppose i^o induces a distribution over posteriors that contains at least two different elements, μ_1 and μ_2, in its support. Then by a standard result, there exist two acts f and g such that f is strictly better than g in μ_1 and g is strictly better than f in μ_2. Therefore, $V^o(\{\{f,g\}\}) > \max\{V^o(\{\{f\}\}), V^o(\{\{g\}\})\}$, violating SRNI. □

The experiment o is not the only null information structure. However, we only need to impose the SRNI axiom on $\succsim^o$ because the rest of our axioms will imply that Blackwell equivalent information structures induce the same conditional preference.

Let $\sigma : \Omega \to \Delta(S)$ be an information structure and F be a menu. A *plan* is a mapping $\gamma : S \to F$, where $\gamma(s)$ is the act chosen by the agent if she observes signal s. Let F^S denote the set of all such plans. Because F is compact and S is finite, F^S is compact. Intuitively, a plan describes the agent's reaction to the information. For example, if $F = \{f, g\}$ and $S = \{s_1, s_2\}$, then the agent has four different plans. The first is to choose f after s_1 and choose g after s_2 (fg in short). The other three are ff, gf and gg.

Given an information structure $\sigma : \Omega \to \Delta(S)$ and a plan $\gamma \in F^S$, the *act induced by σ and γ*, denoted by γ_σ, is an act defined by $\gamma_\sigma(\omega) := \sum_{s\in S} \sigma(s \mid \omega)[\gamma(s)](\omega)$. That is, γ_σ is obtained by reducing the "compound act" described by γ and σ by averaging over different signals. To illustrate, if $F = \{f, g\}$ and $S = \{s_1, s_2\}$, the information structure $\sigma : \Omega \to \Delta(S)$ together with the plan fg describe the following compound act: Contingent on ω being the state of the world, the agent will obtain $f(\omega)$ (when the signal is s_1 and she chooses f according to the plan fg) with probability $\sigma(s_1 \mid \omega)$ and obtain $g(\omega)$ with probability $\sigma(s_2 \mid \omega)$. The induced act $(fg)_\sigma$ is obtained by the convex combination $\sigma(s_1 \mid \omega)f(\omega) + \sigma(s_2 \mid \omega)g(\omega)$.

DEFINITION 5: Fix a menu F and some information structure σ, *the menu induced by F and σ* is defined by $F_\sigma := \left\{\gamma_\sigma \mid \gamma \in F^S\right\}$.

That is, F_σ is obtained by collecting all the acts that can be induced by σ and some plan $\gamma \in F^S$. For singleton menus like $F = \{f\}$, the set of plans F^S is a singleton containing only one plan (choosing act f after any signal). Therefore, the induced menu is $F_\sigma = \{f\}$ for any information structure σ.

For any menu F and any $\sigma \in \mathcal{I}$, the induced menu F_σ is compact.[15]

[15]The space of plans F^S, endowed with the product topology, is compact. Define $T_\sigma : F^S \to \mathcal{F}$ by $[T_\sigma(\gamma)](\omega) = \sum_{s\in S} \sigma(s \mid \omega)\,[\gamma(s)](\omega)$. Each coordinate of $T_\sigma(\gamma)$ is a finite linear combination of the coordinates of γ, so T_σ is continuous. By construction, $F_\sigma = T_\sigma(F^S)$. Thus F_σ is the continuous image of a compact space and is therefore compact.

AXIOM 12—Reduction: $(\{F\}, \sigma) \sim (\{F_\sigma\}, o)$ *for any* F *and any* σ.

Axiom 12 is motivated by the common notion that information carries only instrumental value (but no intrinsic value) when there is no regret. It states that the agent only cares about the state-contingent distributions over outcomes generated by the information structure.

An important implication of Axiom 12 is that $(f, \sigma) \sim (f, o)$ for any act f and information structure σ.[16] Transitivity then gives $(f, \sigma) \sim (f, \sigma')$ for any σ, σ'. Thus, the agent is indifferent between information structures when committed to a single act, and her ranking of acts is independent of information.

Our last axiom concerns mixtures of early-commitment objects $D(F)$ and their corresponding full-flexibility objects $\{F\}$. Their interaction has a clear implication for the agent's regret intensity.

AXIOM 13—Balance: *If* $(\{F\}, \sigma_1) \succ (\{F\}, o)$, $(\{G\}, \sigma_2) \succ (\{G\}, o)$, $\alpha \in (0, 1]$,

$$\Big(\alpha D(F) + (1-\alpha)\{F\}, \sigma_1\Big) \sim (\{F\}, o) \iff \Big(\alpha D(G) + (1-\alpha)\{G\}, \sigma_2\Big) \sim (\{G\}, o).$$

If the agent strictly prefers σ_1 to o when facing $\{F\}$, then the agent weakly prefers o to σ_1 when facing $D(F)$. In the representation this preference is strict when $K > 0$, because expected regret for $D(F)$ is proportional to the instrumental value for $\{F\}$; when $K = 0$, the agent is indifferent. The mixture $\alpha D(F) + (1-\alpha)\{F\}$ effectively describes a scenario where the agent needs to first choose an act from F, but has uncertainty about whether she has to stick to this choice (corresponding to $D(F)$) or she has the chance to re-optimize within F (corresponding to $\{F\}$). Axiom 13 implies that when facing such an uncertainty, there is a weight α that precisely balances out the expected regret generated from facing $D(F)$ and the expected benefit from facing $\{F\}$. At this balance point, the agent is indifferent between facing $\alpha D(F) + (1-\alpha)\{F\}$ with information σ and facing $\{F\}$ with no information at all.

Axiom 13 requires the same balance point $\alpha^* \in (0, 1]$ whenever the relevant strict information comparisons exist, where $\Big(\alpha^* D(F) + (1-\alpha^*)\{F\}, \sigma\Big) \sim (\{F\}, o)$ for all such menus $F \in \mathcal{M}$ and information structures $\sigma \in \mathcal{I}$. Its invariance across the consumption and information objects is the behavioral content that imposes a single regret intensity in every conditional problem.

THEOREM 3: *A binary relation* $\succsim$ *over* $\mathcal{D} \times \mathcal{I}$ *has an informational tradeoff representation if and only if: (1)* $\succsim$ *satisfies Weak Order, Act Independence, SRNI, Reduction, and Balance; and (2)* $\succsim^\sigma$ *satisfies Axioms 2-8 for each* $\sigma \in \mathcal{I}$.

The proof of Theorem 3 is relegated to Appendix C.2.

[16] As before, we abuse notation a little by writing (f, σ) for the pair $(\{\{f\}\}, \sigma)$.

4.1 Uniqueness of the IT Representation

As we have seen, non-variation between the agent's prior and posterior beliefs is a threat to identification. Since the modeler can observe the agent's information choice in the IT representation, such non-variation could only happen when the agent's prior belief is degenerate.

We say an agent *has a trivial preference over information* if $(\mathbb{F}, \sigma) \sim (\mathbb{F}, \sigma')$ for all $\mathbb{F} \in \mathcal{D}$ and $\sigma, \sigma' \in \mathcal{I}$. That is, the agent is indifferent between any two information structures regardless of the menu of menus. The agent *has a non-trivial preference over information* if she does not have a trivial preference over information.

LEMMA 2: *Suppose $\succsim$ has an IT representation with parameters (π, u, K). This preference $\succsim$ has a trivial preference over information if and only if the prior belief π is degenerate.*

Proof. "If." Suppose π is degenerate, that is, there exists exactly one $\omega \in \Omega$ such that $\pi(\omega) = 1$. Then $\mu_s^\sigma = \pi$ for any information structure σ and every signal s with $\sigma_s^\pi > 0$. Hence $W(\mathbb{F}, \sigma) = W(\mathbb{F}, \sigma')$ for all $\mathbb{F} \in \mathcal{D}$ and $\sigma, \sigma' \in \mathcal{I}$, and $\succsim$ has a trivial preference over information.

"Only if." Suppose $\succsim$ has a trivial preference over information, and suppose by contradiction that π is non-degenerate. That is, there exist distinct states $\omega, \omega' \in \Omega$ such that $\pi(\omega), \pi(\omega') > 0$. Since the taste function u is nonconstant, it follows from a standard result that we can construct two acts f, g such that the agent strictly prefers a fully informative experiment to a fully uninformative experiment when facing $\{\{f, g\}\}$. □

As long as the agent has a non-trivial preference over information, all parameters for the IT representation can be uniquely identified.

THEOREM 4: *Suppose $\succsim$ has an informational tradeoff representation (π, u, K) and $\succsim$ has a non-trivial preference over information. Then (π', u', K') also represents $\succsim$ if and only if $\pi' = \pi$, $u' = au + b$ for some $a > 0$ and $b \in \mathbb{R}$, and $K' = K$.*

Proof of Theorem 4. The "if" direction is straightforward.

"Only if." Suppose (π', u', K') also represents $\succsim$. Then (π, u) and (π', u') agree on their preference for acts. By the standard result from the Anscombe-Aumann framework, $\pi' = \pi$ and $u' = au + b$ for some $a > 0$ and $b \in \mathbb{R}$. Since $\succsim$ has a non-trivial preference over information, the prior π is non-degenerate (Lemma 2). Choose a fully informative experiment. Since π is non-degenerate, its induced distribution over posteriors is non-degenerate. Applying Theorem 2 to the conditional preference for this experiment gives $K' = K$. □

4.2 Comparing Information Aversion Attitudes

In this section, we focus our attention on agents with non-trivial preferences over information. Suppose $\succsim_1$, $\succsim_2$ each has an informational tradeoff representation with parameters

(π_1, u_1, K_1) and (π_2, u_2, K_2), respectively.

DEFINITION 6: We say that $\succsim_1$ is *more information averse than* $\succsim_2$ if

$$(\mathbb{F}, \sigma) \succsim_2 (\mathbb{F}, \sigma') \Longrightarrow (\mathbb{F}, \sigma) \succsim_1 (\mathbb{F}, \sigma') \tag{9}$$

for any $\mathbb{F} \in \mathcal{D}$ and any $\sigma, \sigma' \in \mathcal{I}$ such that σ' is Blackwell more informative than σ.

That is, our notion of being "more information averse" requires that whenever agent 2 prefers a less informative experiment to a more informative one given a certain menu of menus, agent 1 will have the same avoidance given the same menu of menus.

We focus on comparing the information aversion attitudes between agents who share the same preference over acts. Formally, we say that $\succsim_1$ and $\succsim_2$ *have the same preference over acts* if $(f, \sigma) \succsim_1 (g, \sigma) \iff (f, \sigma) \succsim_2 (g, \sigma)$ for every $f, g \in \mathcal{F}$ and any fixed $\sigma \in \mathcal{I}$.

THEOREM 5: *If $\succsim_1$ and $\succsim_2$ have the same preference over acts, then $\succsim_1$ is more information averse than $\succsim_2$ if and only if $K_1 \geq K_2$.*

Proof of Theorem 5. Since $\succsim_1$ and $\succsim_2$ have the same preference over acts, $\pi_1 = \pi_2 =: \pi$, $u_2 = au_1 + b$ for some $a > 0$ and $b \in \mathbb{R}$. Therefore, (π_1, u_1, K_2) also represents $\succsim_2$. Let W_1, W_2 be the functionals representing $\succsim_1$ and $\succsim_2$, respectively.

"*If.*" Suppose $K_1 \geq K_2$. Fix any $\mathbb{F} \in \mathcal{D}$ and $\sigma, \sigma' \in \mathcal{I}$ such that $\sigma' \sqsupseteq \sigma$. We want to show

$$W_2(\mathbb{F}, \sigma) - W_2(\mathbb{F}, \sigma') \geq 0 \Longrightarrow W_1(\mathbb{F}, \sigma) - W_1(\mathbb{F}, \sigma') \geq 0 \tag{10}$$

It suffices to show that $\frac{W_1(\mathbb{F},\sigma) - W_1(\mathbb{F},\sigma')}{1+K_1} - \frac{W_2(\mathbb{F},\sigma) - W_2(\mathbb{F},\sigma')}{1+K_2} \geq 0$, which holds because

$$\begin{aligned}
&\frac{W_1(\mathbb{F}, \sigma) - W_1(\mathbb{F}, \sigma')}{1 + K_1} - \frac{W_2(\mathbb{F}, \sigma) - W_2(\mathbb{F}, \sigma')}{1 + K_2} \\
= &\left(\frac{W_1(\mathbb{F}, \sigma)}{1 + K_1} - \frac{W_2(\mathbb{F}, \sigma)}{1 + K_2} \right) - \left(\frac{W_1(\mathbb{F}, \sigma')}{1 + K_1} - \frac{W_2(\mathbb{F}, \sigma')}{1 + K_2} \right) \\
= &-\frac{K_1}{1 + K_1} \sum_{s \in S} \sigma_s^\pi \max_{G \in \mathbb{F}} U(G, \mu_s^\sigma) + \frac{K_2}{1 + K_2} \sum_{s \in S} \sigma_s^\pi \max_{G \in \mathbb{F}} U(G, \mu_s^\sigma) \\
&+ \frac{K_1}{1 + K_1} \sum_{s' \in S'} {\sigma'}_{s'}^{\pi} \max_{G \in \mathbb{F}} U(G, \mu_{s'}^{\sigma'}) - \frac{K_2}{1 + K_2} \sum_{s' \in S'} {\sigma'}_{s'}^{\pi} \max_{G \in \mathbb{F}} U(G, \mu_{s'}^{\sigma'}) \\
= &\left(\frac{K_1}{1 + K_1} - \frac{K_2}{1 + K_2} \right) \left[\sum_{s' \in S'} {\sigma'}_{s'}^{\pi} \max_{G \in \mathbb{F}} U(G, \mu_{s'}^{\sigma'}) - \sum_{s \in S} \sigma_s^\pi \max_{G \in \mathbb{F}} U(G, \mu_s^\sigma) \right] \geq 0,
\end{aligned}$$

where the second equality follows by substituting the equivalent expression for W, and the last inequality follows from $K_1 \geq K_2$ and the fact that σ' is more informative than σ.

"*Only if.*" Suppose $\succsim_1$ is more information averse than $\succsim_2$. Suppose by contradiction that $K_2 > K_1$, we want to construct some $\mathbb{F}$ such that $(\mathbb{F}, o) \succsim_2 (\mathbb{F}, i)$ but $(\mathbb{F}, i) \succ_1 (\mathbb{F}, o)$, where o is the null experiment and i is the fully informative experiment.

By assumption, $\pi_1 = \pi_2 =: \pi$ and π is not degenerate. Thus, there are two states ω and ω' such that $\pi(\omega) = p$ and $\pi(\omega') = q$ with $p, q > 0$.

By a positive affine normalization of the common utility function, assume that its range contains $[0, (p+q)/p]$. Consider the following three acts $f, g, h \in \mathcal{F}$:

	ω	ω'	$\hat{\omega} \in \Omega \setminus \{\omega, \omega'\}$
$u \circ f$	$(p+q)/p$	0	0
$u \circ g$	0	y	0
$u \circ h$	1	1	0

(11)

where $y \in \left(\frac{K_1}{1+K_1}, \frac{K_2}{1+K_2}\right)$. Let $\mathbb{F} = \{\{f, g\}, \{h\}\}$. Then it is straightforward to verify that $\{f, g\}$ attains the maximum under both o and i. Therefore,

$$W_1(\mathbb{F}, o) = \max\Big\{p+q, yq, p+q\Big\} = p+q \tag{12}$$

$$W_2(\mathbb{F}, o) = \max\Big\{p+q, yq, p+q\Big\} = p+q \tag{13}$$

$$W_1(\mathbb{F}, i) = p+q+q[(1+K_1)y - K_1] > p+q = W_1(\mathbb{F}, o) \tag{14}$$

$$W_2(\mathbb{F}, i) = p+q+q[(1+K_2)y - K_2] < p+q = W_2(\mathbb{F}, o) \tag{15}$$

Such an $\mathbb{F}$ can always be constructed when $K_2 > K_1$. This completes the proof. □

5 Discussion and Extensions

We conclude by discussing related literature and three extensions. The first extension allows regret about the initial choice of a menu of menus. The second allows the agent to regret her final act when she later learns the state. The third adds rejoicing from comparison with the worst available menu.

5.1 Related Literature

We are not the first to model regret as the difference between what the agent obtains and the best counterfactual outcome from a forgone choice.[17] Sarver (2008) first identifies anticipated regret from preferences over menus. His dominance axiom separates regret from other motives for desiring a smaller menu, such as temptation. Our paper follows this approach but studies a larger choice domain with an emphasis on the role of information. Buturak and Evren (2017) axiomatize a related representation of choice overload in which an agent tends to adhere more strongly to a default when more options are available.

[17] Earlier examples include Bell (1982), Loomes and Sugden (1982, 1987) and Sugden (1993).

In both papers, the only relevant choice by the agent is made before their subjective uncertainty about their taste is resolved. Thus, this resolution is not beneficial for their future choices. Our model explicitly allows the co-existence of regret for past choices and benefit for future choices.

We are not the first to attempt to build a theoretical model that can account for information avoidance either. Golman, Hagmann, and Loewenstein (2017) provide a survey on information avoidance from multiple disciplines. Caplin and Leahy (2001) introduce a psychological state into the standard expected utility framework and build anticipatory feelings about the future (represented by a belief) directly into the agent's utility function. This framework, even though non-axiomatic, is very general and allows the possibility for information avoidance driven by different anticipatory feelings. Kőszegi (2003) utilizes a similar framework with a focus on patient behavior and builds a model where patients avoid relevant medical information in order to avoid the anxiety from the anticipation of a potential bad outcome. Brunnermeier and Parker (2005) model economic agents who can optimally set their own beliefs facing uncertainty, and such an agent might want to avoid information that breaks their unwarranted optimistic beliefs. Dillenberger (2010) builds an axiomatic model that features a preference for one-shot over gradual resolution of uncertainty. As a result, the agent in his model might behave as if she is averse to some information if the information represents a partial resolution of uncertainty.

Our paper differs from all these works due to our focus on rationalizing the linkage between information avoidance and past decisions.

Finally, our work is broadly related to theories on cognitive dissonance (Festinger 1957) where past decisions could affect subsequent decisions. Yang (2024) axiomatically characterizes a multi-self model where an agent distorts her belief to reconcile it with past decisions. Eyster et al. (2021) study an agent who distorts future decisions to justify past decisions. Our explanation for information avoidance (as a specific form of cognitive dissonance) does not involve any belief or choice distortion.

5.2 Regretting the Choice of Menu of Menus

The IT representation attaches regret to the menu choice, but not the choice of menu of menus. This seems to be a reasonable approximation when only the decision closest to information arrival is salient for comparison. In the college example, the student chooses her college and then her major before receiving information about the labor market. Because the major choice is closer to the arrival of this information, she may compare her chosen major with the other majors available at that college without reconsidering the earlier college choice. This does not imply that regret about the earlier choice is impossible.

It is natural to ask whether the student in the example may regret choosing College 2

({Econ}) rather than College 1 ({Econ, CS}) after observing a signal in favor of the CS major, or generally, if an agent may regret her choice of menu of menus. For each fixed information structure, the baseline model's consumption choices can also be consistent with positive "outer regret", that is, the regret associated with the choice of menu of menus. The following extension, adapted to the present environment from Section 4.3 of Sarver (2008), establishes this limited equivalence.

Before describing the extension, we should notice that the fundamentals of our model only allow for the identification of regret associated with the choice of menu (e.g., choice of major from a college). To identify the outer regret (e.g., choice of the college from a set of colleges), we would need a richer primitive.

Specifically, we can look at an agent's preference over menus of menus of menus (e.g., a finite set of colleges a student can choose from). Recall that $\mathcal{D}$ is the set of menus of menus. Let $\mathbf{F}$ be a nonempty compact set of elements of $\mathcal{D}$, interpreted as an outer menu from which $\mathbb{F}$ is chosen. Suppose the agent has outer and inner regret intensities $K_0, K_1 \geq 0$ and evaluates $(\mathbf{F}, \sigma)$ by

$$W_0(\mathbf{F},\sigma) = \max_{\mathbb{F}\in\mathbf{F}} \max_{F\in\mathbb{F}} \sum_{s\in S} \sigma_s^\pi \Bigg\{ U(F,\mu_s^\sigma) - K_0 \left[\max_{\mathbb{G}\in\mathbf{F}} \max_{G\in\mathbb{G}} U(G,\mu_s^\sigma) - U(F,\mu_s^\sigma) \right] - K_1 \left[\max_{G\in\mathbb{F}} U(G,\mu_s^\sigma) - U(F,\mu_s^\sigma) \right] \Bigg\}. \tag{16}$$

The first bracket compares the chosen menu with every menu that could have been reached through an outer alternative; the second compares it with the menus in the chosen $\mathbb{F}$.

Fix an information structure σ. Let $\mathcal{C}_0^\sigma(\mathbf{F})$ denote the set of $\mathbb{F}$'s selected from $\mathbf{F}$ by (16). For a fixed $\mathbf{F}$, the term $-K_0 \sum_{s\in S} \sigma_s^\pi \max_{\mathbb{G}\in\mathbf{F}} \max_{G\in\mathbb{G}} U(G,\mu_s^\sigma)$ is constant across $\mathbb{F}$. Hence, after dropping that constant, the agent chooses a menu of menus $\mathbb{F}$ from an outer menu $\mathbf{F}$ to maximize

$$(1+K_0+K_1) \max_{F\in\mathbb{F}} \sum_{s\in S} \sigma_s^\pi U(F,\mu_s^\sigma) - K_1 \sum_{s\in S} \sigma_s^\pi \max_{G\in\mathbb{F}} U(G,\mu_s^\sigma).$$

Dividing by $1+K_0$ and defining $K = \dfrac{K_1}{1+K_0}$ gives exactly the IT functional W. In particular,

$$\mathcal{C}_0^\sigma(\mathbf{F}) = \arg\max_{\mathbb{F}\in\mathbf{F}} W(\mathbb{F},\sigma) \text{ for every } \sigma\in\mathcal{I} \text{ and every menu of menus of menus } \mathbf{F}. \tag{17}$$

For every fixed information structure σ, the extended model therefore generates exactly the same choices of a menu of menus as the baseline representation with $K = K_1/(1+K_0)$. Thus, these choices are compatible with regret about the college choice in the example. The identified parameter K measures regret about the menu choice relative to the scale induced by regret about the earlier choice, rather than identifying the coefficient K_1 in

isolation. This equivalence concerns the menu-of-menus choice with information held fixed; the omitted outer-regret term may affect comparisons across information structures.

5.3 Regretting the Choice of Act

If the signal only partially resolves uncertainty, the agent may regret her final act when she later learns the state. The following extension makes this possibility explicit and shows that it need not change information rankings. Consider

$$W_1(\mathbb{F},\sigma) := \max_{F\in\mathbb{F}} \sum_{s\in S} \sigma_s^{\pi}\Big[\widetilde{U}(F,\mathbb{F},\mu_s^{\sigma}) - R(F,\mathbb{F},\mu_s^{\sigma})\Big] \tag{18}$$

where σ_s^{π}, μ_s^{σ}, and the regret term $R(F,\mathbb{F},\mu_s^{\sigma})$ are defined as before. That is,

$$R(F,\mathbb{F},\mu_s^{\sigma}) = K_2\left[\max_{G\in\mathbb{F}}\max_{g\in G}\sum_{\omega\in\Omega}\mu_s^{\sigma}(\omega)u\big(g(\omega)\big) - \max_{f\in F}\sum_{\omega\in\Omega}\mu_s^{\sigma}(\omega)u\big(f(\omega)\big)\right]$$

where $K_2 \geq 0$ represents the regret intensity. However, the "material utility" of a menu F under a posterior belief μ_s^{σ}, denoted by $\widetilde{U}$, is now also dependent on the menu of menus $\mathbb{F}$ from which it is chosen. Formally,

$$\widetilde{U}(F,\mathbb{F},\mu_s^{\sigma}) := \max_{f\in F}\sum_{\omega\in\Omega}\mu_s^{\sigma}(\omega)\Big[u\big(f(\omega)\big) - \widetilde{R}(f,F,\mathbb{F},\omega)\Big] \tag{19}$$

where $\widetilde{R}(f,F,\mathbb{F},\omega)$ is the regret for choosing f if the state turns out to be ω. Formally,

$$\widetilde{R}(f,F,\mathbb{F},\omega) := K_3\left[\max_{G\in\mathbb{F}}\max_{g\in G}u\big(g(\omega)\big) - u\big(f(\omega)\big)\right] + K_4\left[\max_{h\in F}u\big(h(\omega)\big) - u\big(f(\omega)\big)\right] \tag{20}$$

where the first term reflects the agent's regret toward the counterfactual outcome she could get if she could re-optimize her choice of menu, while the second term reflects the agent's regret toward the counterfactual outcome she could get if she could only re-optimize her choice of act from the chosen menu F. Parameters $K_3, K_4 \geq 0$ represent the respective regret intensities. Thus, utility W_1 in the form of equation (18) has five parameters, (π, u, K_2, K_3, K_4).

We are interested in comparing two agents' attitudes toward information. The first one is an agent who can be represented by an IT representation W with parameters (π, u, K), and the second one is an agent who can be represented by W_1 with parameters (π, u, K_2, K_3, K_4). The following lemma states that, under the indicated restrictions on regret intensities, the possibility of experiencing regret about the act choice does not change the agent's preference over information given any menu of menus.

LEMMA 3: *If representations W and W_1 share the same parameters (π, u) and the regret intensity parameters satisfy $K = \frac{K_2}{1+K_3}$ and $K_4 = 0$, then*

$$W(\mathbb{F},\sigma) \geq W(\mathbb{F},\sigma') \iff W_1(\mathbb{F},\sigma) \geq W_1(\mathbb{F},\sigma')$$

for any menu of menus $\mathbb{F}$ and information structures $\sigma, \sigma' \in \mathcal{I}$.

Proof. See Appendix C.5. □

When $K_4 = 0$, regret after the act choice compares the chosen act with the best act available from the entire menu of menus $\mathbb{F}$. After taking expectations, the counterfactual benchmark is independent of the information structure, while the term involving the chosen act rescales material utility. With $K = K_2/(1+K_3)$, the extended representation is therefore a positive affine transformation of the baseline representation for every fixed $\mathbb{F}$. The two representations consequently generate the same preferences over information. These restrictions matter: regret relative to the best act in the chosen menu, or different relative regret intensities, may change information preferences. An axiomatization of regret associated with the choice of an act is provided by Cheung and Wang (2026).

5.4 Possibility of Rejoicing

The baseline uses the best available menu under the updated belief as the reference point for evaluating the chosen menu. Regret is therefore nonnegative, and this specification leaves no room for rejoicing or negative regret. Rejoicing can arise with a different reference point, such as the certainty equivalent of available alternatives or the worst available menu. The former is similar in spirit to the model of disappointment aversion in Gul (1991). To illustrate the latter possibility, consider adding

$$J(F, \mathbb{F}, \mu) = L\left[U(F, \mu) - \min_{G \in \mathbb{F}} U(G, \mu)\right], \; L \geq 0, \tag{21}$$

to the bracket in the IT representation. Adding this term changes the behavioral foundations of the model. A menu that is never chosen, and may even be dominated, can increase utility by providing a worse alternative for comparison. Ex-Ante Regret and Domination would therefore need to be modified, and the early-commitment objects $D(F)$ would no longer isolate regret. A joint axiomatization would require additional behavioral conditions separating regret and rejoicing.

The information-avoidance mechanism is nevertheless robust to moderate rejoicing. In the numerical example, adding (21) changes the value of full information from $100 - 10K$ to $100 - 10K + 20L$ and the value of no information from 95 to $95 + 5L$. The student avoids full information if and only if $K > \frac{1}{2} + \frac{3}{2}L$.

A The Framework with Lotteries

In this section, we present some preliminary results in the framework with lotteries.

Let Z be a finite set of outcomes. Let $\Delta(Z)$ denote the set of lotteries over Z, with typical elements p, q. Endow $\Delta(Z)$ with the standard Euclidean metric.[18] A menu is a nonempty compact subset of $\Delta(Z)$. Let $\widehat{\mathcal{M}}$ denote the set of all menus, with typical elements A, B. Endow $\widehat{\mathcal{M}}$ with the Hausdorff metric. Let $\mathcal{V} := \left\{v \in \mathbb{R}^Z \mid \sum_{z\in Z} v_z = 0\right\}$ denote the set of normalized expected utilities over $\Delta(Z)$. The following lemma will be useful for later proofs.

LEMMA 4: *Let $\{u_i\}_{i\in I}$ be a finite collection of normalized expected utility functions over $\Delta(Z)$, and let $U : \widehat{\mathcal{M}} \to \mathbb{R}$ be defined by $U(A) := \sum_{i\in I} \max_{p\in A} u_i(p)$. Then, $U(A) = 0$ for all $A \in \widehat{\mathcal{M}}$ if and only if $u_i(p) = 0$ for all $i \in I$ and all $p \in \Delta(Z)$.*

Proof of Lemma 4. The "if" direction is immediate. For the "only if" direction, suppose that $U(A) = 0$ for every $A \in \widehat{\mathcal{M}}$. Taking $A = \Delta(Z)$ gives

$$0 = U(\Delta(Z)) = \sum_{i\in I} \max_{p\in\Delta(Z)} u_i(p) = \sum_{i\in I} \max_{z\in Z} u_i(z).$$

For every $i \in I$, $\sum_{z\in Z} u_i(z) = 0$. Thus, $\max_{z\in Z} u_i(z) \geq 0$ (otherwise the sum is negative). Thus, $\max_{z\in Z} u_i(z) = 0$ for each i, which further indicates that $u_i(z) \leq 0$ for every $z \in Z$. But $\sum_{z\in Z} u_i(z) = 0$, so $u_i(z) = 0$ for each $z \in Z$, indicating $u_i(p) = 0$ for each $p \in \Delta(Z)$. □

A.1 The Partial Regret Representation

A menu of menus is a nonempty compact subset of $\widehat{\mathcal{M}}$. Let $\widehat{\mathcal{D}}$ denote the set of all menus of menus, with typical elements $\mathbb{A}, \mathbb{B}$. Endow $\widehat{\mathcal{D}}$ with the Hausdorff metric. Our primitive is a binary relation $\succsim$ over $\widehat{\mathcal{D}}$.

Let $\mathcal{U} := \{u \in \mathbb{R}^Z \mid \sum_{z\in Z} u_z = 0,\ \sum_{z\in Z} u_z^2 = 1\}$ denote the set of doubly normalized expected-utility functions on $\Delta(Z)$.

DEFINITION 7: A binary relation $\succsim$ over $\widehat{\mathcal{D}}$ has a *partial regret (PR) representation* if there exists a finitely-supported probability measure μ over $\mathcal{U}$ and a scalar $K \geq 0$ such that $\succsim$ is represented by the function $\widehat{V} : \widehat{\mathcal{D}} \to \mathbb{R}$ defined by

$$\widehat{V}(\mathbb{A}) = \max_{A\in\mathbb{A}} \sum_{u\in\text{supp}(\mu)} \mu(u) \left[\max_{p\in A} u(p) - R(A, \mathbb{A}, u)\right] \tag{22}$$

where $R(A, \mathbb{A}, u) := K\left[\max_{B\in\mathbb{A}} \max_{p\in B} u(p) - \max_{p\in A} u(p)\right]$.

The interpretation resembles that for the SIT representation: the agent has subjective uncertainty about her taste over lotteries, which resolves after her committing to a menu but

[18] To save notation, we do not explicitly give this metric a symbol. We do not explicitly name the following Hausdorff metrics building on this metric, either. They are defined in the standard way.

before her choosing a lottery from the menu. Given a taste, she can choose the best lottery from the chosen menu but suffers from regret if another menu contains a better lottery.

A useful equivalent expression for the PR representation is

$$\widehat{V}(\mathbb{A}) = (1+K)\max_{A\in\mathbb{A}} \sum_{u\in\text{supp}(\mu)} \mu(u)\max_{p\in A} u(p) - K \sum_{u\in\text{supp}(\mu)} \mu(u)\max_{B\in\mathbb{A}}\max_{q\in B} u(q) \tag{23}$$

Following are the axioms for the PR representation.

Axiom A.1—Weak Order: $\succsim$ *is complete and transitive.*

Axiom A.2—Continuity: *For any* $\mathbb{A}$*, the sets* $\{\mathbb{B} : \mathbb{B} \succsim \mathbb{A}\}$, $\{\mathbb{B} : \mathbb{A} \succsim \mathbb{B}\}$ *are closed.*

Axiom A.3—Independence: *If* $\mathbb{A} \succ \mathbb{B}$*, then* $\alpha\mathbb{A} + (1-\alpha)\mathbb{C} \succ \alpha\mathbb{B} + (1-\alpha)\mathbb{C}$ *for any* $\mathbb{C}$ *and* $\alpha \in (0,1]$.

Definition 8: Let $\mathbb{A}$ be a menu of menus. Say that $\mathbb{B} \in \widehat{\mathcal{D}}$ is *critical for* $\mathbb{A}$ if $\mathbb{B} \subseteq \mathbb{A}$ and $\mathbb{C} \sim \mathbb{A}$ for all $\mathbb{C} \in \widehat{\mathcal{D}}$ satisfying $\mathbb{B} \subseteq \mathbb{C} \subseteq \mathbb{A}$. If $\mathbb{A}$ is finite, say that $B \in \widehat{\mathcal{M}}$ *is critical for* A *in* $\mathbb{A}$ if $B \subseteq A \in \mathbb{A}$ and $(\mathbb{A} \setminus \{A\}) \cup \{C\} \sim \mathbb{A}$ for all $C \in \widehat{\mathcal{M}}$ satisfying $B \subseteq C \subseteq A$.

Axiom A.4—Finiteness: *There exists a natural number* N *such that: (4.1) For every* $\mathbb{A}$*, there exists* $\mathbb{B}$ *with* $|\mathbb{B}| < N$ *such that* $\mathbb{B}$ *is critical for* $\mathbb{A}$*; and (4.2) For every finite* $\mathbb{A}$ *and every* $A \in \mathbb{A}$*, there exists* B *with* $|B| < N$ *such that* B *is critical for* A *in* $\mathbb{A}$.

Axiom A.5—Ex-Ante Regret: *If* $\{A\} \succsim \{B\}$ *and* $A \in \mathbb{A}$*, then* $\mathbb{A} \succsim \mathbb{A} \cup \{B\}$.

Axiom A.6—Interim Preference for Flexibility: *For any menu of menus* $\mathbb{A}$ *and menus* A, B, $\mathbb{A} \cup \{A \cup B\} \succsim \mathbb{A} \cup \{A, B\}$.

Axiom A.7—Inclusion: *If* $B \subseteq A$ *and* $A \in \mathbb{A}$*, then* $\mathbb{A} \cup \{B\} \succsim \mathbb{A}$.

Axiom A.8—Nontriviality: *There exist* $\mathbb{A}$ *and* $\mathbb{B}$ *such that* $\mathbb{B} \subseteq \mathbb{A}$ *with* $\mathbb{A} \succ \mathbb{B}$.

Axioms A.1-A.6 are adapted from Axioms 1-6, respectively. The inclusion axiom (Axiom A.7) corresponds to a weakened version of the domination axiom (Axiom 7), and the nontriviality axiom (Axiom A.8) is stated slightly differently from Axiom 8.

Theorem 6: *A binary relation* $\succsim$ *over* $\widehat{\mathcal{D}}$ *has a partial regret representation if and only if it satisfies Axioms A.1-A.8.*

A.2 Proof of Theorem 6

We start by establishing a nested DLR representation from Axioms A.1-A.4. DLR refers to Dekel, Lipman, and Rustichini (2009) who establish a finite version of Dekel et al. (2001) for representations with subjective state spaces.

First, we introduce the notion of redundancy for collections of linear functions.

DEFINITION 9: Let $\{U_1, \ldots, U_m\}$ be a collection of continuous linear functions from $\widehat{\mathcal{M}}$ to $\mathbb{R}$. We say this collection is *redundant* if there exists U_i that is constant or if there exists $i \neq j$ such that $U_j = \alpha U_i + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$. We say a collection is *non-redundant* if it is not redundant.

Similarly, if $\{u_1, \ldots, u_n\}$ is a collection of expected utility functions over $\Delta(Z)$, then we say this collection is *redundant* if there exists u_i that is constant or if there exists $i \neq j$ such that $u_j = \alpha u_i + \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$. We take the convention that an empty collection is not redundant.

DEFINITION 10: A binary relation $\succsim$ over $\widehat{\mathcal{D}}$ has a *nested DLR representation* if:

1. $\succsim$ can be represented by

$$V_{DLR}(\mathbb{A}) = \sum_{i \in I^+} \max_{A \in \mathbb{A}} U_i(A) - \sum_{i \in I^-} \max_{A \in \mathbb{A}} U_i(A) \tag{24}$$

   where I^+ and I^- are finite index sets for the positive and negative states, respectively, and $\{U_i\}_{i \in I^+ \cup I^-}$ is a non-redundant collection (it is without loss to assume $I^+ \cap I^- = \emptyset$).

2. Let $I := I^+ \cup I^-$. For each $i \in I$, the function $U_i : \widehat{\mathcal{M}} \to \mathbb{R}$ has a DLR representation

$$U_i(A) = \sum_{k \in P_i} \max_{p \in A} w_{ik}(p) - \sum_{j \in N_i} \max_{p \in A} v_{ij}(p) \tag{25}$$

   where P_i is a finite index set for the positive substates for U_i and N_i is a finite index set of negative substates for U_i (it is without loss to assume $P_i \cap N_i = \emptyset$) and $\{w_{ik}\}_{k \in P_i} \cup \{v_{ij}\}_{j \in N_i}$ is a non-redundant collection of normalized expected utilities over $\Delta(Z)$.

The next lemma establishes the equivalence between the nested DLR representation and Axioms A.1-A.4.

LEMMA 5: *A binary relation $\succsim$ over $\widehat{\mathcal{D}}$ has a nested DLR representation if and only if it satisfies Axioms A.1-A.4.*

A closely related result is contained in the proof of Theorem 5 of Stovall (2018). However, our notion of criticality is slightly different from Stovall's, so we have to build the proof of Lemma 5 from scratch. The proof builds on standard results from Dekel et al. (2009), Kopylov (2009), and Stovall (2018). To save space, it is relegated to Section S1 of the supplementary appendix.

The rest of the proof for Theorem 6 mainly focuses on fine-tuning the states and substates in the nested DLR representation through the other axioms.

AXIOM A.9—Strong Ex-Ante Regret: *If for any $B \in \mathbb{B}$, there exists $A \in \mathbb{A}$ such that $\{A\} \succsim \{B\}$, then $\mathbb{A} \succsim \mathbb{A} \cup \mathbb{B}$.*

Axioms A.5 and A.9 are equivalent when Axioms A.1 and A.2 are satisfied.

LEMMA 6: *If $\succsim$ satisfies Axioms A.1 (Weak Order) and A.2 (Continuity), then it satisfies Axiom A.5 if and only if it satisfies Axiom A.9.*

Proof. The "*if*" part is straightforward.
"*Only if.*" Suppose $\succsim$ satisfies Axiom A.5, and suppose two menus of menus of lotteries $\mathbb{A}$ and $\mathbb{B}$ are such that for all $B \in \mathbb{B}$, there exists $A \in \mathbb{A}$ such that $\{A\} \succsim \{B\}$.

First note that $\widehat{\mathcal{M}}$ is compact with the Hausdorff metric. Therefore, $\widehat{\mathcal{M}}$ is separable. We can thus choose a countable dense subset of $\mathbb{B}$, say $\mathbb{B}^* = \{B_i \mid i \in \mathbb{N}\}$. Define a sequence of menus of menus $(\mathbb{B}_n)_{n\in\mathbb{N}}$ by $\mathbb{B}_n = \{B_1, B_2, \ldots, B_n\}$. Then $\mathbb{B}_n \subseteq \mathbb{B}_{n+1}$ for all n and $\mathbb{B}^* = \bigcup_{n=1}^{\infty} \mathbb{B}_n$. By a standard result, $(\mathbb{B}_n)_{n\in\mathbb{N}}$ converges to the closure of $\mathbb{B}^*$, that is, $\mathbb{B}_n \to \overline{\mathbb{B}^*} = \mathbb{B}$ as n goes to infinity. Note that $B_1 \in \mathbb{B}^* \subseteq \mathbb{B}$, so there exists $A_1 \in \mathbb{A}$ such that $\{A_1\} \succsim \{B_1\}$ by assumption, and Axiom A.5 implies that $\mathbb{A} \succsim \mathbb{A} \cup \{B_1\}$. Let $\mathbb{A}_1 = \mathbb{A} \cup \{B_1\} = \mathbb{A} \cup \mathbb{B}_1$. Similarly, $B_2 \in \mathbb{B}^* \subseteq \mathbb{B}$, so there exists $A_2 \in \mathbb{A}$ such that $\{A_2\} \succsim \{B_2\}$, and Axiom A.5 implies that $\mathbb{A}_1 \succsim \mathbb{A}_1 \cup \{B_2\}$. Let $\mathbb{A}_2 = \mathbb{A}_1 \cup \{B_2\} = \mathbb{A} \cup \mathbb{B}_2$. By transitivity, $\mathbb{A} \succsim \mathbb{A}_1$ and $\mathbb{A}_1 \succsim \mathbb{A}_2$ imply that $\mathbb{A} \succsim \mathbb{A}_2 = \mathbb{A} \cup \mathbb{B}_2$. By an induction argument, $\mathbb{A} \succsim \mathbb{A} \cup \mathbb{B}_n$ for all n. Therefore, $\mathbb{A} \cup \mathbb{B}_n$ is in the lower contour set of $\mathbb{A}$ for all n, that is, $\mathbb{A} \cup \mathbb{B}_n \in \{\mathbb{C} \in \widehat{\mathcal{D}} \mid \mathbb{A} \succsim \mathbb{C}\}$. Since $\succsim$ satisfies Axiom A.2 (Continuity), this lower contour set is closed, $\mathbb{A} \succsim \lim_{n\to\infty} \mathbb{A} \cup \mathbb{B}_n = \mathbb{A} \cup \mathbb{B}$. □

The next lemma shows the effect of imposing Axiom A.9 on a nested DLR representation.

LEMMA 7: *A binary relation $\succsim$ over $\widehat{\mathcal{D}}$ satisfies Axioms A.1-A.4 and Axiom A.9 if and only if the following two conditions hold:*

1. *$\succsim$ has a nested DLR representation with at most one positive state; and*
2. *Let $\widehat{v}(A) := V_{DLR}(\{A\}) = \sum_{i\in I^+} U_i(A) - \sum_{i\in I^-} U_i(A)$. There exists a non-negative scalar $\alpha \geq 0$ such that $\sum_{i\in I^+} U_i(A) = \alpha \widehat{v}(A)$ for all $A \in \widehat{\mathcal{M}}$.*

Proof of Lemma 7. "If." Since $\succsim$ has a nested DLR representation, it satisfies Axioms A.1-A.4 (by Lemma 5). If $|I^+| = 0$, that is, $I^+ = \emptyset$, then there is no positive state. Thus, $\mathbb{A} \succsim \mathbb{A} \cup \mathbb{B}$ for any two menus of menus $\mathbb{A}, \mathbb{B} \in \widehat{\mathcal{D}}$. Axiom A.9 is satisfied.

If $|I^+| = 1$, then $V_{DLR}(\mathbb{A}) = \max_{A\in\mathbb{A}} U_0(A) - \sum_{i\in I^-} \max_{A\in\mathbb{A}} U_i(A)$, where $\{U_0\} \cup \{U_i\}_{i\in I^-}$ is non-redundant. Suppose for any $B \in \mathbb{B}$, there exists $A \in \mathbb{A}$ such that $\widehat{v}(A) \geq \widehat{v}(B)$. Condition 2 implies that $\alpha > 0$, and $U_0(A) = \alpha[U_0(A) - \sum_{i\in I^-} U_i(A)]$ for all $A \in \widehat{\mathcal{M}}$ where $U_0(A) - \sum_{i\in I^-} U_i(A) = \widehat{v}(A) = V_{DLR}(\{A\})$. Therefore, condition 2 implies that for any $B \in \mathbb{B}$, there exists $A \in \mathbb{A}$ such that $U_0(A) \geq U_0(B)$, which further indicates that $\max_{A\in\mathbb{A}\cup\mathbb{B}} U_0(A) = \max_{A\in\mathbb{A}} U_0(A)$. Since $\mathbb{A} \cup \mathbb{B} \supseteq \mathbb{A}$, $\max_{A\in\mathbb{A}\cup\mathbb{B}} U_i(A) \geq \max_{A\in\mathbb{A}} U_i(A)$ for any $i \in I^-$. Thus, $V_{DLR}(\mathbb{A}) \geq V_{DLR}(\mathbb{A} \cup \mathbb{B})$, and $\mathbb{A} \succsim \mathbb{A} \cup \mathbb{B}$. Axiom A.9 is satisfied.

"Only if." By Lemma 5, Axioms A.1-A.4 imply the existence of a nested DLR representation for $\succsim$ as in Definition 10. If $|I^+| = 0$, then $I^+ = \emptyset$ and there is no positive state, condition 1 is satisfied. And we can set $\alpha = 0$ so that condition 2 is satisfied.

If $|I^+| > 0$, then $I^+ \neq \emptyset$ and there is at least one positive state. It suffices to show that: If Axiom A.9 is satisfied, then for any positive state $i \in I^+$, there exists $\alpha > 0$ such that $U_i(A) = \alpha\widehat{v}(A)$. Suppose by contradiction that for some $i^* \in I^+$, U_{i^*} does not represent the same preference over $\widehat{\mathcal{M}}$ as $\widehat{v}$. Consider three cases:

Case 1: $\widehat{v}$ represents a trivial preference over $\widehat{\mathcal{M}}$. That is, $\widehat{v}(A) = 0$ for all $A \in \widehat{\mathcal{M}}$. Then there is at least another state in $I^+ \cup I^-$ that is different from i^* (otherwise $\widehat{v}(A) = U_{i^*}(A)$, contradiction). So there are at least two elements in $\{U_k\}_{k\in I^+\cup I^-}$. A standard result (e.g., Lemma A.1 of Kopylov, 2009) guarantees the existence of a collection of menus $\mathbb{A} := \{A_k\}_{k\in I^+\cup I^-}$ such that A_k is the unique maximizer of U_k in $\mathbb{A}$ for each $k \in I^+ \cup I^-$, and $|\mathbb{A}| \geq 2$. Let $\mathbb{A}_1 := \mathbb{A} \setminus \{A_{i^*}\}$ and $\mathbb{B}_1 := \{A_{i^*}\}$. Then $\mathbb{A}_1 \neq \emptyset$ and for any $B \in \mathbb{B}_1$, there exists $A \in \mathbb{A}_1$ such that $\{A\} \succsim \{B\}$ (since $\widehat{v}(A) = 0 = \widehat{v}(B)$ for any $A, B \in \widehat{\mathcal{M}}$). Moreover, $\max_{A\in\mathbb{A}_1\cup\mathbb{B}_1} U_{i^*}(A) > \max_{A\in\mathbb{A}_1} U_{i^*}(A)$ and $\max_{A\in\mathbb{A}_1\cup\mathbb{B}_1} U_k(A) = \max_{A\in\mathbb{A}_1} U_k(A)$ for any $k \neq i^*$. Therefore, $V_{DLR}(\mathbb{A}_1 \cup \mathbb{B}_1) > V_{DLR}(\mathbb{A}_1)$, indicating $\mathbb{A}_1 \cup \mathbb{B}_1 \succ \mathbb{A}_1$, violating Axiom A.9. Contradiction.

Case 2: $\{\widehat{v}\} \cup \{U_k\}_{k\in I^+\cup I^-}$ is non-redundant. There are at least two elements ($\widehat{v}$ and U_{i^*}) in this collection, a standard result (e.g., Lemma A.1 of Kopylov, 2009) guarantees the existence of a collection of menus $\mathbb{A}' := \{B\} \cup \{A_k\}_{k\in I^+\cup I^-}$ such that B is the unique maximizer of $\widehat{v}$ in $\mathbb{A}'$ and A_k is the unique maximizer of U_k in $\mathbb{A}'$ for each $k \in I^+ \cup I^-$. Consider $\mathbb{A}'_1 := \mathbb{A}' \setminus \{A_{i^*}\}$ and $\mathbb{B}'_1 := \{A_{i^*}\}$. Then $B \in \mathbb{A}'_1$ and $\{B\} \succ \{A_{i^*}\}$ because B uniquely maximizes $\widehat{v}$ in $\mathbb{A}'$. Thus, the premise of Axiom A.9 is satisfied. Moreover, $\max_{A\in\mathbb{A}'_1\cup\mathbb{B}'_1} U_{i^*}(A) > \max_{A\in\mathbb{A}'_1} U_{i^*}(A)$ and $\max_{A\in\mathbb{A}'_1\cup\mathbb{B}'_1} U_k(A) = \max_{A\in\mathbb{A}'_1} U_k(A)$ for any $k \neq i^*$. Therefore, $V_{DLR}(\mathbb{A}'_1 \cup \mathbb{B}'_1) > V_{DLR}(\mathbb{A}'_1)$, indicating $\mathbb{A}'_1 \cup \mathbb{B}'_1 \succ \mathbb{A}'_1$, violating Axiom A.9. Contradiction.

Case 3: There exists some $j \in I^+ \cup I^-$ such that $j \neq i^*$ but $\widehat{v}$ represents the same preference over $\widehat{\mathcal{M}}$ as U_j. Then $\{\widehat{v}\} \cup \{U_k\}_{k\in(I^+\cup I^-)\setminus\{j\}}$ is non-redundant. Similar arguments as in Case 2 above will lead to a violation of Axiom A.9. □

The following lemma is useful to further fine-tune the states and substates.

LEMMA 8: *Let $\{U_i\}_{i\in I}$ be a finite and non-redundant collection of continuous linear functions. If each $U_i : \widehat{\mathcal{M}} \to \mathbb{R}$ has a minimal finite DLR representation $U_i(A) = \sum_{k\in P_i} \max_{p\in A} w_{ik}(p) - \sum_{j\in N_i} \max_{p\in A} v_{ij}(p)$ where for each $i \in I$, $\{w_{ik}\}_{k\in P_i} \cup \{v_{ij}\}_{j\in N_i}$ is a non-redundant collection of normalized expected utilities on $\Delta(Z)$, then there exists a collection of finite menus $\{A_i\}_{i\in I}$ (all in the interior of $\Delta(Z)$) such that:*

1. *$U_i(A_i) > U_i(A_j)$ for any $i \in I$ and any $j \neq i$.*
2. *$\left|\arg\max_{p\in A_i} w_{ik}(p)\right| = 1$, $\left|\arg\max_{q\in A_i} v_{ij}(q)\right| = 1$ for any $i \in I$, any $k \in P_i$ and any $j \in N_i$. For each i, these maximizers are all distinct from each other.*

Proof. See the proof of Lemma 3 of Stovall (2018). □

The next lemma states that imposing the Interim Preference for Flexibility axiom is equivalent to requiring there to be no negative substates in any positive state and at most one positive substate in any negative state. Formally,

LEMMA 9: *If $\succsim$ has a nested DLR representation as in Definition 10, then $\succsim$ satisfies Axiom A.6 if and only if $|N_i| = 0$ for all $i \in I^+$ and $|P_i| \leq 1$ for all $i \in I^-$.*

Proof of Lemma 9. "If." For any $i \in I^+$, $U_i(A \cup B) \geq \max\{U_i(A), U_i(B)\}$ since $|N_i| = 0$. Therefore, the positive terms in $V_{DLR}(\mathbb{A} \cup \{A \cup B\})$ are weakly larger than the positive terms in $V_{DLR}(\mathbb{A} \cup \{A, B\})$.

For any negative state $i \in I^-$: If $|P_i| = 0$, then $U_i(C) = -\sum_{j \in N_i} \max_{p \in C} v_{ij}(p)$, $-U_i(A \cup B) \geq \max\{-U_i(A), -U_i(B)\}$. If $|P_i| = 1$, then $U_i(C) = \max_{p \in C} w_{i0}(p) - \sum_{j \in N_i} \max_{p \in C} v_{ij}(p)$. Choose $C \in \{A, B\}$ that attains the largest value of the positive term. Then

$$U_i(A \cup B) = \max_{p \in C} w_{i0}(p) - \sum_{j \in N_i} \max_{p \in A \cup B} v_{ij}(p) \leq U_i(C) \leq \max\{U_i(A), U_i(B)\}.$$

Thus, in either case, $\max_{C \in \mathbb{A} \cup \{A \cup B\}} U_i(C) \leq \max_{C \in \mathbb{A} \cup \{A,B\}} U_i(C)$. Each negative term in $V_{DLR}(\mathbb{A} \cup \{A \cup B\})$ is therefore weakly larger than the corresponding term in $V_{DLR}(\mathbb{A} \cup \{A, B\})$. Combining the positive and negative terms proves Axiom A.6.

"Only if." By assumption, $\succsim$ can be represented by $V_{DLR}(\mathbb{A}) = \sum_{i \in I^+} \max_{A \in \mathbb{A}} U_i(A) - \sum_{i \in I^-} \max_{A \in \mathbb{A}} U_i(A)$ where $\{U_i\}_{i \in I^+ \cup I^-}$ is a non-redundant collection of continuous linear functions. Let $I := I^+ \cup I^-$. For each $i \in I$, $U_i(A) = \sum_{k \in P_i} \max_{p \in A} w_{ik}(p) - \sum_{j \in N_i} \max_{p \in A} v_{ij}(p)$, where $\{w_{ik}\}_{k \in P_i} \cup \{v_{ij}\}_{j \in N_i}$ is a non-redundant collection of normalized expected-utilities on $\Delta(Z)$. Thus, all assumptions of Lemma 8 are satisfied. Therefore, there exists a collection of menus $\{A_i\}_{i \in I}$ in the interior of $\Delta(Z)$ such that both conditions 1 and 2 in Lemma 8 are satisfied.

Let $\mathbb{A} = \{A_i\}_{i \in I}$. We first show that $|P_i| \leq 1$ for all $i \in I^-$. Suppose by contradiction that $|P_{i^*}| \geq 2$ for some $i^* \in I^-$ (i.e., U_{i^*} has two or more positive substates). We want to find A, B such that $V_{DLR}(\mathbb{A} \cup \{A \cup B\}) < V_{DLR}(\mathbb{A} \cup \{A, B\})$.

For ease of exposition, we write U_* for U_{i^*} and A_* for A_{i^*}. Similarly, we write P_* for P_{i^*} (the index set of positive substates in i^*), N_* for N_{i^*}, w_{*k} for w_{i^*k} for any $k \in P_*$ and v_{*j} for v_{i^*j} for any $j \in N_*$. For $a, b \in \mathbb{R}$, write $a \vee b$ to denote $\max\{a, b\}$.

Let p_k be the unique maximizer of w_{*k} in A_*. Since $|P_*| \geq 2$, we can pick k and k' with $k \neq k'$ from P_*. For any $\varepsilon > 0$, define $A^\varepsilon := A_* \cup \{p_k + \varepsilon w_{*k}\}$ and $B^\varepsilon := A_* \cup \{p_{k'} + \varepsilon w_{*k'}\}$. Since w_{*k} is normalized and p_k is in the interior of $\Delta(Z)$, a standard result guarantees the existence of an ε small enough so that $p_k + \varepsilon w_{*k}$ is also in the interior of $\Delta(Z)$.

Note that for any $\varepsilon > 0$, $w_{*k}(p_k + \varepsilon w_{*k}) = w_{*k}(p_k) + \varepsilon \|w_{*k}\|^2 > w_{*k}(p_k)$. Therefore, $\max_{p \in A^\varepsilon} w_{*k}(p) = w_{*k}(p_k) \vee w_{*k}(p_k + \varepsilon w_{*k}) = w_{*k}(p_k) + \varepsilon \|w_{*k}\|^2$, and

$$\max_{p \in B^\varepsilon} w_{*k}(p) = w_{*k}(p_k) \vee \Big(w_{*k}(p_{k'}) + \varepsilon w_{*k}(w_{*k'}) \Big)$$

Since $w_{*k}(p_{k'}) < w_{*k}(p_k)$, there exists $\varepsilon_1 > 0$ with $w_{*k}(p_{k'}) + \varepsilon_1 w_{*k}(w_{*k'}) < w_{*k}(p_k)$, which further indicates that $\max_{p \in B^{\varepsilon_1}} w_{*k}(p) = w_{*k}(p_k)$. With similar arguments, we could find some $\varepsilon_2 > 0$ such that $\max_{p \in B^{\varepsilon_2}} w_{*k'}(p) > w_{*k'}(p_{k'}) = \max_{p \in A^{\varepsilon_2}} w_{*k'}(p)$. Let $\varepsilon_3 := \min\{\varepsilon_1, \varepsilon_2\}$,

$$\max_{p \in A^{\varepsilon_3}} w_{*k}(p) > w_{*k}(p_k) = \max_{p \in B^{\varepsilon_3}} w_{*k}(p)$$

$$\max_{p \in A^{\varepsilon_3}} w_{*k'}(p) = w_{*k'}(p_{k'}) < \max_{p \in B^{\varepsilon_3}} w_{*k'}(p).$$

Note that $U_*(A^{\varepsilon_3} \cup B^{\varepsilon_3}) = \sum_{k \in P_*} \max_{p \in A^{\varepsilon_3} \cup B^{\varepsilon_3}} w_{*k}(p) - \sum_{j \in N_*} \max_{p \in A^{\varepsilon_3} \cup B^{\varepsilon_3}} v_{*j}(p)$. Moreover, we have $U_*(A^{\varepsilon_3}) = \sum_{k \in P_*} \max_{p \in A^{\varepsilon_3}} w_{*k}(p) - \sum_{j \in N_*} \max_{p \in A^{\varepsilon_3}} v_{*j}(p)$ and $U_*(B^{\varepsilon_3}) = \sum_{k \in P_*} \max_{p \in B^{\varepsilon_3}} w_{*k}(p) - \sum_{j \in N_*} \max_{p \in B^{\varepsilon_3}} v_{*j}(p)$. For each $j \in N_*$,

$$\max_{p \in A^{\varepsilon_3} \cup B^{\varepsilon_3}} v_{*j}(p) = \max_{p \in A_*} v_{*j}(p) \vee v_{*j}(p_k + \varepsilon_3 w_{*k}) \vee v_{*j}(p_{k'} + \varepsilon_3 w_{*k'}).$$

Since A_* is finite and v_{*j} has a unique maximizer in A_* that is not p_k or $p_{k'}$, we have a small number $\varepsilon_j > 0$ with $\varepsilon_j < \varepsilon_3$ such that

$$\max_{p \in A^{\varepsilon_j} \cup B^{\varepsilon_j}} v_{*j}(p) = \max_{p \in A^{\varepsilon_j}} v_{*j}(p) = \max_{p \in B^{\varepsilon_j}} v_{*j}(p) = \max_{p \in A_*} v_{*j}(p).$$

If $N_* \neq \emptyset$, let $\varepsilon := \min_{j \in N_*} \varepsilon_j$; if $N_* = \emptyset$, let $\varepsilon := \varepsilon_3$. Then

$$U_*(A^{\varepsilon} \cup B^{\varepsilon}) > \max\{U_*(A_*), U_*(A^{\varepsilon}), U_*(B^{\varepsilon})\}$$

because: (i) they all have the same negative term, and (ii) $A^{\varepsilon} \cup B^{\varepsilon}$ must have a strictly larger positive term because A^{ε} contains a unique maximizer $p_k + \varepsilon w_{*k}$ in substate k and B^{ε} contains a unique maximizer $p_{k'} + \varepsilon w_{*k'}$ in substate k'.

Finally, we can further scale down ε to guarantee that for any $i \in I$ with $i \neq i^*$,

$$U_i(A_i) > \max\{U_i(A^{\varepsilon} \cup B^{\varepsilon}), U_i(A^{\varepsilon}), U_i(B^{\varepsilon})\}$$

because: (i) A_i is the unique maximizer of U_i in $\mathbb{A}$ and (ii) $A^{\varepsilon} \cup B^{\varepsilon}$, A^{ε} and B^{ε} can all be made arbitrarily close to A_*. Therefore, $V_{DLR}(\mathbb{A} \cup \{A^{\varepsilon} \cup B^{\varepsilon}\}) < V_{DLR}(\mathbb{A} \cup \{A^{\varepsilon}, B^{\varepsilon}\})$, contradicting Axiom A.6.

We will then show that $|N_i| = 0$ for any $i \in I^+$. Suppose by contradiction that $|N_{i^*}| \geq 1$ for some $i^* \in I^+$. Since $|N_*| \geq 1$, we can fix some $j \in N_*$. Let q_j be the unique maximizer of v_{*j} in A_*. For any $\varepsilon > 0$, define $A^{\varepsilon} := A_*$ and $B^{\varepsilon} = (A_* \setminus \{q_j\}) \cup \{q_j - \varepsilon v_{*j}\}$. We can find ε small enough such that $q_j - \varepsilon v_{*j}$ is in the interior of $\Delta(Z)$. For any such ε, $v_{*j}(q_j - \varepsilon v_{*j}) = v_{*j}(q_j) - \varepsilon \|v_{*j}\|^2 < v_{*j}(q_j) = \max_{p \in A_*} v_{*j}(p)$. If $A_* \setminus \{q_j\}$ is nonempty, we can make ε small enough so that $v_{*j}(q_j - \varepsilon v_{*j}) = v_{*j}(q_j) - \varepsilon \|v_{*j}\|^2 > \max_{p \in A_* \setminus \{q_j\}} v_{*j}(p)$. We can also choose ε small enough that

$$v_{*j'}(q_j - \varepsilon v_{*j}) = v_{*j'}(q_j) - \varepsilon v_{*j'}(v_{*j}) < \max_{p \in A_*} v_{*j'}(p), \ \forall j' \in N_* \text{ and } j' \neq j$$

$$w_{*k}(q_j - \varepsilon v_{*j}) = w_{*k}(q_j) - \varepsilon w_{*k}(v_{*j}) < \max_{p \in A_*} w_{*k}(p), \ \forall k \in P_*$$

If $A_* = \{q_j\}$, condition 2 of Lemma 8 implies $P_* = \emptyset$ and $N_* = \{j\}$, so the remaining component-maximizer conditions are vacuous. For such an ε, we have $U_*(B^\varepsilon) > U_*(A^\varepsilon) = U_*(A_*) = U_*(A^\varepsilon \cup B^\varepsilon)$. We can further scale down ε so that $U_i(A_i) > U_i(A^\varepsilon), U_i(B^\varepsilon)$ for any $i \in I$ and $i \neq i^*$. Then $V_{DLR}(\mathbb{A} \cup \{A^\varepsilon, B^\varepsilon\}) > V_{DLR}(\mathbb{A} \cup \{A^\varepsilon \cup B^\varepsilon\})$, violating Axiom A.6. Contradiction. □

Lemma 10 states that imposing the Inclusion axiom is equivalent to requiring there to be no negative substates in any negative state. Formally,

LEMMA 10: *If $\succsim$ has a nested DLR representation as in Definition 10, then $\succsim$ satisfies Axiom A.7 (Inclusion) if and only if $|N_i| = 0$ for any $i \in I^-$.*

Proof of Lemma 10. "If." Fix A, B with $B \subseteq A$. For any $i \in I^-$, $U_i(A) \geq U_i(B)$ since $|N_i| = 0$. Since $A \in \mathbb{A}$, we must have $-\max_{C \in \mathbb{A}} U_i(C) = -\max_{C \in \mathbb{A} \cup \{B\}} U_i(C)$. For any positive state $i \in I^+$, $\max_{C \in \mathbb{A} \cup \{B\}} U_i(C) \geq \max_{C \in \mathbb{A}} U_i(C)$. Thus, $V_{DLR}(\mathbb{A} \cup \{B\}) \geq V_{DLR}(\mathbb{A})$.

"Only if." Let $\mathbb{A} = \{A_i\}_{i \in I}$ be constructed the same way as in the proof of Lemma 9. Suppose by contradiction that $|N_{i^*}| \geq 1$ for some $i^* \in I^-$. For ease of exposition, write U_* for U_{i^*}, A_* for A_{i^*}, P_* for P_{i^*}, N_* for N_{i^*}, w_{*k} for w_{i^*k} for any $k \in P_*$, and v_{*j} for v_{i^*j} for any $j \in N_*$.

Since $|N_*| \geq 1$, we can fix some $j \in N_*$. Let q_j be the unique maximizer of v_{*j} in A_*. For any $\varepsilon > 0$, define $A^\varepsilon := A_* \cup \{q_j - \varepsilon v_{*j}\}$ and $B^\varepsilon = (A_* \setminus \{q_j\}) \cup \{q_j - \varepsilon v_{*j}\}$. We can find ε small enough such that $q_j - \varepsilon v_{*j}$ is in the interior of $\Delta(Z)$. For any such ε, $v_{*j}(q_j - \varepsilon v_{*j}) = v_{*j}(q_j) - \varepsilon \|v_{*j}\|^2 < v_{*j}(q_j) = \max_{p \in A_*} v_{*j}(p)$. If $A_* \setminus \{q_j\}$ is nonempty, we can make ε small enough so that $v_{*j}(q_j - \varepsilon v_{*j}) = v_{*j}(q_j) - \varepsilon \|v_{*j}\|^2 > \max_{p \in A_* \setminus \{q_j\}} v_{*j}(p)$. We can also choose ε small enough that

$$v_{*j'}(q_j - \varepsilon v_{*j}) = v_{*j'}(q_j) - \varepsilon v_{*j'}(v_{*j}) < \max_{p \in A_*} v_{*j'}(p), \ \forall j' \in N_* \text{ and } j' \neq j$$
$$w_{*k}(q_j - \varepsilon v_{*j}) = w_{*k}(q_j) - \varepsilon w_{*k}(v_{*j}) < \max_{p \in A_*} w_{*k}(p), \ \forall k \in P_*$$

If $A_* = \{q_j\}$, condition 2 of Lemma 8 implies $P_* = \emptyset$ and $N_* = \{j\}$, so the remaining component-maximizer conditions are vacuous. For such an ε, we have $U_*(B^\varepsilon) > U_*(A^\varepsilon) = U_*(A_*)$. We can further scale down ε so that $U_i(A_i) > U_i(A^\varepsilon), U_i(B^\varepsilon)$ for any $i \in I$ and $i \neq i^*$. Then $V_{DLR}(\mathbb{A} \cup \{A^\varepsilon, B^\varepsilon\}) < V_{DLR}(\mathbb{A} \cup \{A^\varepsilon\})$ (since $i^* \in I^-$), which violates Axiom A.7. Contradiction. □

Now we are ready to present the proof for Theorem 6.

Proof of Theorem 6. ***Necessity of the axioms.*** Lemmas 5, 7, 9, 10 imply the necessity of Axioms A.1-A.7. To verify Axiom A.8, let p^0 be the uniform lottery and $D = \Delta(Z)$. Then

$$\widehat{V}(\{\{p^0\}, D\}) = \widehat{V}(\{D\}) > 0 = \widehat{V}(\{\{p^0\}\})$$

because every utility in the support is normalized and nonconstant.

Sufficiency of the axioms. Axioms A.1-A.4 deliver a nested DLR representation (Lemma 5). Axiom A.5 guarantees that there is at most one positive state (part 1 of Lemma 7). Axiom A.6 guarantees that there are no negative substates in any positive state and at most one positive substate in any negative state (Lemma 9). Axiom A.7 guarantees that there are no negative substates in any negative state (Lemma 10). Thus, there can be at most one positive state (which contains only positive substates) and there might be multiple negative states (each of which contains at most one positive substate and no negative substates).

Therefore, Axioms A.1-A.7 imply that there exist two non-redundant collections of normalized expected utilities over $\Delta(Z)$, $\{w_k\}_{k\in P}$ and $\{v_j\}_{j\in N}$, such that $\succsim$ can be represented by the function $V_S : \widehat{\mathcal{D}} \to \mathbb{R}$ defined by

$$V_S(\mathbb{A}) = \max_{A\in\mathbb{A}} \sum_{k\in P} \max_{p\in A} w_k(p) - \sum_{j\in N} \max_{A\in\mathbb{A}} \max_{p\in A} v_j(p). \tag{26}$$

For convenience, define $U_0(A) := \sum_{k\in P} \max_{p\in A} w_k(p)$, $U_j(A) := \max_{p\in A} v_j(p)$ for each $j \in N$ and $\widehat{v}(A) := V_S(\{A\}) = \sum_{k\in P} \max_{p\in A} w_k(p) - \sum_{j\in N} \max_{p\in A} v_j(p)$. If $\succsim$ can be represented by V_S as defined in equation (26) and $\succsim$ satisfies Axiom A.8, then $|P| \geq 1$ and U_0 is non-constant. Otherwise there will be no positive state in the representation and $\mathbb{A} \succsim \mathbb{B}$ for any $\mathbb{A} \subseteq \mathbb{B}$, violating Axiom A.8.

Since $\{v_j\}_{j\in N}$ is non-redundant, $\{U_j\}_{j\in N}$ must also be non-redundant.

LEMMA 11: *If $\succsim$ can be represented by V_S defined in equation (26) with $|P| \geq 1$ and $\succsim$ satisfies Axioms A.5 and A.8, then there exists $\alpha > 0$ such that $U_0 = \alpha\widehat{v}$.*

Proof of Lemma 11. We discuss two possible cases.

Case 1: Suppose $\{U_0\} \cup \{U_j\}_{j\in N}$ is non-redundant. Then $V_S(\mathbb{A}) := \max_{A\in\mathbb{A}} U_0(A) - \sum_{j\in N} \max_{A\in\mathbb{A}} U_j(A)$ is a nested DLR representation with exactly one positive state. By part 2 of Lemma 7, this implies that there exists $\alpha > 0$ such that $U_0 = \alpha\widehat{v}$.

Case 2: Suppose $\{U_0\} \cup \{U_j\}_{j\in N}$ is redundant. By our previous results, $\{U_j\}_{j\in N}$ is not redundant and U_0 is non-constant. Therefore, there exists some $\beta > 0$ and exactly one $j \in N$ such that $U_0 = \beta U_j$. (N cannot be empty, otherwise U_0 is the only function in the collection and $\{U_0\}$ is not redundant.) It must be that $\beta > 1$, otherwise there will be no positive state (U_0 is absorbed by U_j) and Axiom A.8 will be violated. $U_0 = \beta U_j$ means that $\sum_{k\in P} \max_{p\in A} w_k(p) = \beta \max_{p\in A} v_j(p)$ for any menu A. Now this can only happen when $|P| = 1$, otherwise the LHS will exhibit strict preference for flexibility in some cases while the RHS will always exhibit strategic rationality. Let $P = \{k\}$, then $w_k = \beta v_j$, and $U_0 = \beta \max_{p\in A} v_j(p)$. Let $U_0' := (\beta - 1)U_j$, then $\{U_0'\} \cup \{U_{j'}\}_{j'\in N,\ j'\neq j}$ is non-redundant.

Then we can apply Lemma 7 again so that there exists $\alpha > 0$ with $U_0' = \alpha\widehat{v}$ where

$$\widehat{v}(A) := V_S(\{A\}) = U_0(A) - \sum_{j\in N} U_j(A) = U_0'(A) - \sum_{j'\in N,\ j'\neq j} U_{j'}(A).$$

Therefore, $U_0 = \frac{\beta}{\beta-1}U_0' = \frac{\beta}{\beta-1}(\alpha\widehat{v})$, and $U_0 = \alpha'\widehat{v}$ where $\alpha' = \frac{\alpha\beta}{\beta-1} > 0$. □

Thus, Axioms A.1-A.8 guarantee the existence of some $\alpha > 0$ such that $U_0 = \alpha\widehat{v}$. That is, $\sum_{k\in P}\max_{p\in A} w_k(p) = \alpha\left[\sum_{k\in P}\max_{p\in A} w_k(p) - \sum_{j\in N}\max_{p\in A} v_j(p)\right]$. Therefore,

$$\sum_{j\in N}\max_{p\in A} v_j(p) = \frac{\alpha-1}{\alpha}\sum_{k\in P}\max_{p\in A} w_k(p). \tag{27}$$

Claim: For equation (27) to hold, it must be that $\alpha \geq 1$.

Proof of the Claim. Suppose by contradiction that $\alpha < 1$, and take $A = \Delta(Z)$. Since $P \neq \emptyset$ and each w_k is normalized and nonconstant, $U_0(A) = \sum_{k\in P}\max_{p\in A} w_k(p) > 0$. Each normalized v_j has a nonnegative maximum on A, so the left-hand side of (27) is nonnegative. The right-hand side is strictly negative because $(\alpha-1)/\alpha < 0$, a contradiction. □

If $\alpha = 1$, then for any menu A, $\sum_{j\in N}\max_{p\in A} v_j(p) = 0$. By Lemma 4, this indicates that $N = \emptyset$. And $V_S(\mathbb{A}) = \max_{A\in\mathbb{A}}\sum_{k\in P}\max_{p\in A} w_k(p)$ with $|P| \geq 1$. We can get a PR representation by setting $K = 0$ and (doubly) normalizing each w_k.

If $\alpha > 1$, then $|N| \geq 1$, and for any menu A,

$$U_0(A) = \sum_{k\in P}\max_{p\in A} w_k(p) = \frac{\alpha}{\alpha-1}\sum_{j\in N}\max_{p\in A} v_j(p),$$

which further indicates that

$$V_S(\mathbb{A}) = \max_{A\in\mathbb{A}}\left[\frac{\alpha}{\alpha-1}\sum_{j\in N}\max_{p\in A} v_j(p)\right] - \sum_{j\in N}\max_{A\in\mathbb{A}}\max_{p\in A} v_j(p).$$

Since $\alpha > 1$, we can get a PR representation by setting $K = \alpha - 1 > 0$, (doubly) normalizing each v_j and scaling by multiplying by K. This completes the proof of Theorem 6. □

A.3 Uniqueness of the PR representation

Let $\widehat{V}$ be a PR representation for $\succsim$ with parameters (μ, K). Define $\widehat{v} : \widehat{\mathcal{M}} \to \mathbb{R}$, for any menu $A \in \widehat{\mathcal{M}}$, by $\widehat{v}(A) := \widehat{V}(\{A\}) = \sum_{u\in\text{supp}(\mu)}\mu(u)\max_{p\in A} u(p)$. That is, $\widehat{v}$ represents a preference over $\widehat{\mathcal{M}}$ generated by $\succsim$ restricting to singleton menus of menus (menus of menus containing only one menu). This is a special case of the representation captured in Dekel et al. (2001) and Dekel et al. (2007) with preference for flexibility. Their identification results indicate that μ is uniquely identified. Formally,

LEMMA 12: *If both* (μ, K) *and* (μ', K') *represent* $\succsim$*, then* $\mu' = \mu$.

To identify K, we follow steps similar to that in Sarver (2008). Define $\widehat{r} : \widehat{\mathcal{D}} \to \mathbb{R}$, for any menu of menus $\mathbb{A} \in \widehat{\mathcal{D}}$, by $\widehat{r}(\mathbb{A}) := \min_{A \in \mathbb{A}} \sum_{u \in \text{supp}(\mu)} \mu(u) R(A, \mathbb{A}, u)$ Thus, $\widehat{r}(\mathbb{A})$ is the agent's minimal expected regret facing $\mathbb{A}$. Given a menu of menus $\mathbb{A}$, the menu A that maximizes $\widehat{v}$ also minimizes expected regret. Therefore, for any $\mathbb{A} \in \widehat{\mathcal{D}}$, the agent will choose $A \in \mathbb{A}$ to maximize $\widehat{v}(A)$, and $\widehat{V}(\mathbb{A}) = \max_{A \in \mathbb{A}} \widehat{v}(A) - \widehat{r}(\mathbb{A})$.

THEOREM 7: *Two PR representations* (μ, K) *and* (μ', K') *represent the same preference* $\succsim$ *if and only if* $\widehat{v}' = \widehat{v}$ *and* $\widehat{r}' = \widehat{r}$.

Proof of Theorem 7. The "if" part is straightforward. For the "only if" part, suppose (μ, K) and (μ', K') represent the same preference. The mixture space theorem guarantees that there exists $\alpha > 0$ such that $\widehat{v}' = \alpha \widehat{v}$ and $\widehat{r}' = \alpha \widehat{r}$. Then by Lemma 12, $\mu' = \mu$ implies that $\widehat{v}' = \widehat{v}$. Thus, $\alpha = 1$, and $\widehat{r}' = \widehat{r}$. □

We then rule out a less interesting case where Axiom A.5 is trivially satisfied.

LEMMA 13: *If* $\succsim$ *has a PR representation with parameters* (μ, K)*, then the following are equivalent: (1)* $\widehat{r}(\mathbb{A}) = 0$ *for all* $\mathbb{A} \in \widehat{\mathcal{D}}$*; (2)* $\succsim$ *satisfies monotonicity. That is, if* $\mathbb{B} \subseteq \mathbb{A}$*, then* $\mathbb{A} \succsim \mathbb{B}$*; (3)* $K = 0$ *or* $\mu = \delta_u$ *for some* $u \in \mathcal{U}$ *(or both).*

Proof of Lemma 13. $1 \implies 2$: Suppose $\widehat{r}(\mathbb{A}) = 0$ for all $\mathbb{A} \in \widehat{\mathcal{D}}$. If $\mathbb{A} \supseteq \mathbb{B}$, then $\widehat{V}(\mathbb{A}) = \max_{A \in \mathbb{A}} \widehat{v}(A) \geq \max_{B \in \mathbb{B}} \widehat{v}(B) = \widehat{V}(\mathbb{B})$. Thus, $\mathbb{A} \succsim \mathbb{B}$.
$2 \implies 1$: Suppose $\widehat{r}(\mathbb{A}) > 0$ for some $\mathbb{A} \in \widehat{\mathcal{D}}$. Let $A \in \arg\max_{B \in \mathbb{A}} \widehat{v}(B)$, then $\{A\} \subseteq \mathbb{A}$ but $\{A\} \succ \mathbb{A}$ since $\widehat{V}(\{A\}) = \widehat{v}(A)$ while $\widehat{V}(\mathbb{A}) = \widehat{v}(A) - \widehat{r}(\mathbb{A})$.
$1 \implies 3$: Suppose, by contradiction, that $K > 0$ and $|\text{supp}(\mu)| \geq 2$. There are lotteries p_u, one for each $u \in \text{supp}(\mu)$, such that each p_u uniquely maximizes u in $A_0 := \{p_v : v \in \text{supp}(\mu)\}$ and these maximizers are distinct. For example, take $p_u = \bar{p} + \varepsilon u$, where $\bar{p}$ is the uniform lottery and $\varepsilon > 0$ is small enough. Double normalization gives $u(\bar{p}) = 0$, $u \cdot u = 1$, and $u \cdot v < 1$ for $v \neq u$, which verifies the claim. Let $\mathbb{A} := \{\{p\} : p \in A_0\}$. The agent must then choose a lottery before the taste is known, and

$$\widehat{r}(\mathbb{A}) = K \min_{p \in A_0} \sum_{u \in \text{supp}(\mu)} \mu(u) \left[\max_{q \in A_0} u(q) - u(p) \right] > 0.$$

Indeed, every summand is nonnegative, and for each $p \in A_0$ at least one taste strictly prefers a different lottery. This contradicts (1).
$3 \implies 1$: If $K = 0$, regret is zero. If $\mu = \delta_u$, fix any $\mathbb{A}$ and choose

$$A^* \in \arg\max_{A \in \mathbb{A}} \max_{p \in A} u(p).$$

Compactness ensures that such a menu exists. The menu A^* incurs no regret, so the minimum expected regret is again zero. □

A PR representation is *nontrivial* if there exist $\mathbb{A}$, $\mathbb{B}$ such that $\mathbb{B} \subseteq \mathbb{A}$ but $\mathbb{B} \succ \mathbb{A}$.

THEOREM 8: *If $\succsim$ has a nontrivial PR representation and $\succsim$ has two PR representations (μ, K) and (μ', K'), then $\mu' = \mu$ and $K' = K$.*

Proof. $\mu' = \mu$ by Lemma 12, and by Theorem 7, $\widehat{v}' = \widehat{v}$, $\widehat{r}' = \widehat{r}$. Since $\succsim$ has a nontrivial PR representation, $K, K' > 0$. Since $\mu' = \mu$, it must be that $\widehat{r}(\mathbb{A})/K = \widehat{r}'(\mathbb{A})/K'$ for all $\mathbb{A} \in \widehat{\mathcal{D}}$. Thus, $\widehat{r}(\mathbb{A}) = \widehat{r}'(\mathbb{A})$ for all $\mathbb{A}$ implies that $K' = K$. □

B Proof of Theorem 1

To save space, we omit the proof for the necessity of these axioms from the main text. Interested readers can find the complete proof in Section S2 of the supplementary appendix. Here we focus on the proof for the sufficiency of Axioms 1-8.

We have characterized the partial regret (PR) representation in Appendix A. The PR representation looks very similar to the SIT representation, only with the caveat that it is in the framework of menus of menus of lotteries. In this proof, we will connect Axioms 1-8 in the main text with Axioms A.1-A.8 in the lottery framework in Appendix A through two translations. The first translation will be from the preference over menus of menus of acts to menus of menus of "utility acts," and the second translation will be from the menus of menus of "utility acts" to menus of menus of lotteries.

A useful equivalent expression of the SIT representation is

$$\begin{aligned} V(\mathbb{F}) = \max_{F \in \mathbb{F}} \Bigg[(1+K) \sum_{s \in S} \max_{f \in F} \sum_{\omega \in \Omega} \pi(\omega) \sigma(s \mid \omega) u(f(\omega)) \Bigg] \\ - K \sum_{s \in S} \max_{G \in \mathbb{F}} \max_{g \in G} \sum_{\omega \in \Omega} \pi(\omega) \sigma(s \mid \omega) u(g(\omega)) \end{aligned} \tag{28}$$

Step 1: From $\succsim$ to a preference over "menus of menus of utility acts."

Note that Axioms 1-3 imply their corresponding axioms (weak order, continuity and independence) over acts (i.e., menu of menus containing only one singleton menu, like $\{\{f\}\}$). Axioms 8 (Nontriviality); and part 1 of Axiom 7 (Domination) imply the nontriviality axiom and the monotonicity axiom in the Anscombe-Aumann framework. Therefore, by the standard result from the Anscombe-Aumann framework (see, for example, Kreps 2018), Axioms 1-3, 7 and 8 imply that there exists a unique probability measure $\bar{\pi} \in \Delta(\Omega)$ and a surjective affine utility index $u : \Delta(X) \to [0,1]$ such that the restriction of $\succsim$ to acts can be represented by $V(f) := \sum_{\omega \in \Omega} \bar{\pi}(\omega) u\big(f(\omega)\big)$. For any act $f \in \mathcal{F}$, the composition $u \circ f : \Omega \to [0,1]$ specifies the utility associated with f in each ω. We call this the *utility act* induced by f, denoted by $u(f)$.

Let $\mathcal{F}_u := u(\mathcal{F}) = \{u(f) \mid f \in \mathcal{F}\}$, that is, $\mathcal{F}_u$ is the set of utility acts induced by AA acts from $\mathcal{F}$. Since u is surjective, $\mathcal{F}_u = [0,1]^{|\Omega|}$. Endow $\mathcal{F}_u$ with the Euclidean metric. Let $\mathcal{M}_u$ be the set of all non-empty compact subsets of $\mathcal{F}_u$, with typical elements F_u, G_u. We call these utility menus. Endow $\mathcal{M}_u$ with the Hausdorff metric. Let $\mathcal{D}_u$ be the set of all non-empty compact subsets of $\mathcal{M}_u$, with typical elements $\mathbb{F}_u, \mathbb{G}_u$. We call these utility menus of menus. Endow $\mathcal{D}_u$ with the Hausdorff metric.

We extend u pointwise to menus and menus of menus.

LEMMA 14: *If $\succsim$ satisfies Axioms 1-3, 7 and 8, then $u(\mathbb{F}) = u(\mathbb{G})$ implies $\mathbb{F} \sim \mathbb{G}$.*

Proof of Lemma 14. We say two AA acts f and g are *indistinguishable* if $\{\{f(\omega)\}\} \sim \{\{g(\omega)\}\}$ for all $\omega \in \Omega$. Note that f and g are indistinguishable if and only if $u(f) = u(g)$ (since $u(f)$ and $u(g)$ are not scalars, but functions from Ω to $[0,1]$).

We say two menus F and G are indistinguishable if for any $f \in F$ there exists $g \in G$ that is indistinguishable from f and for any $g' \in G$ there exists $f' \in F$ that is indistinguishable from g'. For a menu $F \in \mathcal{M}$, let $u(F) := \{u(f) \mid f \in F\}$. Note that F and G are indistinguishable if and only if $u(F) = u(G)$. (The "only if" part is easy. For the "if" part: If $u(F) = u(G)$, then for any $f \in F$ there exists $g \in G$ such that $u(f) = u(g)$, so f and g are indistinguishable.)

We say two menus of menus $\mathbb{F}$ and $\mathbb{G}$ are indistinguishable if for any $F \in \mathbb{F}$ there exists $G \in \mathbb{G}$ that is indistinguishable to F and for any $G' \in \mathbb{G}$ there exists $F' \in \mathbb{F}$ that is indistinguishable to G'. For a menu of menus $\mathbb{F} \in \mathcal{D}$, let $u(\mathbb{F}) := \{u(F) \mid F \in \mathbb{F}\}$. Then $\mathbb{F}$ and $\mathbb{G}$ are indistinguishable if and only if $u(\mathbb{F}) = u(\mathbb{G})$. If $\mathbb{F}$ and $\mathbb{G}$ are indistinguishable, then $\mathbb{F}$ dominates $\mathbb{G}$ and $\mathbb{G}$ dominates $\mathbb{F}$. By part (7.2) of Axiom 7, $\mathbb{F} \sim \mathbb{F} \cup \mathbb{G}$ and $\mathbb{G} \sim \mathbb{F} \cup \mathbb{G}$. Therefore, $\mathbb{F} \sim \mathbb{G}$ by transitivity. □

Let $\succsim_u$ over $\mathcal{D}_u$ be defined by $\mathbb{F}_u \succsim_u \mathbb{G}_u$ if and only if $\mathbb{F} \succsim \mathbb{G}$ where $\mathbb{F} \in u^{-1}(\mathbb{F}_u)$ and $\mathbb{G} \in u^{-1}(\mathbb{G}_u)$. Lemma 14 guarantees that $\succsim_u$ is a preference.

For utility acts $f_u, g_u \in \mathcal{F}_u$, we say that f_u *dominates* g_u if $f_u(\omega) \geq g_u(\omega)$ for every $\omega \in \Omega$. A utility menu F_u dominates G_u if for every $g_u \in G_u$ there exists $f_u \in F_u$ that dominates g_u. A utility menu of menus $\mathbb{F}_u$ dominates $\mathbb{G}_u$ if for every $G_u \in \mathbb{G}_u$, there exists $F_u \in \mathbb{F}_u$ that dominates G_u. The following property of $\succsim_u$ will be important for later proofs.

AXIOM †—Utility-act version of Domination: (1) If f_u dominates g_u, then $f_u \succsim_u g_u$ and $\{\{f_u, g_u\}\} \sim_u \{\{f_u\}\}$. (2) If $\mathbb{F}_u$ dominates $\mathbb{G}_u$, then $\mathbb{F}_u \sim_u \mathbb{F}_u \cup \mathbb{G}_u$.

It is easy to show that if $\succsim$ satisfies Axioms 1-8, then the induced $\succsim_u$ satisfies Axiom †.

Step 2: From $\succsim_u$ to a preference over menus of menus of lotteries.

Recall that $\mathcal{F}_u$ is the collection of all utility acts. Thus, we can identify $\mathcal{F}_u$ with $[0,1]^{|\Omega|}$. For convenience, let $\Omega = \{\omega_1, \ldots, \omega_n\}$, and introduce an artificial state ω_0 and a new space

$\mathcal{F}'$ defined by $\mathcal{F}' := \left\{ f' \in [0,n] \times [0,1]^n \;\middle|\; \sum_{i=0}^n f'(\omega_i) = n \right\}$. We suppress the subscript u for ease of notation, but these are still utility acts.

Endow $\mathcal{F}'$ with the standard Euclidean metric. Consider $r : \mathcal{F}' \to \mathcal{F}_u$ where $r(f')$ is the vector in $\mathcal{F}_u$ that agrees with the last n components of f'. That is, $r(f') \in \mathcal{F}_u$ and for any $i = 1, 2, \ldots, n$, $[r(f')](\omega_i) = f'(\omega_i)$. It is easy to verify that r is a homeomorphism between $\mathcal{F}'$ and $\mathcal{F}_u$. Let r^{-1} denote its inverse. Let $\mathcal{M}'$ denote the set of nonempty compact subsets of $\mathcal{F}'$, with typical elements F', G'. Endow $\mathcal{M}'$ with the Hausdorff metric. Let $\mathcal{D}'$ denote the set of nonempty compact subsets of $\mathcal{M}'$, with typical elements $\mathbb{F}', \mathbb{G}'$. Endow $\mathcal{D}'$ with the Hausdorff metric. We slightly abuse notation and let r also denote the homeomorphism from $\mathcal{M}'$ to $\mathcal{M}_u$ and $\mathcal{D}'$ to $\mathcal{D}_u$, with $r(F') = \{r(f') \mid f' \in F'\} \in \mathcal{M}_u$ and $r(\mathbb{F}') = \{r(F') \mid F' \in \mathbb{F}'\} \in \mathcal{D}_u$. With this construction, we can define a preference $\succsim_*$ over $\mathcal{D}'$ by $\mathbb{F}' \succsim_* \mathbb{G}'$ if $r(\mathbb{F}') \succsim_u r(\mathbb{G}')$.

To move on, consider $\mathcal{F}'' := \left\{ f'' \in [0,n]^{n+1} \;\middle|\; \sum_{i=0}^n f''(\omega_i) = n \right\}$. Endow $\mathcal{F}''$ with the standard Euclidean metric. Let $\mathcal{M}''$ denote the set of nonempty compact subsets of $\mathcal{F}''$, with typical elements F'', G''. Endow $\mathcal{M}''$ with the Hausdorff metric. Let $\mathcal{D}''$ denote the set of nonempty compact subsets of $\mathcal{M}''$, with typical elements $\mathbb{F}'', \mathbb{G}''$. Endow $\mathcal{D}''$ with the Hausdorff metric.

Fix a menu F^{n+1} defined by $F^{n+1} := \left\{ \left(\frac{n}{n+1}, \ldots, \frac{n}{n+1} \right) \right\}$. Then $F^{n+1} \in \mathcal{M}'$. Also observe that for any $F'' \in \mathcal{M}''$ and $0 \leq \varepsilon \leq \frac{1}{n^2}$,

$$\varepsilon F'' + (1-\varepsilon)F^{n+1} = \left\{ \varepsilon f'' + (1-\varepsilon) \left(\frac{n}{n+1}, \ldots, \frac{n}{n+1} \right) \;\middle|\; f'' \in F'' \right\} \in \mathcal{M}'.$$

Finally, define a relation $\succsim_{**}$ on $\mathcal{D}''$ by: $\mathbb{F}'' \succsim_{**} \mathbb{G}''$ if

$$\varepsilon \mathbb{F}'' + (1-\varepsilon)\{F^{n+1}\} \succsim_* \varepsilon \mathbb{G}'' + (1-\varepsilon)\{F^{n+1}\} \text{ for all } 0 < \varepsilon < \frac{1}{n^2}.$$

Step 3: Verify that $\succsim_{**}$ is the unique extension of $\succsim_*$ to $\mathcal{D}''$ satisfying adapted versions of Axioms A.1-A.8.

The construction of $\succsim_*$ and $\succsim_{**}$ in Step 2 is the same as that in DLST.

Axiom 1*—Weak Order: *$\succsim_{**}$ is complete and transitive.*

Axiom 2*—Continuity: *For any $\mathbb{F}''$, $\{\mathbb{G}'' : \mathbb{G}'' \succsim_{**} \mathbb{F}''\}$, $\{\mathbb{G}'' : \mathbb{F}'' \succsim_{**} \mathbb{G}''\}$ are closed.*

Axiom 3*—Independence: *For any $\mathbb{F}'', \mathbb{G}'', \mathbb{H}''$ and any $\alpha \in (0,1)$, $\mathbb{F}'' \succsim_{**} \mathbb{G}''$ if and only if $\alpha\mathbb{F}'' + (1-\alpha)\mathbb{H}'' \succsim_{**} \alpha\mathbb{G}'' + (1-\alpha)\mathbb{H}''$.*

We can define the notion of critical subsets with respect to $\succsim_{**}$ in a similar way as before.

Axiom 4*—Finiteness: *There exists a natural number N such that: (4.1) For every $\mathbb{F}''$, there exists $\mathbb{G}''$ with $|\mathbb{G}''| < N$ such that $\mathbb{G}''$ is critical for $\mathbb{F}''$; and (4.2) For every finite $\mathbb{F}'' \in \mathcal{D}''$ and every $F'' \in \mathbb{F}''$, there exists G'' with $|G''| < N$ such that G'' is critical for F'' in $\mathbb{F}''$.*

Axiom 5*—Ex-Ante Regret: *If $\{F''\} \succsim_{**} \{G''\}$ and $F'' \in \mathbb{F}''$, then $\mathbb{F}'' \succsim_{**} \mathbb{F}'' \cup \{G''\}$.*

AXIOM 6*—Interim Preference for Flexibility: $\mathbb{F}'' \cup \{F'' \cup G''\} \succsim_{**} \mathbb{F}'' \cup \{F'', G''\}$.

AXIOM 7*—Inclusion: *If* $G'' \subseteq F''$ *and* $F'' \in \mathbb{F}''$, *then* $\mathbb{F}'' \cup \{G''\} \succsim_{**} \mathbb{F}''$.

AXIOM 8*—Nontriviality: *There exist* $\mathbb{F}'', \mathbb{G}'' \in \mathcal{D}''$ *such that* $\mathbb{F}'' \supseteq \mathbb{G}''$ *and* $\mathbb{F}'' \succ_{**} \mathbb{G}''$.

LEMMA 15: *If a binary relation* $\succsim$ *over* $\mathcal{D}$ *satisfies Axioms 1–8, then the induced binary relation* $\succsim_{**}$ *over* $\mathcal{D}''$ *satisfies Axioms* 1^*–8^* *as displayed above.*

The proof of Lemma 15 is relegated to Section S3 of the supplementary appendix.

Step 4: Apply Theorem 6 in Appendix A and translate the resulting PR representation to a SIT representation.

To proceed, we rescale every element of $\mathcal{F}''$ with factor $\frac{1}{n}$. This gives us a unit simplex and makes $\mathcal{D}''$ formally equivalent to the choice domain in Appendix A. The resulting common factor $1/n$ in the utility functional is omitted below because it does not affect the represented preference.

Through steps 1-3, we have verified that if a binary relation $\succsim$ over the menu of menus of acts satisfies Axioms 1-8, then the induced $\succsim_u$ satisfies Axiom † and the induced binary relation $\succsim_{**}$ defined over $\mathcal{D}''$ is unique and satisfies Axioms 1^*–8^*. Therefore, Theorem 6 guarantees the existence of a finitely-supported probability measure $\widehat{\mu}$ over $\widehat{\mathcal{U}}$ (the set of doubly normalized expected utilities over $\Delta(\widehat{\Omega})$ where $\widehat{\Omega} = \Omega \cup \{\omega_0\}$) and a scalar $\widehat{K} \geq 0$ such that $\succsim_{**}$ can be represented by

$$\widehat{W}(\mathbb{F}'') = \max_{F'' \in \mathbb{F}''} \left[(1+\widehat{K}) \sum_{\widehat{u} \in \text{supp}(\widehat{\mu})} \widehat{\mu}(\widehat{u}) \max_{f'' \in F''} \widehat{u}(f'') \right] - \widehat{K} \sum_{\widehat{u} \in \text{supp}(\widehat{\mu})} \widehat{\mu}(\widehat{u}) \max_{F'' \in \mathbb{F}''} \max_{f'' \in F''} \widehat{u}(f'') \quad (29)$$

where $\widehat{u}(f'') = \sum_{\widehat{\omega} \in \widehat{\Omega}} \widehat{u}(\widehat{\omega}) f''(\widehat{\omega})$ is the expected utility for lottery f'' under taste $\widehat{u}$. Moreover, $\widehat{\mu}$ is uniquely identified (Lemma 12), and $\widehat{K}$ is uniquely identified when $|\text{supp}(\widehat{\mu})| > 1$ (Lemma 13 and Theorem 8). Note that $\widehat{W}$ also represents $\succsim_*$ when restricting to $\mathcal{D}'$.

To obtain a SIT representation, we will transform each $\widehat{u}$ into a posterior belief and adjust its probability weight. We first use the utility-act version of Domination (Axiom †) to restrict the support of $\widehat{\mu}$.

LEMMA 16: *If* $\succsim_{**}$ *has a representation as defined in equation* (29) *and* $\succsim_u$ *satisfies Axiom* †, *then* $\widehat{u}(\omega) \geq \widehat{u}(\omega_0)$ *for any* $\widehat{u} \in \text{supp}(\widehat{\mu})$ *and any* $\omega \in \Omega$.

Proof of Lemma 16. Suppose by contradiction that $\succsim_{**}$ can be represented as in equation (29), $\succsim_u$ satisfies Axiom †, but there exists $\widehat{u}_* \in \text{supp}(\widehat{\mu})$ and $\omega_* \in \Omega$ such that $\widehat{u}_*(\omega_0) > \widehat{u}_*(\omega_*)$. Consider $f' := (n - \varepsilon, 0, \ldots, 0, \varepsilon, 0, \ldots, 0)$ where $n - \varepsilon$ is assigned to state ω_0, ε is assigned to state ω_* and 0 is assigned to any other state. We can find ε small enough so that $f' \in \mathcal{F}'$. Let $g' := (n, 0, \ldots, 0)$ where n is assigned to state ω_0. Therefore, $r(f')$ dominates $r(g')$ (recall that the notion of domination is defined over $\mathcal{F}_u$ and thus only concerns the last

n coordinates). Thus, part 1 of Axiom † dictates that $\{\{r(f')\}\} \sim_u \{\{r(f'), r(g')\}\}$, and $\{\{f'\}\} \sim_* \{\{f', g'\}\}$ by construction.

Note that for any $\widehat{u} \in \text{supp}(\widehat{\mu})$, $\widehat{u}(f') = (n-\varepsilon)\widehat{u}(\omega_0) + \varepsilon\widehat{u}(\omega_*)$ and $\widehat{u}(g') = n\widehat{u}(\omega_0)$. In particular, $\widehat{u}_*(f') - \widehat{u}_*(g') = \varepsilon\Big(\widehat{u}_*(\omega_*) - \widehat{u}_*(\omega_0)\Big) < 0$. Therefore,

$$\widehat{W}(\{\{f', g'\}\}) = \sum_{\widehat{u}\in\text{supp}(\widehat{\mu})} \widehat{\mu}(\widehat{u})\Big[\widehat{u}(f') \vee \widehat{u}(g')\Big] = \sum_{\widehat{u}\neq\widehat{u}_*} \widehat{\mu}(\widehat{u})\Big[\widehat{u}(f') \vee \widehat{u}(g')\Big] + \widehat{\mu}(\widehat{u}_*)\widehat{u}_*(g')$$

while $\widehat{W}(\{\{f'\}\}) = \sum_{\widehat{u}\neq\widehat{u}_*} \widehat{\mu}(\widehat{u})\widehat{u}(f') + \widehat{\mu}(\widehat{u}_*)\widehat{u}_*(f') < \widehat{W}(\{\{f', g'\}\})$, contradiction. □

Given our construction, there are two normalizations we can do that will allow us to replace the finitely-supported probability measure $\widehat{\mu}$ on $\widehat{\mathcal{U}}$ with a unique finitely-supported probability measure ν on $\Delta(\Omega)$.

For every $\widehat{u} \in \text{supp}(\widehat{\mu})$, define $\xi_{\widehat{u}}(\omega) := \widehat{u}(\omega) - \widehat{u}(\omega_0)$ for each $\omega \in \widehat{\Omega}$. Lemma 16 gives $\xi_{\widehat{u}}(\omega) \geq 0$ for every $\omega \in \Omega$. Since $\widehat{u}$ is nonconstant, at least one of these inequalities is strict, so $c_{\widehat{u}} := \sum_{\omega\in\Omega} \xi_{\widehat{u}}(\omega) > 0$. Moreover, $\xi_{\widehat{u}}(\omega_0) = 0$ and $\sum_{i=0}^{n} f''(\omega_i) = n$. Thus

$$\sum_{i=0}^{n} f''(\omega_i)\xi_{\widehat{u}}(\omega_i) = \widehat{u}(f'') - n\widehat{u}(\omega_0)$$

Subtracting the constant $n\widehat{u}(\omega_0)$ leaves the maximizing acts unchanged. Let

$$\mu_{\widehat{u}}(\omega) := \frac{\xi_{\widehat{u}}(\omega)}{c_{\widehat{u}}}, \text{ for each } \omega \in \Omega.$$

Then $\mu_{\widehat{u}}$ is a belief over Ω. For $f' \in \mathcal{F}'$, this gives

$$\widehat{u}(f') = n\widehat{u}(\omega_0) + c_{\widehat{u}} \sum_{\omega\in\Omega} \mu_{\widehat{u}}(\omega)[r(f')](\omega).$$

Let $C := \sum_{\widehat{u}\in\text{supp}(\widehat{\mu})} \widehat{\mu}(\widehat{u})c_{\widehat{u}}$, then $C > 0$ and let $\nu(\mu) := \frac{1}{C}\sum_{\{\widehat{u}|\mu_{\widehat{u}}=\mu\}} \widehat{\mu}(\widehat{u})c_{\widehat{u}}$.

The weights of ν are positive and sum to one. Substituting the preceding identity into (29), subtracting the constant $n\sum_{\widehat{u}} \widehat{\mu}(\widehat{u})\widehat{u}(\omega_0)$, and dividing by C gives

$$V(\mathbb{F}) = (1+\widehat{K}) \max_{F\in\mathbb{F}} \sum_{\mu\in\text{supp}(\nu)} \nu(\mu)U(F,\mu) - \widehat{K} \sum_{\mu\in\text{supp}(\nu)} \nu(\mu) \max_{F\in\mathbb{F}} U(F,\mu).$$

This functional represents $\succsim$. Its restriction to acts is expected utility with prior $\pi := \sum_{\mu} \nu(\mu)\mu$ and the utility index u fixed in Step 1.

Finally, enumerate $\text{supp}(\nu) = \{\mu_1, \ldots, \mu_m\}$ and choose distinct signals $s_j = \frac{j}{m} \in [0,1]$. Write $\nu_j := \nu(\mu_j)$. For every state with $\pi(\omega) > 0$, set $\sigma(s_j \mid \omega) := \frac{\nu_j\mu_j(\omega)}{\pi(\omega)}$. These probabilities sum to one because $\sum_j \nu_j\mu_j = \pi$. If $\pi(\omega) = 0$, then every $\mu_j(\omega) = 0$, and we may assign any probability distribution over these signals in that state. The ex-ante probability of s_j is ν_j, and Bayes' rule gives posterior μ_j. Hence $(\pi, u, \widehat{K}, \sigma)$ is a SIT representation.

C Other Omitted Proofs

C.1 Proof of Theorem 2

Proof. Suppose $(\pi_0, u_0, K_0, \sigma_0)$ and (π, u, K, σ) represent the same preference. Restricting this preference to acts and applying Anscombe–Aumann uniqueness gives $\pi_0 = \pi$ and $u_0 = au + b$ for some $a > 0$. We may therefore use the same utility index u, normalized to have range $[0, 1]$, in both representations. Let V_0, V be the resulting functionals and let ν_0, ν be their distributions over posteriors.

The two functionals agree on every constant act and are bounded above by one because regret is nonnegative. They also agree on every menu of menus. To see this, fix $\mathbb{F}$ and let h be a constant act with utility one. Choose $\alpha \in (0, 1)$ small enough that both $\alpha V_0(\mathbb{F}) + 1 - \alpha$ and $\alpha V(\mathbb{F}) + 1 - \alpha$ belong to $[0, 1]$. Mixture-linearity gives these values to $\alpha\mathbb{F} + (1-\alpha)\{\{h\}\}$. Its indifference to a constant act under one representation must hold under the other. Since every number in $[0, 1]$ is the utility of a constant act, the two values are equal. It follows that $V_0(\mathbb{F}) = V(\mathbb{F})$.

On full-flexibility objects, the common functional is

$$V(\{F\}) = \sum_{\mu \in \text{supp}(\nu)} \nu(\mu) U(F, \mu).$$

The identification result for the subjective learning representation in DLST therefore gives $\nu_0 = \nu$. If this distribution is nondegenerate, there are posteriors $\mu_1 \neq \mu_2$ in its support. Choose a vector x with $\sum_\omega \mu_1(\omega)x(\omega) > \sum_\omega \mu_2(\omega)x(\omega)$ and a number t strictly between these two sums. Let g be a constant act with utility $1/2$, and choose f with utility act $u(f(\omega)) = 1/2 + \delta[x(\omega) - t]$, where $\delta > 0$ is small enough to keep all these utilities in $[0, 1]$. Then f is strictly preferred to g at μ_1 and g is strictly preferred to f at μ_2. For $F = \{f, g\}$, define

$$A(F) := \sum_\mu \nu(\mu) U(F, \mu) - \max_{h \in F} \sum_\omega \pi(\omega) u(h(\omega)).$$

The strict reversal implies $A(F) > 0$: neither fixed act attains the posterior maximum at every positive-probability posterior. On the early-commitment object $D(F)$,

$$V(D(F)) = \max_{h \in F} \sum_\omega \pi(\omega) u(h(\omega)) - KA(F),$$

and the same formula holds for V_0 with K_0. Since $V_0(D(F)) = V(D(F))$ and $A(F) > 0$, we obtain $K_0 = K$, including when $K = 0$. If ν is degenerate, it puts probability one on π, and expected regret is zero for every $\mathbb{F}$. Any nonnegative K then gives the same functional. □

C.2 Proof of Theorem 3

DEFINITION 11: A binary relation $\succsim$ over $\mathcal{D} \times \mathcal{I}$ has an *aligned informational tradeoff representation* if there exists a tuple $\left(\pi, u, (K^\sigma)_{\sigma\in\mathcal{I}}, (i^\sigma)_{\sigma\in\mathcal{I}}\right)$ that consists of a prior π on Ω, a non-constant affine function $u : \Delta(X) \to \mathbb{R}$, a collection of information structures $i^\sigma : \Omega \to \Delta(S^\sigma)$ and a collection of non-negative scalars K^σ such that $\succsim$ can be represented by the function $W : \mathcal{D} \times \mathcal{I} \to \mathbb{R}$ defined by

$$\begin{aligned} W(\mathbb{F}, \sigma) = \max_{F\in\mathbb{F}} \Bigg[(1 + K^\sigma) \sum_{t\in S^\sigma} \max_{f\in F} \sum_{\omega\in\Omega} \pi(\omega) i^\sigma(t \mid \omega) u(f(\omega)) \Bigg] \\ - K^\sigma \sum_{t\in S^\sigma} \max_{G\in\mathbb{F}} \max_{g\in G} \sum_{\omega\in\Omega} \pi(\omega) i^\sigma(t \mid \omega) u(g(\omega)) \end{aligned} \tag{30}$$

This is called the "aligned" IT representation because the collection of IT representations for each σ is aligned together by the same π and u.

AXIOM 14—Stable Preference over Acts: *For any act f and any information structures σ, σ', $(f, \sigma) \sim (f, \sigma')$.*

Axiom 14 is motivated by the role of information in the decision process. When there is only one act to be chosen, any information is irrelevant. Together with Weak Order, it also implies that the ranking of acts is the same under every information structure.

LEMMA 17: *A binary relation $\succsim$ over $\mathcal{D} \times \mathcal{I}$ has an aligned informational tradeoff representation if and only if: (1) $\succsim$ satisfies Weak Order, Stable Preference over Acts and Act Independence; and (2) $\succsim^\sigma$ satisfies Axioms 2-8 for each $\sigma \in \mathcal{I}$.*

The proof of Lemma 17 is relegated to Appendix C.3.

In sum, Stable Preference over Acts and Act Independence guarantee the existence of a utility representation for a preference over the two dimensional choice domain such that the identified prior belief and taste over outcomes are information-independent.

DEFINITION 12: A binary relation $\succsim$ over $\mathcal{D} \times \mathcal{I}$ has a *regret-varying IT representation* if there exists a tuple $\left(\pi, u, (K^\sigma)_{\sigma\in\mathcal{I}}\right)$ that consists of a probability measure π on Ω, a non-constant affine function $u : \Delta(X) \to \mathbb{R}$ and a collection of non-negative scalars K^σ such that $\succsim$ can be represented by the function $W : \mathcal{D} \times \mathcal{I} \to \mathbb{R}$ defined by

$$\begin{aligned} W(\mathbb{F}, \sigma) = \max_{F\in\mathbb{F}} \Bigg[(1 + K^\sigma) \sum_{s\in S} \max_{f\in F} \sum_{\omega\in\Omega} \pi(\omega) \sigma(s \mid \omega) u(f(\omega)) \Bigg] \\ - K^\sigma \sum_{s\in S} \max_{G\in\mathbb{F}} \max_{g\in G} \sum_{\omega\in\Omega} \pi(\omega) \sigma(s \mid \omega) u(g(\omega)) \end{aligned} \tag{31}$$

Compared with an aligned IT representation, a regret-varying IT representation has the feature that σ is no longer merely a label: it is the information structure anticipated by the

agent. It is not yet the IT representation because the regret intensity level could still vary across different information structures.

LEMMA 18: *A binary relation $\succsim$ over $\mathcal{D} \times \mathcal{I}$ has a regret-varying informational tradeoff representation if and only if: (1) $\succsim$ satisfies Weak Order, Act Independence, SRNI and Reduction; and (2) $\succsim^\sigma$ satisfies Axioms 2-8 for each $\sigma \in \mathcal{I}$.*

The proof of Lemma 18 is relegated to Appendix C.4.

Finally, the Balance axiom guarantees that the parameters for regret intensity identified from all conditional preferences over menus of menus are the same.

Proof of Theorem 3. "Only if." Suppose $\succsim$ has an IT representation, then $\succsim$ has a regret-varying IT representation with $K^\sigma = K$ for all $\sigma \in \mathcal{I}$. By Lemma 18, $\succsim$ satisfies Weak Order, Act Independence, SRNI and Reduction, and $\succsim^\sigma$ satisfies Axioms 2-8 for each $\sigma \in \mathcal{I}$. As for the Balance axiom, note that $W(D(F), \sigma) = (1+K)W(\{F\}, o) - K \cdot W(\{F\}, \sigma)$ for any F and any σ. Rearranging, we have

$$W(\{F\}, o) = \frac{1}{1+K} \cdot W(D(F), \sigma) + \frac{K}{1+K} \cdot W(\{F\}, \sigma). \tag{32}$$

Suppose $(\{F\}, \sigma) \succ (\{F\}, o)$, then $W(\{F\}, \sigma) > W(\{F\}, o)$, which further indicates that $W(\{F\}, \sigma) > W(D(F), \sigma)$, otherwise the above equation cannot hold. Further suppose that $\Big(\alpha D(F) + (1-\alpha)\{F\}, \sigma\Big) \sim (\{F\}, o)$ for some $\alpha \in (0, 1]$, then

$$W(\{F\}, o) = \alpha \cdot W(D(F), \sigma) + (1-\alpha) \cdot W(\{F\}, \sigma). \tag{33}$$

For equations (32) and (33) to hold at the same time, it must be that $\alpha = \frac{1}{1+K}$. Then $\succsim$ must satisfy Balance by the fact that equation (32) is an identity.

"If." First, we can apply Lemma 18 to conclude that $\succsim$ has a regret-varying IT representation with parameters $\big(\pi, u, (K^\sigma)_{\sigma \in \mathcal{I}}\big)$.

If some σ induces a degenerate distribution over posteriors, then we can set K^σ to be any non-negative scalar without affecting the value of W.

Suppose by contradiction that $K^\sigma \neq K^{\sigma'}$ for some $\sigma, \sigma' \in \mathcal{I}$ such that σ and σ' each induces a non-degenerate distribution over posteriors. We want to show that Balance must be violated. By assumption, we can find F, G such that $(\{F\}, \sigma) \succ (\{F\}, o)$ and $(\{G\}, \sigma') \succ (\{G\}, o)$. Let $\alpha = 1/(1+K^\sigma)$ and $\alpha' = 1/(1+K^{\sigma'})$, then $\alpha \neq \alpha'$. Without loss of generality, suppose $\alpha > \alpha'$. Follow a computation similar to that in the proof for the necessity of Balance,

$$\begin{aligned} \Big(\alpha' D(F) + (1-\alpha')\{F\}, \sigma\Big) \succ \Big(\alpha D(F) + (1-\alpha)\{F\}, \sigma\Big) &\sim (\{F\}, o) \\ \Big(\alpha' D(G) + (1-\alpha')\{G\}, \sigma'\Big) &\sim (\{G\}, o) \end{aligned}$$

which is a direct violation of Balance. This completes the proof of the sufficiency of the axioms for the existence of an IT representation. □

C.3 Proof of Lemma 17

Proof. *"Only if."* Suppose $\succsim$ has an aligned IT representation, then $\succsim$ must be complete and transitive. Moreover, each $\succsim^\sigma$ has a SIT representation with parameters $(\pi, u, K^\sigma, i^\sigma)$. By Theorem 1, $\succsim^\sigma$ satisfies Axioms 2-8 for each $\sigma \in \mathcal{I}$.

For any act f and any $\sigma \in \mathcal{I}$, $W(f, \sigma) = \sum_{\omega\in\Omega} \pi(\omega)u(f(\omega))$. That is, the value of f is independent of σ. Stable Preference over Acts is thus satisfied.

To show that $\succsim$ satisfies Act Independence, note that W is affine in its first argument. That is, $W(\alpha\mathbb{F} + (1-\alpha)\mathbb{G}, \sigma) = \alpha W(\mathbb{F}, \sigma) + (1-\alpha)W(\mathbb{G}, \sigma)$ for any $\mathbb{F}, \mathbb{G} \in \mathcal{D}$ and any $\sigma \in \mathcal{I}$. The rest follows from applying this linearity and the fact that $W(h, \sigma) = W(h, \sigma')$ for any act h and any pair (σ, σ').

"If." As mentioned in the main text, $\succsim$ being complete and transitive implies that $\succsim^\sigma$ is complete and transitive for each $\sigma \in \mathcal{I}$. By Theorem 1, each $\succsim^\sigma$ has a SIT representation V^σ with parameters $(\pi^\sigma, u^\sigma, K^\sigma, i^\sigma)$.

LEMMA 19: *If $\succsim$ satisfies Stable Preference over Acts, then there exist π and u such that $\pi^\sigma = \pi$ and $u^\sigma = \alpha^\sigma u + b^\sigma$ for some $\alpha^\sigma > 0$ and $b^\sigma \in \mathbb{R}$ for all $\sigma \in \mathcal{I}$.*

Proof of Lemma 19. For any act $f \in \mathcal{F}$, $V^\sigma(f) = \sum_{\omega\in\Omega} \pi^\sigma(\omega)u^\sigma(f(\omega))$. Thus, each conditional preference restricted to acts has a subjective expected utility representation. By Stable Preference over Acts, $(f, \sigma) \sim (f, \sigma')$ and $(g, \sigma) \sim (g, \sigma')$ for any acts f, g and information structures σ, σ'. Weak Order therefore implies $f \succsim^\sigma g \iff f \succsim^{\sigma'} g$. The uniqueness result in the Anscombe–Aumann framework then gives a common prior π and utility u, with $\pi^\sigma = \pi$ and $u^\sigma = \alpha^\sigma u + b^\sigma$ for some $\alpha^\sigma > 0$ and $b^\sigma \in \mathbb{R}$ for every σ. □

Back to the proof of Lemma 17. The Act Independence axiom helps to establish that $\succsim$ actually has a utility representation. Let $\widetilde{V}^\sigma$ be a SIT representation with parameters $(\pi, u, K^\sigma, i^\sigma)$ where π, u are as characterized in Lemma 19. Then $\widetilde{V}^\sigma$ represents $\succsim^\sigma$ for each $\sigma \in \mathcal{I}$. Consider a function $W : \mathcal{D} \times \mathcal{I} \to \mathbb{R}$ defined by $W(\mathbb{F}, \sigma) := \widetilde{V}^\sigma(\mathbb{F})$. Suppose $\succsim$ also satisfies Act Independence, we will now show that $\succsim$ can be represented by W.

LEMMA 20: *For any $(\mathbb{F}, \sigma)$ and $(\mathbb{G}, \sigma')$, there exist acts f, g, h and $\alpha \in (0, 1]$ such that $(\alpha\mathbb{F} + (1-\alpha)h, \sigma) \sim (f, \sigma)$ and $(\alpha\mathbb{G} + (1-\alpha)h, \sigma') \sim (g, \sigma')$*

Proof of Lemma 20. Normalize the common utility index u to have range $[0, 1]$. Since regret is nonnegative, $\widetilde{V}^\sigma(\mathbb{F}) \leq 1$ and $\widetilde{V}^{\sigma'}(\mathbb{G}) \leq 1$. Let h be a constant act with $u(h(\omega)) = 1$ in every state. Then $\widetilde{V}^\sigma(h) = \widetilde{V}^{\sigma'}(h) = 1$.

Write $a = \widetilde{V}^\sigma(\mathbb{F})$ and $b = \widetilde{V}^{\sigma'}(\mathbb{G})$. Mixture linearity gives

$$\widetilde{V}^\sigma\big(\alpha\mathbb{F} + (1-\alpha)h\big) = 1 + \alpha(a-1), \text{ and } \widetilde{V}^{\sigma'}\big(\alpha\mathbb{G} + (1-\alpha)h\big) = 1 + \alpha(b-1).$$

Both quantities are at most 1. Because a and b are finite, we can choose a common $\alpha \in (0,1)$ small enough that both are nonnegative. Since u is affine and has range $[0,1]$, there are constant acts f and g whose utilities equal these two quantities. The conditional representations then give the two indifferences stated in the lemma. □

Now we can complete the proof of Lemma 17. Fix $(\mathbb{F}, \sigma)$ and $(\mathbb{G}, \sigma')$. Let f, g, h and $\alpha \in (0,1]$ satisfy Lemma 20. Then

$$\begin{aligned}
(\mathbb{F}, \sigma) \succsim (\mathbb{G}, \sigma') &\iff (\alpha\mathbb{F} + (1-\alpha)h, \sigma) \succsim (\alpha\mathbb{G} + (1-\alpha)h, \sigma') \\
&\iff (f, \sigma) \succsim (g, \sigma') \\
&\iff (f, \sigma) \succsim (g, \sigma) \\
&\iff \sum_{\omega \in \Omega} \pi(\omega) u(f(\omega)) \geq \sum_{\omega \in \Omega} \pi(\omega) u(g(\omega)) \\
&\iff W(\alpha\mathbb{F} + (1-\alpha)h, \sigma) \geq W(\alpha\mathbb{G} + (1-\alpha)h, \sigma') \\
&\iff W(\mathbb{F}, \sigma) \geq W(\mathbb{G}, \sigma')
\end{aligned}$$

The first equivalence holds since $\succsim$ satisfies Act Independence. The second equivalence holds by the construction of f, g, h and α. The third equivalence follows from $(g, \sigma') \sim (g, \sigma)$ and transitivity. The fourth equivalence holds since $\widetilde{V}^\sigma$ represents $\succsim^\sigma$. The fifth equivalence holds since $W(\mathbb{F}, \sigma) = \widetilde{V}^\sigma(\mathbb{F})$ so $W(\cdot, \sigma)$ can represent $\succsim^\sigma$ and $(\alpha\mathbb{F} + (1-\alpha)h, \sigma) \sim (f, \sigma)$ by construction. The last equivalence holds since W is affine in the menus of menus. This completes the proof of the sufficiency of the axioms since W is by definition an aligned informational tradeoff representation. □

C.4 Proof of Lemma 18

Proof. *"Only if."* Suppose $\succsim$ has a regret-varying IT representation, then $\succsim$ has an aligned IT representation with $i^\sigma = \sigma$ for each $\sigma \in \mathcal{I}$. $\succsim$ must satisfy Reduction since

$$\begin{aligned}
W(\{F_\sigma\}, o) = \max_{f \in F_\sigma} \sum_{\omega \in \Omega} \pi(\omega) u(f(\omega)) &= \max_{\gamma \in F^S} \sum_{\omega \in \Omega} \pi(\omega) u\Big(\gamma_\sigma(\omega)\Big) \\
&= \max_{\gamma \in F^S} \sum_{\omega \in \Omega} \pi(\omega) u\left(\sum_{s \in S} \sigma(s \mid \omega)[\gamma(s)](\omega)\right) \\
&= \max_{\gamma \in F^S} \sum_{\omega \in \Omega} \pi(\omega) \sum_{s \in S} \sigma(s \mid \omega) u\Big([\gamma(s)](\omega)\Big) \\
&= \sum_{s \in S} \max_{f \in F} \sum_{\omega \in \Omega} \pi(\omega)\sigma(s \mid \omega) u(f(\omega)) = W(\{F\}, \sigma)
\end{aligned}$$

where the second equality follows from the construction of F_σ, the third equality follows from the definition of an induced act γ_σ, the fourth equality follows from u being an affine

function, and the fifth equality follows from the fact that maximizing over the set of all plans is equivalent to maximizing over the menu of acts contingent on each signal realization.

By Lemma 17, $\succsim$ satisfies Weak Order and Act Independence, and $\succsim^\sigma$ satisfies Axioms 2-8 for each $\sigma \in \mathcal{I}$. In addition, $\succsim^o$ can be represented by (π, u, K^o, o). By Lemma 1, $\succsim$ satisfies SRNI.

"If." For a singleton menu $\{f\}$, the induced menu is $\{f\}_\sigma = \{f\}$. Reduction therefore gives $(f, \sigma) \sim (f, o)$ for every act f and information structure σ. Transitivity gives $(f, \sigma) \sim (f, \sigma')$, so Stable Preference over Acts holds. By Lemma 17, $\succsim$ has an aligned IT representation with parameters $\left(\pi, u, \left(K^\sigma\right)_{\sigma\in\mathcal{I}}, \left(i^\sigma\right)_{\sigma\in\mathcal{I}}\right)$, we want to show that if $\succsim$ satisfies SRNI and Reduction, then i^σ and σ induce the same distribution over posteriors for any σ. By Lemma 1, $W(\mathbb{F}, o) = \max_{F\in\mathbb{F}} \max_{f\in F} \sum_{\omega\in\Omega} \pi(\omega) u(f(\omega))$ since SRNI implies that i^o and o induce the same degenerate distribution over posteriors. By Reduction, $W(\{F\}, \sigma) = W(\{F_\sigma\}, o)$ for any menu $F \in \mathcal{M}$ and information structure $\sigma \in \mathcal{I}$. Note that

$$W(\{F\}, \sigma) = \sum_{t\in S^\sigma} \max_{f\in F} \sum_{\omega\in\Omega} \pi(\omega) i^\sigma(t \mid \omega) u(f(\omega))$$
$$W(\{F_\sigma\}, o) = \sum_{s\in S} \max_{f\in F} \sum_{\omega\in\Omega} \pi(\omega) \sigma(s \mid \omega) u(f(\omega))$$

where the first equation follows from applying W to the pair $(\{F\}, \sigma)$ and S^σ is the set of signals for the identified information structure i^σ, that is, $i^\sigma : \Omega \to \Delta(S^\sigma)$. The second equation follows from the previous derivation in the proof for the necessity of Reduction (S is the set of signals for σ and generally different from S^σ). By the identification result for subjective learning in DLST, these equalities for all menus F imply that σ and i^σ induce the same distribution over posteriors. This completes the proof. □

C.5 Proof of Lemma 3

Proof. Combining equations (18), (19) and (20), we can write W_1 as

$$\begin{aligned} W_1(\mathbb{F}, \sigma) = \max_{F\in\mathbb{F}} \Bigg[& \sum_{s\in S} (1 + K_2 + K_3 + K_4) \max_{f\in F} \sum_{\omega\in\Omega} \pi(\omega) \sigma(s \mid \omega) u(f(\omega)) \\ & - K_4 \sum_{\omega\in\Omega} \pi(\omega) \max_{h\in F} u(h(\omega)) \Bigg] \\ & - K_2 \sum_{s\in S} \max_{G\in\mathbb{F}} \max_{g\in G} \sum_{\omega\in\Omega} \pi(\omega) \sigma(s \mid \omega) u(g(\omega)) \\ & - K_3 \sum_{\omega\in\Omega} \pi(\omega) \max_{G\in\mathbb{F}} \max_{g\in G} u(g(\omega)) \end{aligned} \tag{34}$$

If $K_4 = 0$, this reduces to

$$\begin{aligned} W_1(\mathbb{F}, \sigma) = (1 + K_2 + K_3) \max_{F \in \mathbb{F}} \sum_{s \in S} \max_{f \in F} \sum_{\omega \in \Omega} \pi(\omega) \sigma(s \mid \omega) u(f(\omega)) \\ - K_2 \sum_{s \in S} \max_{G \in \mathbb{F}} \max_{g \in G} \sum_{\omega \in \Omega} \pi(\omega) \sigma(s \mid \omega) u(g(\omega)) \\ - K_3 \sum_{\omega \in \Omega} \pi(\omega) \max_{G \in \mathbb{F}} \max_{g \in G} u(g(\omega)) \end{aligned} \tag{35}$$

Note that the last term does not depend on σ. For convenience, let

$$\begin{aligned} U_1(\mathbb{F}, \sigma) &:= \max_{F \in \mathbb{F}} \sum_{s \in S} \max_{f \in F} \sum_{\omega \in \Omega} \pi(\omega) \sigma(s \mid \omega) u(f(\omega)) \\ U_2(\mathbb{F}, \sigma) &:= \sum_{s \in S} \max_{G \in \mathbb{F}} \max_{g \in G} \sum_{\omega \in \Omega} \pi(\omega) \sigma(s \mid \omega) u(g(\omega)) \end{aligned}$$

Therefore,

$$\begin{aligned} & W_1(\mathbb{F}, \sigma) \geq W_1(\mathbb{F}, \sigma') \\ \iff & (1 + K_2 + K_3) \cdot U_1(\mathbb{F}, \sigma) - K_2 \cdot U_2(\mathbb{F}, \sigma) \geq (1 + K_2 + K_3) \cdot U_1(\mathbb{F}, \sigma') - K_2 \cdot U_2(\mathbb{F}, \sigma') \\ \iff & \frac{1 + K_2 + K_3}{1 + K_3} \cdot U_1(\mathbb{F}, \sigma) - \frac{K_2}{1 + K_3} \cdot U_2(\mathbb{F}, \sigma) \geq \frac{1 + K_2 + K_3}{1 + K_3} \cdot U_1(\mathbb{F}, \sigma') - \frac{K_2}{1 + K_3} \cdot U_2(\mathbb{F}, \sigma') \\ \iff & (1 + K) \cdot U_1(\mathbb{F}, \sigma) - K \cdot U_2(\mathbb{F}, \sigma) \geq (1 + K) \cdot U_1(\mathbb{F}, \sigma') - K \cdot U_2(\mathbb{F}, \sigma') \\ \iff & W(\mathbb{F}, \sigma) \geq W(\mathbb{F}, \sigma') \end{aligned}$$

where the third equivalence follows from the assumption that $K = K_2/(1 + K_3)$. □

References

Aliprantis, Charalambos D. and Border, Kim C. *Infinite Dimensional Analysis: A Hitchhiker's Guide*. Springer, 2006.

Anscombe, Francis J. and Aumann, Robert J. A definition of subjective probability. *Annals of Mathematical Statistics*, 34(1):199–205, 1963.

Bell, David E. Regret in decision making under uncertainty. *Operations Research*, 30(5): 961–981, 1982.

Brunnermeier, Markus K. and Parker, Jonathan A. Optimal expectations. *American Economic Review*, 95(4):1092–1118, 2005.

Buturak, Gökhan and Evren, Özgür. Choice overload and asymmetric regret. *Theoretical Economics*, 12(3):1029–1056, 2017.

Caplin, Andrew and Leahy, John. Psychological expected utility theory and anticipatory feelings. *The Quarterly Journal of Economics*, 116(1):55–79, 2001.

Cheung, Paul H.Y. and Wang, Zichang. Scrutiny and conservatism. Working paper, 2026.

Dekel, Eddie, Lipman, Barton L., and Rustichini, Aldo. Representing preferences with a unique subjective state space. *Econometrica*, 69(4):891–934, 2001.

Dekel, Eddie, Lipman, Barton L., Rustichini, Aldo, and Sarver, Todd. Representing preferences with a unique subjective state space: A corrigendum. *Econometrica*, 75(2):591–600, 2007.

Dekel, Eddie, Lipman, Barton L., and Rustichini, Aldo. Temptation-driven preferences. *The Review of Economic Studies*, 76(3):937–971, 2009.

Dillenberger, David. Preferences for one-shot resolution of uncertainty and Allais-type behavior. *Econometrica*, 78(6):1973–2004, 2010.

Dillenberger, David, Lleras, Juan Sebastián, Sadowski, Philipp, and Takeoka, Norio. A theory of subjective learning. *Journal of Economic Theory*, 153:287–312, 2014.

Ehrlich, Danuta, Guttman, Isaiah, Schönbach, Peter, and Mills, Judson. Postdecision exposure to relevant information. *The Journal of Abnormal and Social Psychology*, 54(1): 98–102, 1957.

Eyster, Erik, Li, Shengwu, and Ridout, Sarah. A theory of ex post rationalization. *arXiv preprint arXiv:2107.07491*, 2021.

Festinger, Leon. *A Theory of Cognitive Dissonance*. Stanford University Press, 1957.

Frey, Dieter. Reversible and irreversible decisions: Preference for consonant information as a function of attractiveness of decision alternatives. *Personality and Social Psychology Bulletin*, 7(4):621–626, 1981.

Frey, Dieter. Recent research on selective exposure to information. *Advances in Experimental Social Psychology*, 19:41–80, 1986.

Frey, Dieter and Wicklund, Robert A. A clarification of selective exposure: The impact of choice. *Journal of Experimental Social Psychology*, 14(1):132–139, 1978.

Golman, Russell, Hagmann, David, and Loewenstein, George. Information avoidance. *Journal of Economic Literature*, 55(1):96–135, 2017.

Gul, Faruk. A theory of disappointment aversion. *Econometrica*, 59(3):667–686, 1991.

Hilbert, Leon P., Noordewier, Marret K., and van Dijk, Wilco W. The prospective associations between financial scarcity and financial avoidance. *Journal of Economic Psychology*, 88:102459, 2022.

Jakobsen, Alexander M. An axiomatic model of persuasion. *Econometrica*, 89(5):2081–2116, 2021.

Jonas, Eva, Schulz-Hardt, Stefan, Frey, Dieter, and Thelen, Norman. Confirmation bias in sequential information search after preliminary decisions: An expansion of dissonance theoretical research on selective exposure to information. *Journal of Personality and Social Psychology*, 80(4):557–571, 2001.

Karlsson, Niklas, Loewenstein, George, and Seppi, Duane. The ostrich effect: Selective attention to information. *Journal of Risk and Uncertainty*, 38(2):95–115, 2009.

Kopylov, Igor. Finite additive utility representations for preferences over menus. *Journal of Economic Theory*, 144(1):354–374, 2009.

Kőszegi, Botond. Health anxiety and patient behavior. *Journal of Health Economics*, 22(6): 1073–1084, 2003.

Kreps, David. *Notes on the Theory of Choice*. Routledge, 2018.

Lerman, Caryn, Hughes, Chanita, Trock, Bruce J., Myers, Ronald E., Main, David, Bonney, Aba, Abbaszadegan, Mohammad R., Harty, Anne E., Franklin, Barbara A., Lynch, Jane F., et al. Genetic testing in families with hereditary nonpolyposis colon cancer. *JAMA*, 281(17):1618–1622, 1999.

Loomes, Graham and Sugden, Robert. Regret theory: An alternative theory of rational choice under uncertainty. *The Economic Journal*, 92(368):805–824, 1982.

Loomes, Graham and Sugden, Robert. Some implications of a more general form of regret theory. *Journal of Economic Theory*, 41(2):270–287, 1987.

Momsen, Katharina and Schneider, Sebastian O. How do defaults affect information acquisition and processing? Working paper, September 2024.

Oster, Emily, Shoulson, Ira, and Dorsey, E. Optimal expectations and limited medical testing: Evidence from Huntington disease. *American Economic Review*, 103(2):804–30, 2013.

Pennesi, Daniele. Subjective timing of information arrival. Working paper, University of Torino, June 2026. June 2026 draft.

Persoskie, Alexander, Ferrer, Rebecca A., and Klein, W.M.P. Association of cancer worry and perceived risk with doctor avoidance: An analysis of information avoidance in a nationally representative US sample. *Journal of Behavioral Medicine*, 37(5):977–987, 2014.

Reb, Jochen and Connolly, Terry. Myopic regret avoidance: Feedback avoidance and learning in repeated decision making. *Organizational Behavior and Human Decision Processes*, 109 (2):182–189, 2009.

Sarver, Todd. Anticipating regret: Why fewer options may be better. *Econometrica*, 76(2): 263–305, 2008.

Schmidt, Robert J., Schwieren, Christiane, and Vollmann, Martin. The value of verbal feedback in allocation decisions. *Journal of Behavioral and Experimental Economics*, 87: 101548, 2020.

Sicherman, Nachum, Loewenstein, George, Seppi, Duane J., and Utkus, Stephen P. Financial attention. *The Review of Financial Studies*, 29(4):863–897, 2016.

Stovall, John E. Temptation with uncertain normative preference. *Theoretical Economics*, 13(1):145–174, 2018.

Sugden, Robert. An axiomatic foundation for regret theory. *Journal of Economic Theory*, 60(1):159–180, 1993.

Yang, Yuzhao. An axiomatic model of cognitive dissonance. *working paper*, 2024.

Supplementary Appendix to "Regret and Information Avoidance"

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Abstract

This supplementary appendix contains proofs omitted from the main text. Section S1 contains the proof for the nested DLR characterization (Lemma 5) used in the main text. Section S2 contains the proof for the necessity of Axioms 1–8 for the subjective informational tradeoff (SIT) representation. Section S3 contains proofs for the sequence of domain translations used in the sufficiency argument.

S1 Proof of the Nested DLR Characterization

This section proves Lemma 5 in the main text. We use the notation of its Appendix A: Z is a finite set of outcomes. $\widehat{\mathcal{M}}$ is the set of nonempty compact sets (menus) of lotteries on Z, and $\widehat{\mathcal{D}}$ is the set of nonempty compact sets of menus (menus of menus). Axioms A.1–A.4 and the nested DLR representation are stated there.

Proof of Lemma 5 in the main text.
"Only if." Suppose that $\succsim$ has a nested DLR representation as expressed in Definition 10 in the main text. It is straightforward to verify that $\succsim$ satisfies axioms A.1-A.3.

We show that $\succsim$ satisfies Finiteness (axiom A.4). If $I = \emptyset$, the preference is trivial and one may take $N = 2$, using arbitrary singleton critical subsets.

Now suppose $I \neq \emptyset$. Let $n_{\text{out}} := |I|$, $n_{\text{in}} := \sum_{i \in I} \big(|P_i| + |N_i|\big)$, and $N := \max\{n_{\text{out}}, n_{\text{in}}\} + 1$. Non-redundancy implies that no U_i is constant, so $n_{\text{in}} \geq 1$. Fix $\mathbb{A} \in \widehat{\mathcal{D}}$. For every $i \in I$, continuity and compactness allow us to choose some $A_i \in \arg\max_{A \in \mathbb{A}} U_i(A)$. Construct another menu of menus by collecting these maximizers together. Let $\mathbb{B} := \{A_i \mid i \in I\}$. Then $|\mathbb{B}| \leq n_{\text{out}} < N$. If $\mathbb{B} \subseteq \mathbb{C} \subseteq \mathbb{A}$ and $\mathbb{C} \in \widehat{\mathcal{D}}$, then, for every $i \in I$, $\max_{A \in \mathbb{C}} U_i(A) = U_i(A_i) = \max_{A \in \mathbb{A}} U_i(A)$. Thus $V_{DLR}(\mathbb{C}) = V_{DLR}(\mathbb{A})$, so $\mathbb{B}$ is critical for $\mathbb{A}$. This proves clause (4.1).

For clause (4.2), fix a *finite* $\mathbb{A} \in \widehat{\mathcal{D}}$ and some $A \in \mathbb{A}$. For every $i \in I$, choose a maximizer for each substate. That is, for each $k \in P_i$, choose $p_{ik} \in \arg\max_{p \in A} w_{ik}(p)$; and for each $j \in N_i$, choose $q_{ij} \in \arg\max_{p \in A} v_{ij}(p)$. Collect these maximizers together and define $B := \{p_{ik} \mid i \in I,\ k \in P_i\} \cup \{q_{ij} \mid i \in I,\ j \in N_i\}$. B must be nonempty because $n_{\text{in}} \geq 1$. Moreover, $B \subseteq A$ and $|B| \leq n_{\text{in}} < N$. If $B \subseteq C \subseteq A$ and $C \in \widehat{\mathcal{M}}$, then every selected maximizer remains available in C, while $C \subseteq A$. Hence

$$\max_{p \in C} w_{ik}(p) = \max_{p \in A} w_{ik}(p), \text{ and } \max_{p \in C} v_{ij}(p) = \max_{p \in A} v_{ij}(p) \text{ for every } i, k, j.$$

Therefore $U_i(C) = U_i(A)$ for every i, and $V_{DLR}\big((\mathbb{A} \setminus \{A\}) \cup \{C\}\big) = V_{DLR}(\mathbb{A})$. Thus B is critical for A in $\mathbb{A}$.

"If." Suppose now that $\succsim$ satisfies Axioms A.1–A.4, and let N_0 be the bound in Axiom A.4. We first obtain a finite additive representation at the outer level. To do so, we invoke the main result of Kopylov (2009):

> Let E be a compact convex metric space and let $\mathcal{K}(E)$ denote the collection of its nonempty compact subsets. A preference $\succsim$ over $\mathcal{K}(E)$ satisfies Kopylov's Finiteness (henceforth *KF*) if there exists an integer n such that, for every $m \geq n$ and every $K_1, \ldots, K_m \in \mathcal{K}(E)$, there is some $r \in \{1, \ldots, m\}$ for which $\bigcup_{\ell=1}^{m} K_\ell \sim \bigcup_{\ell \neq r} K_\ell$. Theorem 2.1 of Kopylov (2009) establishes that Weak Order, Continuity, Independence, and KF imply the existence of a finite additive representation
>
> $$W(K) = \sum_{i \in I^+} \max_{e \in K} \phi_i(e) - \sum_{i \in I^-} \max_{e \in K} \phi_i(e),$$
>
> where the ϕ_i's are continuous affine functions on E, and the representation can be chosen so that the collection of the ϕ_i's is non-redundant.

We first show that Kopylov's Finiteness (KF) on $\widehat{\mathcal{D}}$ is implied by Axioms A.1-A.4.

Lemma S1. *If $\succsim$ satisfies axioms A.1-A.4, then it satisfies KF on $\widehat{\mathcal{D}}$.*

Proof of Lemma S1. Fix $m \geq N_0$ and $\mathbb{A}_1, \ldots, \mathbb{A}_m \in \widehat{\mathcal{D}}$. Set $\mathbb{A}^{\cup} := \bigcup_{\ell=1}^{m} \mathbb{A}_\ell$, $\mathbb{A}^{-r} := \bigcup_{\ell \neq r} \mathbb{A}_\ell$. Because these are finite unions of compact sets, both $\mathbb{A}^{\cup}$ and every $\mathbb{A}^{-r}$ belong to $\widehat{\mathcal{D}}$. By clause (4.1), there is a critical $\mathbb{B} \subseteq \mathbb{A}^{\cup}$ such that $|\mathbb{B}| < N_0$.

There must be some $r^* \in \{1, \ldots, m\}$ such that $\mathbb{B} \subseteq \mathbb{A}^{-r^*}$ (If not, for every r we could select some $B_r \in \mathbb{B} \setminus \mathbb{A}^{-r}$. Since $B_r \in \mathbb{B}$ but $B_r \notin \mathbb{A}_\ell$ for every $\ell \neq r$, then $B_r \in \mathbb{A}_r$ and the B_r's would be pairwise distinct. This would imply $|\mathbb{B}| \geq m \geq N_0$, a contradiction.) Criticality of $\mathbb{B}$ now gives $\mathbb{A}^{-r^*} \sim \mathbb{A}^{\cup}$. Thus $\succsim$ satisfies KF on $\widehat{\mathcal{D}}$. □

Before we can apply Kopylov's theorem, we account for the fact that menus need not be convex. For a menu $A \in \widehat{\mathcal{M}}$, write $c(A) := \text{co}(A)$, the convex hull of A. For a menu of menus write $c(\mathbb{A}) := \{c(A) \mid A \in \mathbb{A}\}$.

Lemma S2. *If $\succsim$ satisfies axioms A.1-A.3, then $\mathbb{A} \sim c(\mathbb{A})$ for every $\mathbb{A} \in \widehat{\mathcal{D}}$.*

Proof of Lemma S2. Let $d = |Z|$ and fix some $A \in \widehat{\mathcal{M}}$. We first want to show that

$$\frac{1}{d+1}A + \frac{d}{d+1}c(A) = c(A) \tag{S1}$$

The left hand side is contained in $c(A)$, because $A \subseteq c(A)$ and $c(A)$ is convex. For the reverse inclusion, fix $p \in c(A)$. Since $p \in \mathbb{R}^d$, Caratheodory's theorem gives $p = \sum_{\ell=1}^{r} \lambda_\ell p_\ell$ where $r \leq d+1$, $p_\ell \in A$ for each ℓ, $\lambda_\ell \geq 0$ for each ℓ, and $\sum_{\ell=1}^{r} \lambda_\ell = 1$. Therefore, at least one coefficient, say λ_k, is at least $\frac{1}{d+1}$. Thus we can write

$$p = \frac{1}{d+1}p_k + \frac{d}{d+1}q, \text{ where } q = \frac{d+1}{d}\left[\left(\lambda_k - \frac{1}{d+1}\right)p_k + \sum_{\ell \neq k} \lambda_\ell p_\ell\right].$$

$q \in c(A)$ by construction, hence p is also an element of the left hand side, proving (S1).

Now suppose $\mathbb{A} = \{A_1, \ldots, A_m\}$ is finite. Set $C = m^{-1}\sum_{j=1}^{m} c(A_j)$ and $\alpha = 1/(md+1)$. So $\frac{1-\alpha}{m} = d\alpha$. Consequently, for each i,

$$\begin{aligned}
\alpha A_i + (1-\alpha)C &= \alpha A_i + d\alpha \sum_{j=1}^{m} c(A_j) \\
&= \alpha(d+1)\left[\frac{1}{d+1}A_i + \frac{d}{d+1}c(A_i)\right] + d\alpha \sum_{j \neq i} c(A_j) \\
&= \alpha(d+1)c(A_i) + d\alpha \sum_{j \neq i} c(A_j) \\
&= \alpha c(A_i) + (1-\alpha)C
\end{aligned}$$

Thus, mixing either A_i or its convex hull with the same menu C gives exactly the same menu. Applying this simultaneously to every i yields

$$\alpha\mathbb{A} + (1-\alpha)\{C\} = \alpha c(\mathbb{A}) + (1-\alpha)\{C\}.$$

If either $\mathbb{A} \succ c(\mathbb{A})$ or $c(\mathbb{A}) \succ \mathbb{A}$, Independence would imply a strict ranking between identical objects. Thus, $\mathbb{A} \sim c(\mathbb{A})$ by Weak Order.

For an arbitrary compact $\mathbb{A}$, there exists a sequence of finite subsets $\mathbb{A}_n \subseteq \mathbb{A}$ converging to $\mathbb{A}$. Since convexification is continuous under Hausdorff metric, $c(\mathbb{A}_n) \to c(\mathbb{A})$. The finite case gives $\mathbb{A}_n \sim c(\mathbb{A}_n)$ for every n. Suppose, toward a contradiction, that $\mathbb{A} \succ c(\mathbb{A})$. By Weak Order and Continuity, $\mathbb{A} \succ c(\mathbb{A}_n)$ and $\mathbb{A}_n \succ c(\mathbb{A})$ for sufficiently large n. Fix one such index k. Since $c(\mathbb{A}_k) \sim \mathbb{A}_k$, we have $\mathbb{A} \succ \mathbb{A}_k \succ c(\mathbb{A})$. Since $\mathbb{A}_n \to \mathbb{A}$ and $c(\mathbb{A}_n) \to c(\mathbb{A})$, Continuity implies that $\mathbb{A}_\ell \succ \mathbb{A}_k \succ c(\mathbb{A}_\ell)$ for all sufficiently large ℓ, contradicting $\mathbb{A}_\ell \sim c(\mathbb{A}_\ell)$. Similar arguments rule out $c(\mathbb{A}) \succ \mathbb{A}$. Thus, $\mathbb{A} \sim c(\mathbb{A})$. □

Let $\widehat{\mathcal{M}}^c$ be the space of nonempty compact convex subsets of $\Delta(Z)$. Map $A \in \widehat{\mathcal{M}}^c$ to its support function $h_A(v) = \max_{p \in A} v \cdot p$. Formally, let $S^{d-1} := \{v \in \mathbb{R}^d \mid \|v\|_2 = 1\}$. Then $h_A : S^{d-1} \to \mathbb{R}$ for each A. This mapping preserves the Hausdorff distance as the uniform distance between support functions, and $h_{\alpha A+(1-\alpha)B} = \alpha h_A + (1-\alpha)h_B$. Let $E := \{h_A \mid A \in \widehat{\mathcal{M}}^c\}$. Then E is a subset of $C(S^{d-1})$, the set of continuous functions defined on S^{d-1}. Endow $C(S^{d-1})$ with the uniform norm.

We identify $\widehat{\mathcal{M}}^c$ with its image under this isometric affine embedding. Thus E is a compact convex subset of a normed linear space. The restriction of $\succsim$ to compact subsets of $\widehat{\mathcal{M}}^c$ inherits Weak Order, Continuity, Independence, and KF. Kopylov's theorem applies to this restriction and gives finite, disjoint index sets I^+, I^- and a non-redundant collection of continuous affine functions $\{\phi_i\}_{i \in I^+ \cup I^-}$ on E. Define $U_i(A) := \phi_i(c(A))$ for every compact menu A. Each U_i is continuous and mixture-linear because convexification commutes with mixtures. Since $\mathbb{A} \sim c(\mathbb{A})$, the function

$$V^0(\mathbb{A}) = \sum_{i \in I^+} \max_{A \in \mathbb{A}} U_i(A) - \sum_{i \in I^-} \max_{A \in \mathbb{A}} U_i(A)$$

represents $\succsim$ on the original domain.

If $I^+ \cup I^- = \emptyset$, this is already the trivial nested DLR representation. Henceforth let $I := I^+ \cup I^- \neq \emptyset$, and define $\varepsilon_i := \mathbf{1}\{i \in I^+\} - \mathbf{1}\{i \in I^-\}$. It remains to show that every U_i has a finite DLR representation over menus of lotteries. For each $i \in I$, define a binary relation $\succsim_i$ over $\widehat{\mathcal{M}}$ by $A \succsim_i B \iff U_i(A) \geq U_i(B)$. Because U_i is continuous and linear, $\succsim_i$ satisfies Weak Order, Continuity, and Independence. The next lemma shows that $\succsim_i$ also satisfies KF with bound N_0.

Lemma S3. *For each $i \in I$, the constructed $\succsim_i$ satisfies KF with bound N_0 on $\widehat{\mathcal{M}}$.*

Proof of Lemma S3. Suppose, toward a contradiction, that $\succsim_{i^*}$ does not satisfy KF with bound N_0 for some $i^* \in I$. Then there exist $m \geq N_0$ and menus $A_1, \ldots, A_m \in \widehat{\mathcal{M}}$ such that

$$U_{i^*}(A^{\cup}) \neq U_{i^*}(A^{-r}) \text{ for every } r = 1, \ldots, m, \text{ where } A^{\cup} := \bigcup_{\ell=1}^{m} A_\ell \text{ and } A^{-r} := \bigcup_{\ell \neq r} A_\ell.$$

Since $\{U_i\}_{i \in I}$ is non-redundant, the standard separating-menu result (Lemma A.1 of Kopylov (2009)) gives menus $\{Z_i\}_{i \in I} \subseteq \widehat{\mathcal{M}}$ satisfying $U_i(Z_i) > U_i(Z_j)$ whenever $i \neq j$.

Let $\mathbb{C} := \{A^{\cup}, A^{-1}, \ldots, A^{-m}\}$. Because all the preceding inequalities are strict and both I and $\mathbb{C}$ are finite, there exists $\eta \in (0,1)$, sufficiently small, such that

$$U_i\big((1-\eta)Z_i + \eta D\big) > U_i\big((1-\eta)Z_j + \eta D'\big) \text{ for every } i \neq j \text{ and every } D, D' \in \mathbb{C}.$$

For each $i \in I$, define $\overline{Z}_i := (1-\eta)Z_i + \eta A^{\cup}$. Moreover, let $\mathbb{Z} := \{\overline{Z}_i : i \in I\}$. The menu of menus $\mathbb{Z}$ is finite. By (4.2) of Finiteness (Axiom A.4), there is a menu $B \subseteq \overline{Z}_{i^*}$, with

$|B| < N_0$, that is critical for $\overline{Z}_{i^*}$ in $\mathbb{Z}$. For every $p \in B$, choose $z(p) \in Z_{i^*}$ and $a(p) \in A^{\cup}$ such that $p = (1-\eta)z(p) + \eta a(p)$, and choose an index $r(p)$ satisfying $a(p) \in A_{r(p)}$. Since $|\{r(p) \mid p \in B\}| \leq |B| < N_0 \leq m$, there is some $r^* \in \{1, \ldots, m\}$ that is different from every $r(p)$. Let $C := (1-\eta)Z_{i^*} + \eta A^{-r^*}$. By construction, $C \in \widehat{\mathcal{M}}$ and $B \subseteq C \subseteq (1-\eta)Z_{i^*} + \eta A^{\cup} = \overline{Z}_{i^*}$. Criticality therefore implies

$$\mathbb{Z}' := \left(\mathbb{Z} \setminus \{\overline{Z}_{i^*}\}\right) \cup \{C\} \sim \mathbb{Z}.$$

The choice of η implies that $\overline{Z}_i$ is the unique maximizer of U_i in $\mathbb{Z}$ for every i. It also implies that, in $\mathbb{Z}'$, $\overline{Z}_i$ remains the unique maximizer of U_i for every $i \neq i^*$, while C is the unique maximizer of U_{i^*}. Hence $0 = V^0(\mathbb{Z}') - V^0(\mathbb{Z}) = \varepsilon_{i^*}\left[U_{i^*}(C) - U_{i^*}(\overline{Z}_{i^*})\right]$. Thus $U_{i^*}(C) = U_{i^*}(\overline{Z}_{i^*})$, which (by linearity) further implies that

$$(1-\eta)U_{i^*}(Z_{i^*}) + \eta U_{i^*}(A^{-r^*}) = (1-\eta)U_{i^*}(Z_{i^*}) + \eta U_{i^*}(A^{\cup}).$$

Since $\eta > 0$, $U_{i^*}(A^{-r^*}) = U_{i^*}(A^{\cup})$, contradicting the construction of $A_1, \ldots, A_m$. Therefore $\succsim_{i^*}$ satisfies KF with bound N_0. Since i^* was arbitrary, this completes the proof. □

Returning to the proof of Lemma 5 in the main text, applying the finite-additive representation theorem to each $\succsim_i$ now gives finite index sets P_i, N_i and expected-utility functions $\widehat{w}_{ik}, \widehat{v}_{ij}$ such that

$$W_i(A) = \sum_{k \in P_i} \max_{p \in A} \widehat{w}_{ik}(p) - \sum_{j \in N_i} \max_{p \in A} \widehat{v}_{ij}(p)$$

represents $\succsim_i$ and the collection $\{\hat{w}_{ik} \mid k \in P_i\} \cup \{\hat{v}_{ij} \mid j \in N_i\}$ is non-redundant. Restrict W_i and U_i to the convex-menu space $\widehat{\mathcal{M}}^c$. Both are nonconstant continuous affine representations of the same preference on this mixture space. Mixture-space uniqueness therefore gives constants $a_i > 0$ and $b_i \in \mathbb{R}$ with $W_i = a_i U_i + b_i$ on $\widehat{\mathcal{M}}^c$. By construction, $U_i(A) = U_i(c(A))$ for every menu A. The finite-additive form of W_i also gives $W_i(A) = W_i(c(A))$, because an expected-utility function has the same maximum on a set and its convex hull. Thus $W_i = a_i U_i + b_i$ on all of $\widehat{\mathcal{M}}$. Let p^0 be the uniform lottery on Z, and define

$$w_{ik} := \frac{\widehat{w}_{ik} - \widehat{w}_{ik}(p^0)}{a_i}, \qquad v_{ij} := \frac{\widehat{v}_{ij} - \widehat{v}_{ij}(p^0)}{a_i}.$$

These expected-utility functions are normalized to sum to zero over Z. Their finite DLR expression equals

$$\sum_{k \in P_i} \max_{p \in A} w_{ik}(p) - \sum_{j \in N_i} \max_{p \in A} v_{ij}(p) = \frac{W_i(A) - W_i(\{p^0\})}{a_i} = U_i(A) - U_i(\{p^0\}).$$

Replacing each U_i by this centered function changes V^0 only by a constant. It preserves non-redundancy and gives the nested DLR representation in Definition 10 in the main text. □

S2 Necessity of Axioms 1–8

Suppose $\succsim$ has a SIT representation with parameters (π, u, K, σ). Let S be the finite signal set of σ. For a menu F, define

$$U_*(F) := \sum_{s\in S} \max_{f\in F} \sum_{\omega\in\Omega} \pi(\omega)\sigma(s \mid \omega)u(f(\omega)), \tag{S2}$$

$$U_s(F) := \max_{f\in F} \sum_{\omega\in\Omega} \pi(\omega)\sigma(s \mid \omega)u(f(\omega)), \qquad s \in S. \tag{S3}$$

Then $U_*(F) = \sum_{s\in S} U_s(F)$, and the representation can be written as

$$V(\mathbb{F}) = (1+K)\max_{F\in\mathbb{F}} U_*(F) - K\sum_{s\in S} \max_{F\in\mathbb{F}} U_s(F). \tag{S4}$$

Each U_i is continuous under the Hausdorff metric and mixture-linear on menus. In particular, for $i \in \{*\} \cup S$,

$$U_i\big(\alpha F + (1-\alpha)G\big) = \alpha U_i(F) + (1-\alpha)U_i(G).$$

Lemma S4. *The SIT representation satisfies Weak Order, Continuity, and Independence.*

Proof. Weak Order follows from the real-valued representation. Continuity follows from continuity of the U_i's and the maximum theorem, applied first over each compact menu and then over each compact menu of menus. Finally,

$$\max_{C\in\alpha\mathbb{F}+(1-\alpha)\mathbb{H}} U_i(C) = \alpha \max_{F\in\mathbb{F}} U_i(F) + (1-\alpha)\max_{H\in\mathbb{H}} U_i(H)$$

for every $i \in \{*\} \cup S$. Substitution in (S4) gives

$$V\big(\alpha\mathbb{F} + (1-\alpha)\mathbb{H}\big) = \alpha V(\mathbb{F}) + (1-\alpha)V(\mathbb{H}),$$

which yields Independence. This is also the finite-additive structure characterized by Theorem 2.1 of Kopylov (2009). □

The second clause of Finiteness is imposed only for a finite $\mathbb{F}$. This restriction ensures that $(\mathbb{F} \setminus \{F\}) \cup \{H\}$ remains compact for every compact H between a proposed critical menu and F.

Lemma S5. *The SIT representation satisfies Finiteness.*

Proof. Let $N = |S| + 2$. For clause (4.1), fix $\mathbb{F} \in \mathcal{D}$. If $|\mathbb{F}| < N$, take $\mathbb{G} = \mathbb{F}$. Otherwise, continuity and compactness allow us to choose

$$F_* \in \arg\max_{F\in\mathbb{F}} U_*(F) \quad \text{and} \quad F_s \in \arg\max_{F\in\mathbb{F}} U_s(F) \quad (s \in S).$$

Set $\mathbb{G} = \{F_*\} \cup \{F_s : s \in S\}$. Then $|\mathbb{G}| \leq |S| + 1 < N$, and every maximum in (S4) is unchanged for any $\mathbb{H}$ satisfying $\mathbb{G} \subseteq \mathbb{H} \subseteq \mathbb{F}$. Thus $\mathbb{G}$ is critical for $\mathbb{F}$.

For clause (4.2), fix a finite $\mathbb{F}$ and $F \in \mathbb{F}$. If $|F| < N$, take $G = F$. Suppose henceforth that $|F| \geq N$, and define

$$S_F := \left\{ s \in S : F \in \arg\max_{H \in \mathbb{F}} U_s(H) \right\}.$$

There are three cases.

First, suppose $F \in \arg\max_{H \in \mathbb{F}} U_*(H)$. For every $s \in S$, select

$$f_s \in \arg\max_{f \in F} \sum_{\omega \in \Omega} \pi(\omega)\sigma(s \mid \omega)u(f(\omega))$$

and let $G = \{f_s : s \in S\}$. If $G \subseteq H \subseteq F$, then $U_s(H) = U_s(F)$ for every s, and hence $U_*(H) = U_*(F)$. Replacing F by H leaves every term of (S4) unchanged.

Second, suppose F does not maximize U_* but $S_F \neq \varnothing$. Select the same maximizing act f_s only for $s \in S_F$ and let $G = \{f_s : s \in S_F\}$. Any H with $G \subseteq H \subseteq F$ preserves $U_s(F)$ for the signals at which F can determine a negative maximum. For every $s \notin S_F$, some menu in the finite set $\mathbb{F}$ has strictly higher U_s than F, and some menu has strictly higher U_* than F; because $H \subseteq F$, those maxima remain unchanged.

Finally, suppose F maximizes neither U_* nor any U_s. Choose any $f^* \in F$ and let $G = \{f^*\}$. For any H between G and F, $U_i(H) \leq U_i(F)$ for every i, while another menu in $\mathbb{F}$ strictly determines each outer maximum. Thus the value is again unchanged.

In every case, G is nonempty, $|G| < N$, and G is critical for F in $\mathbb{F}$. Because $\mathbb{F}$ is finite, every replacement object is a nonempty finite set of compact menus and therefore belongs to $\mathcal{D}$. □

Lemma S6. *The SIT representation satisfies Ex-Ante Regret.*

Proof. For every menu F, equation (S4) gives $V(\{F\}) = U_*(F)$. Hence $\{F\} \succsim \{G\}$ implies $U_*(F) \geq U_*(G)$. If $F \in \mathbb{F}$, adding G to $\mathbb{F}$ leaves the positive maximum in (S4) unchanged and can only increase its negative maxima. Therefore $V(\mathbb{F}) \geq V(\mathbb{F} \cup \{G\})$. □

Lemma S7. *The SIT representation satisfies Interim Preference for Flexibility.*

Proof. The menus of menus $\mathbb{F} \cup \{F \cup G\}$ and $\mathbb{F} \cup \{F, G\}$ have the same union of feasible acts. Their negative terms in (S4) are therefore equal. Moreover,

$$U_*(F \cup G) = \sum_{s \in S} \max\{U_s(F), U_s(G)\} \geq \max\{U_*(F), U_*(G)\}.$$

The positive term is weakly larger for $\mathbb{F} \cup \{F \cup G\}$, proving the axiom. □

Lemma S8. *The SIT representation satisfies Nontriviality and Domination.*

Proof. For a constant act identified with lottery ℓ, $V(\{\{\ell\}\}) = u(\ell)$. Nontriviality follows because u is nonconstant.

If f dominates g, statewise monotonicity gives $u(f(\omega)) \geq u(g(\omega))$ for every ω. Thus $f \succsim g$, and g never raises a maximum when added to $\{f\}$, so $\{\{f, g\}\} \sim \{\{f\}\}$. If $\mathbb{F}$ dominates $\mathbb{G}$, each menu in $\mathbb{G}$ is weakly dominated, signal by signal, by some menu in $\mathbb{F}$. Adding $\mathbb{G}$ changes none of the maxima in (S4); hence $\mathbb{F} \sim \mathbb{F} \cup \mathbb{G}$. □

S3 Proofs for the Domain Translations

This section proves Lemma 15 in the main text. We follow the notation and the three domain transformations used there. The map u sends an act to its utility act and is extended pointwise to menus and menus of menus. Let $n = |\Omega|$, so that $\mathcal{F}_u = [0, 1]^n$, and let $\mathcal{M}_u$ and $\mathcal{D}_u$ be the associated spaces of nonempty compact menus and menus of menus, endowed with their Hausdorff metrics. We identify an act with the menu of menus containing only its singleton menu when writing preferences.

The notions of criticality in each translated domain are defined in the same way as in the main text. Thus, $\mathbb{G}_u$ is critical for $\mathbb{F}_u$ if $\mathbb{G}_u \subseteq \mathbb{F}_u$ and $\mathbb{H}_u \sim_u \mathbb{F}_u$ whenever $\mathbb{G}_u \subseteq \mathbb{H}_u \subseteq \mathbb{F}_u$ and $\mathbb{H}_u \in \mathcal{D}_u$. For a finite $\mathbb{F}_u$, G_u is critical for F_u in $\mathbb{F}_u$ if $G_u \subseteq F_u \in \mathbb{F}_u$ and

$$(\mathbb{F}_u \setminus \{F_u\}) \cup \{H_u\} \sim_u \mathbb{F}_u \quad \text{whenever } G_u \subseteq H_u \subseteq F_u, \quad H_u \in \mathcal{M}_u.$$

The definitions for the primed and double-primed domains are identical, with the corresponding notation. In particular, the second clause of Finiteness always applies only to a *finite* menu of menus, so every replacement remains compact.

S3.1 From acts to utility acts

The first step is to translate the axioms from $\succsim$ to $\succsim_u$.

Axiom S1 (Weak Order)**.** $\succsim_u$ is complete and transitive.

Axiom S2 (Continuity)**.** For every $\mathbb{F}_u \in \mathcal{D}_u$, the sets $\{\mathbb{G}_u : \mathbb{G}_u \succsim_u \mathbb{F}_u\}$ and $\{\mathbb{G}_u : \mathbb{F}_u \succsim_u \mathbb{G}_u\}$ are closed.

Axiom S3 (Independence)**.** For every $\mathbb{F}_u, \mathbb{G}_u, \mathbb{H}_u \in \mathcal{D}_u$ and every $\alpha \in (0, 1)$,

$$\mathbb{F}_u \succsim_u \mathbb{G}_u \iff \alpha\mathbb{F}_u + (1-\alpha)\mathbb{H}_u \succsim_u \alpha\mathbb{G}_u + (1-\alpha)\mathbb{H}_u.$$

Axiom S4 (Finiteness)**.** There exists a natural number N such that:

(4.1) For every $\mathbb{F}_u \in \mathcal{D}_u$, there exists $\mathbb{G}_u$ with $|\mathbb{G}_u| < N$ that is critical for $\mathbb{F}_u$.

(4.2) For every finite $\mathbb{F}_u \in \mathcal{D}_u$ and every $F_u \in \mathbb{F}_u$, there exists G_u with $|G_u| < N$ that is critical for F_u in $\mathbb{F}_u$.

Axiom S5 (Ex-Ante Regret)**.** If $\{F_u\} \succsim_u \{G_u\}$ and $F_u \in \mathbb{F}_u$, then $\mathbb{F}_u \succsim_u \mathbb{F}_u \cup \{G_u\}$.

Axiom S6 (Interim Preference for Flexibility)**.** For every $\mathbb{F}_u \in \mathcal{D}_u$ and every $F_u, G_u \in \mathcal{M}_u$,

$$\mathbb{F}_u \cup \{F_u \cup G_u\} \succsim_u \mathbb{F}_u \cup \{F_u, G_u\}.$$

For utility acts, f_u dominates g_u if $f_u(\omega) \geq g_u(\omega)$ for every $\omega \in \Omega$. A utility menu F_u dominates G_u if every $g_u \in G_u$ is dominated by some $f_u \in F_u$. A utility menu of menus $\mathbb{F}_u$ dominates $\mathbb{G}_u$ if every $G_u \in \mathbb{G}_u$ is dominated by some $F_u \in \mathbb{F}_u$.

Axiom S7 (Domination)**.** (7.1) If f_u dominates g_u, then $f_u \succsim_u g_u$ and $\{\{f_u, g_u\}\} \sim_u \{\{f_u\}\}$. (7.2) If $\mathbb{F}_u$ dominates $\mathbb{G}_u$, then $\mathbb{F}_u \sim_u \mathbb{F}_u \cup \mathbb{G}_u$.

Axiom S8 (Nontriviality)**.** There exist $\mathbb{F}_u, \mathbb{G}_u \in \mathcal{D}_u$ such that $\mathbb{F}_u \supseteq \mathbb{G}_u$ and $\mathbb{F}_u \succ_u \mathbb{G}_u$.

Proposition S1. *If a binary relation $\succsim$ over $\mathcal{D}$ satisfies Axioms 1–8 in the main text, then the induced binary relation $\succsim_u$ satisfies Axioms S1–S8.*

Proof. The operation $u(\cdot)$ that maps an act, menu, or menu of menus into its utility counterpart commutes with mixtures and unions and preserves inclusion. To ensure that every compact utility object has a compact preimage, choose lotteries ℓ^+, ℓ^- with $u(\ell^+) = 1$ and $u(\ell^-) = 0$, and define

$$s(f_u)(\omega) := f_u(\omega)\ell^+ + (1 - f_u(\omega))\ell^-.$$

The map s is continuous, affine, and injective, and $u(s(f_u)) = f_u$. Extend s pointwise to menus and menus of menus. These extensions are continuous under the Hausdorff metrics, map compact sets to compact sets, and satisfy $u \circ s = \mathrm{id}$. By the utility-equivalence result in Step 1 of the main text, the definition of $\succsim_u$ is independent of the preimages chosen. In particular,

$$\mathbb{F}_u \succsim_u \mathbb{G}_u \quad \Longleftrightarrow \quad s(\mathbb{F}_u) \succsim s(\mathbb{G}_u).$$

Thus Weak Order follows from its counterpart for $\succsim$. Both contour sets of $\succsim_u$ are inverse images under the continuous map s of the corresponding closed contour sets of $\succsim$, proving Continuity. Because s preserves mixtures and unions, Independence, Ex-Ante Regret, and Interim Preference for Flexibility follow from their counterparts for $\succsim$. Coordinatewise domination of utility acts is equivalent to statewise preference between their canonical lifts. The same is true of domination at the two set levels. Axiom 7 in the main text therefore implies Axiom S7.

For Finiteness, let N be a bound for $\succsim$. Fix $\mathbb{F}_u \in \mathcal{D}_u$ and apply clause (4.1) to $s(\mathbb{F}_u)$. Let $\mathbb{G} \subseteq s(\mathbb{F}_u)$ be a critical subset with $|\mathbb{G}| < N$, and set $\mathbb{G}_u = u(\mathbb{G})$. Since s is injective and $\mathbb{G} \subseteq s(\mathbb{F}_u)$, $s(\mathbb{G}_u) = \mathbb{G}$. Every $\mathbb{H}_u \in \mathcal{D}_u$ with $\mathbb{G}_u \subseteq \mathbb{H}_u \subseteq \mathbb{F}_u$ therefore satisfies

$$\mathbb{G} \subseteq s(\mathbb{H}_u) \subseteq s(\mathbb{F}_u),$$

so $s(\mathbb{H}_u) \sim s(\mathbb{F}_u)$ and hence $\mathbb{H}_u \sim_u \mathbb{F}_u$. Thus $\mathbb{G}_u$ is critical and $|\mathbb{G}_u| \leq |\mathbb{G}| < N$.

For clause (4.2), fix a finite $\mathbb{F}_u$ and $F_u \in \mathbb{F}_u$. Its lift $s(\mathbb{F}_u)$ is finite. Choose $G \subseteq s(F_u)$, with $|G| < N$, critical for $s(F_u)$ in $s(\mathbb{F}_u)$, and let $G_u = u(G)$. If $G_u \subseteq H_u \subseteq F_u$, then $G = s(G_u) \subseteq s(H_u) \subseteq s(F_u)$. Moreover, injectivity of s gives

$$s\big((\mathbb{F}_u \setminus \{F_u\}) \cup \{H_u\}\big) = \big(s(\mathbb{F}_u) \setminus \{s(F_u)\}\big) \cup \{s(H_u)\}.$$

Criticality of G and the definition of $\succsim_u$ imply that this replacement is indifferent to $\mathbb{F}_u$. Hence G_u is critical for F_u in $\mathbb{F}_u$, with $|G_u| < N$.

The statement of Axiom S8 differs slightly from Nontriviality in the main text. By Axiom 8 there, choose lotteries $\ell \succ \ell'$. Then ℓ dominates ℓ', so part (7.2) of Domination and transitivity imply

$$\{\{\ell\}, \{\ell'\}\} \sim \{\{\ell\}\} \succ \{\{\ell'\}\}.$$

Let $\mathbb{F} = \{\{\ell\}, \{\ell'\}\}$ and $\mathbb{G} = \{\{\ell'\}\}$. Then $\mathbb{F} \supseteq \mathbb{G}$ and $\mathbb{F} \succ \mathbb{G}$. Applying u gives $u(\mathbb{F}) \supseteq u(\mathbb{G})$ and $u(\mathbb{F}) \succ_u u(\mathbb{G})$, as required by Axiom S8. □

S3.2 The homeomorphic translation

The second step is to translate the axioms from $\succsim_u$ to $\succsim_*$.

As in the main text, let

$$\mathcal{F}' := \left\{ f' \in [0,n] \times [0,1]^n : \sum_{i=0}^{n} f'(\omega_i) = n \right\}.$$

Let $r : \mathcal{F}' \to \mathcal{F}_u$ drop the artificial-state coordinate ω_0. Its inverse adds the coordinate $n - \sum_{i=1}^{n} f_u(\omega_i)$. Both maps are affine homeomorphisms and induce homeomorphisms on the compact-menu and compact-menu-of-menus spaces. We continue to denote these maps by r and r^{-1}. Define

$$\mathbb{F}' \succsim_* \mathbb{G}' \quad \Longleftrightarrow \quad r(\mathbb{F}') \succsim_u r(\mathbb{G}').$$

Axiom S1* (Weak Order)**.** $\succsim_*$ is complete and transitive.

Axiom S2* (Continuity)**.** For every $\mathbb{F}' \in \mathcal{D}'$, the sets $\{\mathbb{G}' : \mathbb{G}' \succsim_* \mathbb{F}'\}$ and $\{\mathbb{G}' : \mathbb{F}' \succsim_* \mathbb{G}'\}$ are closed.

Axiom S3* (Independence)**.** For every $\mathbb{F}', \mathbb{G}', \mathbb{H}' \in \mathcal{D}'$ and every $\alpha \in (0,1)$,

$$\mathbb{F}' \succsim_* \mathbb{G}' \iff \alpha\mathbb{F}' + (1-\alpha)\mathbb{H}' \succsim_* \alpha\mathbb{G}' + (1-\alpha)\mathbb{H}'.$$

Axiom S4* (Finiteness)**.** There exists a natural number N such that:

(4.1) For every $\mathbb{F}' \in \mathcal{D}'$, there exists $\mathbb{G}'$ with $|\mathbb{G}'| < N$ that is critical for $\mathbb{F}'$.

(4.2) For every finite $\mathbb{F}' \in \mathcal{D}'$ and every $F' \in \mathbb{F}'$, there exists G' with $|G'| < N$ that is critical for F' in $\mathbb{F}'$.

Axiom S5* (Ex-Ante Regret)**.** If $\{F'\} \succsim_* \{G'\}$ and $F' \in \mathbb{F}'$, then $\mathbb{F}' \succsim_* \mathbb{F}' \cup \{G'\}$.

Axiom S6* (Interim Preference for Flexibility)**.** For every $\mathbb{F}' \in \mathcal{D}'$ and every $F', G' \in \mathcal{M}'$,

$$\mathbb{F}' \cup \{F' \cup G'\} \succsim_* \mathbb{F}' \cup \{F', G'\}.$$

Axiom S7* (Inclusion)**.** If $G' \subseteq F'$ and $F' \in \mathbb{F}'$, then $\mathbb{F}' \cup \{G'\} \succsim_* \mathbb{F}'$.

Axiom S8* (Nontriviality)**.** There exist $\mathbb{F}', \mathbb{G}' \in \mathcal{D}'$ such that $\mathbb{F}' \supseteq \mathbb{G}'$ and $\mathbb{F}' \succ_* \mathbb{G}'$.

Although Inclusion has no separately stated counterpart on $\mathcal{D}_u$, it follows from part (7.2) of Domination.

Proposition S2. *If $\succsim_u$ satisfies Axioms S1–S8, then $\succsim_*$ satisfies Axioms S1*–S8*.*

Proof that Axiom S1 implies Axiom S1.* Fix any $\mathbb{F}', \mathbb{G}' \in \mathcal{D}'$. Completeness of $\succsim_u$ gives $r(\mathbb{F}') \succsim_u r(\mathbb{G}')$ or $r(\mathbb{G}') \succsim_u r(\mathbb{F}')$, or both. Thus $\mathbb{F}' \succsim_* \mathbb{G}'$ or $\mathbb{G}' \succsim_* \mathbb{F}'$, or both.

For transitivity, suppose $\mathbb{F}' \succsim_* \mathbb{G}'$ and $\mathbb{G}' \succsim_* \mathbb{H}'$. Then $r(\mathbb{F}') \succsim_u r(\mathbb{G}')$ and $r(\mathbb{G}') \succsim_u r(\mathbb{H}')$. Transitivity of $\succsim_u$ implies $r(\mathbb{F}') \succsim_u r(\mathbb{H}')$, and hence $\mathbb{F}' \succsim_* \mathbb{H}'$. □

Proof that Axiom S2 implies Axiom S2.* Fix any $\mathbb{F}' \in \mathcal{D}'$. By definition,

$$\{\mathbb{G}' : \mathbb{F}' \succsim_* \mathbb{G}'\} = r^{-1}\big(\{\mathbb{G}_u : r(\mathbb{F}') \succsim_u \mathbb{G}_u\}\big).$$

The set on the right inside the inverse image is closed by Axiom S2. Since r is a homeomorphism, the lower contour set on the left is closed. The argument for the upper contour set $\{\mathbb{G}' : \mathbb{G}' \succsim_* \mathbb{F}'\}$ is the same. □

Proof that Axiom S3 implies Axiom S3.* First note that r and its inverse are mixture-linear. In particular,

$$r\big(\alpha\mathbb{F}' + (1-\alpha)\mathbb{H}'\big) = \alpha r(\mathbb{F}') + (1-\alpha) r(\mathbb{H}').$$

Therefore, for any $\alpha \in (0,1)$,

$$\begin{aligned}
\mathbb{F}' \succsim_* \mathbb{G}' &\iff r(\mathbb{F}') \succsim_u r(\mathbb{G}') \\
&\iff \alpha r(\mathbb{F}') + (1-\alpha) r(\mathbb{H}') \succsim_u \alpha r(\mathbb{G}') + (1-\alpha) r(\mathbb{H}') \\
&\iff r\big(\alpha \mathbb{F}' + (1-\alpha)\mathbb{H}'\big) \succsim_u r\big(\alpha \mathbb{G}' + (1-\alpha)\mathbb{H}'\big) \\
&\iff \alpha \mathbb{F}' + (1-\alpha)\mathbb{H}' \succsim_* \alpha \mathbb{G}' + (1-\alpha)\mathbb{H}'.
\end{aligned}$$

□

Proof that Axiom S4 implies Axiom S4.* Let N be a bound satisfying Axiom S4. Fix $\mathbb{F}' \in \mathcal{D}'$ and let $\mathbb{G}_u$ be critical for $r(\mathbb{F}')$, with $|\mathbb{G}_u| < N$. Set $\mathbb{G}' = r^{-1}(\mathbb{G}_u)$. Then $|\mathbb{G}'| = |\mathbb{G}_u| < N$. For every $\mathbb{H}' \in \mathcal{D}'$ with $\mathbb{G}' \subseteq \mathbb{H}' \subseteq \mathbb{F}'$, inclusion preservation gives

$$r(\mathbb{G}') \subseteq r(\mathbb{H}') \subseteq r(\mathbb{F}').$$

Criticality implies $r(\mathbb{H}') \sim_u r(\mathbb{F}')$, and hence $\mathbb{H}' \sim_* \mathbb{F}'$. Thus $\mathbb{G}'$ is critical for $\mathbb{F}'$.

For the inner clause, fix a finite $\mathbb{F}'$ and $F' \in \mathbb{F}'$. The set $r(\mathbb{F}')$ is finite, so Axiom S4(4.2) supplies a critical $G_u \subseteq r(F')$ with $|G_u| < N$. Let $G' = r^{-1}(G_u)$. If $G' \subseteq H' \subseteq F'$, then $G_u \subseteq r(H') \subseteq r(F')$. Since r is injective,

$$r\big((\mathbb{F}' \setminus \{F'\}) \cup \{H'\}\big) = \big(r(\mathbb{F}') \setminus \{r(F')\}\big) \cup \{r(H')\}.$$

The right side is indifferent to $r(\mathbb{F}')$. Hence G' is critical for F' in $\mathbb{F}'$, with $|G'| = |G_u| < N$. □

Proof that Axiom S5 implies Axiom S5.* For every $F' \in \mathcal{M}'$, $r(\{F'\}) = \{r(F')\}$. Thus

$$\begin{aligned}
\{F'\} \succsim_* \{G'\}, \quad F' \in \mathbb{F}' &\Longrightarrow \{r(F')\} \succsim_u \{r(G')\}, \quad r(F') \in r(\mathbb{F}') \\
&\Longrightarrow r(\mathbb{F}') \succsim_u r(\mathbb{F}') \cup \{r(G')\} \\
&\Longrightarrow r(\mathbb{F}') \succsim_u r(\mathbb{F}' \cup \{G'\}) \\
&\Longrightarrow \mathbb{F}' \succsim_* \mathbb{F}' \cup \{G'\}.
\end{aligned}$$

□

Proof that Axiom S6 implies Axiom S6.* The pointwise map r satisfies

$$\begin{aligned}
r(\mathbb{F}' \cup \{F' \cup G'\}) &= r(\mathbb{F}') \cup \{r(F') \cup r(G')\}, \\
r(\mathbb{F}' \cup \{F', G'\}) &= r(\mathbb{F}') \cup \{r(F'), r(G')\}.
\end{aligned}$$

Axiom S6 compares the two objects on the right. The definition of $\succsim_*$ therefore gives the desired comparison on the left. □

Proof that Axiom S7 implies Axiom S7.* Suppose $G' \subseteq F'$ and $F' \in \mathbb{F}'$. Then $r(G') \subseteq r(F')$, so $r(F')$ dominates $r(G')$. Consequently, $r(\mathbb{F}')$ dominates $\{r(G')\}$. Part (7.2) of Axiom S7 implies

$$r(\mathbb{F}') \sim_u r(\mathbb{F}') \cup \{r(G')\} = r(\mathbb{F}' \cup \{G'\}).$$

Hence $\mathbb{F}' \sim_* \mathbb{F}' \cup \{G'\}$, which implies Inclusion. □

Proof that Axiom S8 implies Axiom S8.* By Axiom S8, choose $\mathbb{F}_u \supseteq \mathbb{G}_u$ with $\mathbb{F}_u \succ_u \mathbb{G}_u$. Since r^{-1} preserves inclusion,

$$r^{-1}(\mathbb{F}_u) \supseteq r^{-1}(\mathbb{G}_u), \qquad r^{-1}(\mathbb{F}_u) \succ_* r^{-1}(\mathbb{G}_u).$$

This proves Nontriviality. □

S3.3 Extension to the full simplex

The last step is to translate the axioms from $\succsim_*$ to $\succsim_{**}$.

Recall that

$$\mathcal{F}'' := \left\{ f'' \in [0,n]^{n+1} : \sum_{i=0}^{n} f''(\omega_i) = n \right\},$$

and let $\mathcal{M}''$ and $\mathcal{D}''$ be its nonempty compact-menu and compact-menu-of-menus spaces. As in the main text, let

$$e := \left(\frac{n}{n+1}, \ldots, \frac{n}{n+1} \right), \qquad F^{n+1} := \{e\}, \qquad \varepsilon_0 := \frac{1}{n^2}.$$

For $0 < \varepsilon \leq \varepsilon_0$, the map $f'' \mapsto \varepsilon f'' + (1-\varepsilon)e$ sends $\mathcal{F}''$ into $\mathcal{F}'$. Indeed, every real-state coordinate of its image lies between zero and

$$\varepsilon n + (1-\varepsilon)\frac{n}{n+1} \leq \varepsilon_0 n + (1-\varepsilon_0)\frac{n}{n+1} = 1.$$

Extend this map pointwise to menus and menus of menus, writing

$$t_\varepsilon(\mathbb{F}'') := \varepsilon \mathbb{F}'' + (1-\varepsilon)\{F^{n+1}\}.$$

The induced maps are continuous and injective. They preserve inclusion, unions, and the cardinality of finite sets. Since both F^{n+1} and $\{F^{n+1}\}$ are singletons, they also preserve mixtures. Define $\succsim_{**}$, as in the main text, by

$$\mathbb{F}'' \succsim_{**} \mathbb{G}'' \quad \Longleftrightarrow \quad t_\varepsilon(\mathbb{F}'') \succsim_* t_\varepsilon(\mathbb{G}'') \quad \text{for every } 0 < \varepsilon < \varepsilon_0.$$

We list all eight adapted axioms on $\mathcal{D}''$. They are the axioms denoted 1^*–8^* in Step 3 of the main text.

Axiom S1** (Weak Order)**.** $\succsim_{**}$ is complete and transitive.

Axiom S2** (Continuity)**.** For every $\mathbb{F}'' \in \mathcal{D}''$, the sets $\{\mathbb{G}'' : \mathbb{G}'' \succsim_{**} \mathbb{F}''\}$ and $\{\mathbb{G}'' : \mathbb{F}'' \succsim_{**} \mathbb{G}''\}$ are closed.

Axiom S3** (Independence)**.** For every $\mathbb{F}'', \mathbb{G}'', \mathbb{H}'' \in \mathcal{D}''$ and every $\alpha \in (0,1)$,

$$\mathbb{F}'' \succsim_{**} \mathbb{G}'' \iff \alpha\mathbb{F}'' + (1-\alpha)\mathbb{H}'' \succsim_{**} \alpha\mathbb{G}'' + (1-\alpha)\mathbb{H}''.$$

Axiom S4** (Finiteness)**.** There exists a natural number N such that:

(4.1) For every $\mathbb{F}'' \in \mathcal{D}''$, there exists $\mathbb{G}''$ with $|\mathbb{G}''| < N$ that is critical for $\mathbb{F}''$.

(4.2) For every finite $\mathbb{F}'' \in \mathcal{D}''$ and every $F'' \in \mathbb{F}''$, there exists G'' with $|G''| < N$ that is critical for F'' in $\mathbb{F}''$.

Axiom S5** (Ex-Ante Regret)**.** If $\{F''\} \succsim_{**} \{G''\}$ and $F'' \in \mathbb{F}''$, then $\mathbb{F}'' \succsim_{**} \mathbb{F}'' \cup \{G''\}$.

Axiom S6** (Interim Preference for Flexibility)**.** For every $\mathbb{F}'' \in \mathcal{D}''$ and every $F'', G'' \in \mathcal{M}''$,

$$\mathbb{F}'' \cup \{F'' \cup G''\} \succsim_{**} \mathbb{F}'' \cup \{F'', G''\}.$$

Axiom S7** (Inclusion)**.** If $G'' \subseteq F''$ and $F'' \in \mathbb{F}''$, then $\mathbb{F}'' \cup \{G''\} \succsim_{**} \mathbb{F}''$.

Axiom S8** (Nontriviality)**.** There exist $\mathbb{F}'', \mathbb{G}'' \in \mathcal{D}''$ such that $\mathbb{F}'' \supseteq \mathbb{G}''$ and $\mathbb{F}'' \succ_{**} \mathbb{G}''$.

The following simplifying result helps us use the definition of $\succsim_{**}$ more easily. Set

$$t(\mathbb{F}'') := t_{\varepsilon_0}(\mathbb{F}'') = \frac{1}{n^2}\mathbb{F}'' + \left(1 - \frac{1}{n^2}\right)\{F^{n+1}\}. \tag{S5}$$

We use the same notation for the corresponding maps at the act and menu levels.

Lemma S9. *If $\succsim_*$ satisfies Axioms S1*–S3*, then*

$$\mathbb{F}'' \succsim_{**} \mathbb{G}'' \iff t(\mathbb{F}'') \succsim_* t(\mathbb{G}'').$$

Moreover, $\succsim_{**}$ agrees with $\succsim_*$ on $\mathcal{D}'$ and is its unique extension to $\mathcal{D}''$ satisfying Independence.

Proof. Fix $0 < \varepsilon < \varepsilon_0$ and put $\alpha = \varepsilon/\varepsilon_0$. Because the anchor is a singleton at both set levels,

$$t_\varepsilon(\mathbb{F}'') = \alpha t(\mathbb{F}'') + (1-\alpha)\{F^{n+1}\},$$

and the analogous identity holds for $\mathbb{G}''$. Independence therefore gives

$$t_\varepsilon(\mathbb{F}'') \succsim_* t_\varepsilon(\mathbb{G}'') \iff t(\mathbb{F}'') \succsim_* t(\mathbb{G}'').$$

Thus the endpoint comparison implies every comparison in the definition of $\succsim_{**}$; conversely, the comparison at $\varepsilon = \varepsilon_0/2$ implies the endpoint comparison. Mixture-linearity of t and Independence of $\succsim_*$ also imply Independence of $\succsim_{**}$, as verified explicitly below.

For $\mathbb{F}', \mathbb{G}' \in \mathcal{D}'$, Independence gives

$$\mathbb{F}' \succsim_* \mathbb{G}' \quad \Longleftrightarrow \quad t_\varepsilon(\mathbb{F}') \succsim_* t_\varepsilon(\mathbb{G}') \quad (0 < \varepsilon < \varepsilon_0),$$

so $\succsim_{**}$ agrees with $\succsim_*$ on $\mathcal{D}'$. Finally, any extension of $\succsim_*$ satisfying Independence must rank $\mathbb{F}''$ and $\mathbb{G}''$ in the same way as their contractions $t_{\varepsilon_0/2}(\mathbb{F}'')$ and $t_{\varepsilon_0/2}(\mathbb{G}'')$. These contractions lie in $\mathcal{D}'$, where their ranking is already fixed by $\succsim_*$. This proves uniqueness. The argument also covers $n = 1$, for which $\varepsilon_0 = 1$. □

Proposition S3. *If $\succsim_*$ satisfies Axioms S1*–S8*, then $\succsim_{**}$ satisfies Axioms S1**–S8**.*

Proof that Axioms S1–S3* imply Axiom S1**.* For any $\mathbb{F}'', \mathbb{G}'' \in \mathcal{D}''$, completeness of $\succsim_*$ gives $t(\mathbb{F}'') \succsim_* t(\mathbb{G}'')$ or $t(\mathbb{G}'') \succsim_* t(\mathbb{F}'')$, or both. By Lemma S9, $\succsim_{**}$ is complete. For transitivity,

$$\begin{aligned} \mathbb{F}'' \succsim_{**} \mathbb{G}'', \quad \mathbb{G}'' \succsim_{**} \mathbb{H}'' &\Longrightarrow t(\mathbb{F}'') \succsim_* t(\mathbb{G}''), \quad t(\mathbb{G}'') \succsim_* t(\mathbb{H}'') \\ &\Longrightarrow t(\mathbb{F}'') \succsim_* t(\mathbb{H}'') \\ &\Longrightarrow \mathbb{F}'' \succsim_{**} \mathbb{H}'', \end{aligned}$$

where the second implication follows from transitivity of $\succsim_*$. □

Proof that Axioms S1–S3* imply Axiom S2**.* We consider the lower contour set; the same argument works for the upper contour set. Fix $\mathbb{F}''$. By Lemma S9,

$$\{\mathbb{G}'' : \mathbb{F}'' \succsim_{**} \mathbb{G}''\} = t^{-1}\big(\{\mathbb{G}' : t(\mathbb{F}'') \succsim_* \mathbb{G}'\}\big).$$

The contour set inside the inverse image is closed by Axiom S2*. Since t is continuous, the lower contour set of $\succsim_{**}$ is closed. Reversing the comparison proves the same for its upper contour set. □

Proof that Axioms S1–S3* imply Axiom S3**.* First note that t is mixture-linear. For every $\alpha \in [0, 1]$,

$$\begin{aligned} t\big(\alpha\mathbb{F}'' + (1-\alpha)\mathbb{H}''\big) &= \varepsilon_0\big(\alpha\mathbb{F}'' + (1-\alpha)\mathbb{H}''\big) + (1-\varepsilon_0)\{F^{n+1}\} \\ &= \alpha t(\mathbb{F}'') + (1-\alpha)t(\mathbb{H}''). \end{aligned}$$

The second equality uses the fact that F^{n+1} is a singleton menu. The decomposition need not hold for a non-singleton menu. Hence, for $\alpha \in (0,1)$,

$$\begin{aligned}
\mathbb{F}'' \succsim_{**} \mathbb{G}'' &\iff t(\mathbb{F}'') \succsim_{*} t(\mathbb{G}'') \\
&\iff \alpha t(\mathbb{F}'') + (1-\alpha) t(\mathbb{H}'') \succsim_{*} \alpha t(\mathbb{G}'') + (1-\alpha) t(\mathbb{H}'') \\
&\iff t\big(\alpha \mathbb{F}'' + (1-\alpha)\mathbb{H}''\big) \succsim_{*} t\big(\alpha \mathbb{G}'' + (1-\alpha)\mathbb{H}''\big) \\
&\iff \alpha \mathbb{F}'' + (1-\alpha)\mathbb{H}'' \succsim_{**} \alpha \mathbb{G}'' + (1-\alpha)\mathbb{H}''.
\end{aligned}$$

The first and last equivalences follow from Lemma S9, the second from Axiom S3*, and the third from mixture-linearity. □

Proof that Axioms S1–S4* imply Axiom S4***. Let N be a bound for Axiom S4*. The map t preserves inclusion and the cardinality of finite sets. Fix any $\mathbb{F}'' \in \mathcal{D}''$. Choose $\mathbb{G}' \subseteq t(\mathbb{F}'')$ critical for $t(\mathbb{F}'')$, with $|\mathbb{G}'| < N$. Injectivity of t gives a unique $\mathbb{G}'' \subseteq \mathbb{F}''$ such that $t(\mathbb{G}'') = \mathbb{G}'$, and $|\mathbb{G}''| = |\mathbb{G}'| < N$. If $\mathbb{G}'' \subseteq \mathbb{H}'' \subseteq \mathbb{F}''$, then

$$\mathbb{G}' = t(\mathbb{G}'') \subseteq t(\mathbb{H}'') \subseteq t(\mathbb{F}'').$$

Criticality gives $t(\mathbb{H}'') \sim_{*} t(\mathbb{F}'')$. Lemma S9 implies $\mathbb{H}'' \sim_{**} \mathbb{F}''$, so $\mathbb{G}''$ is critical for $\mathbb{F}''$.

For clause (4.2), fix a finite $\mathbb{F}''$ and $F'' \in \mathbb{F}''$. The set $t(\mathbb{F}'')$ is finite. Choose $G' \subseteq t(F'')$ critical for $t(F'')$ in $t(\mathbb{F}'')$, with $|G'| < N$. Its inverse image $G'' \subseteq F''$ under the act-level map t is compact and has the same cardinality. If $G'' \subseteq H'' \subseteq F''$, then $G' \subseteq t(H'') \subseteq t(F'')$, and injectivity gives

$$t\big((\mathbb{F}'' \setminus \{F''\}) \cup \{H''\}\big) = \big(t(\mathbb{F}'') \setminus \{t(F'')\}\big) \cup \{t(H'')\}.$$

Criticality on $\mathcal{D}'$ and Lemma S9 therefore make this replacement indifferent to $\mathbb{F}''$. Thus G'' is critical for F'' in $\mathbb{F}''$, with $|G''| < N$. □

Proof that Axioms S1–S3* and S5* imply Axiom S5***. First note that t preserves unions and singletons:

$$t(\mathbb{F}'' \cup \{G''\}) = t(\mathbb{F}'') \cup \{t(G'')\}, \qquad t(\{F''\}) = \{t(F'')\}.$$

Suppose $\{F''\} \succsim_{**} \{G''\}$ and $F'' \in \mathbb{F}''$. Lemma S9 gives $\{t(F'')\} \succsim_{*} \{t(G'')\}$, and $t(F'') \in t(\mathbb{F}'')$. By Axiom S5*,

$$t(\mathbb{F}'') \succsim_{*} t(\mathbb{F}'') \cup \{t(G'')\} = t(\mathbb{F}'' \cup \{G''\}).$$

Applying Lemma S9 once more proves $\mathbb{F}'' \succsim_{**} \mathbb{F}'' \cup \{G''\}$. □

Proof that Axioms S1–S3* and S6* imply Axiom S6**.* At the menu level, $t(F'' \cup G'') = t(F'') \cup t(G'')$. Hence Axiom S6* gives

$$t(\mathbb{F}'') \cup \{t(F'') \cup t(G'')\} \succsim_* t(\mathbb{F}'') \cup \{t(F''), t(G'')\}.$$

The left and right sides are respectively $t(\mathbb{F}'' \cup \{F'' \cup G''\})$ and $t(\mathbb{F}'' \cup \{F'', G''\})$. Lemma S9 yields the required comparison. □

Proof that Axioms S1–S3* and S7* imply Axiom S7**.* Suppose $G'' \subseteq F''$ and $F'' \in \mathbb{F}''$. Then $t(G'') \subseteq t(F'')$ and $t(F'') \in t(\mathbb{F}'')$. By Axiom S7*,

$$t(\mathbb{F}'') \cup \{t(G'')\} \succsim_* t(\mathbb{F}'').$$

Since the left side equals $t(\mathbb{F}'' \cup \{G''\})$, Lemma S9 implies $\mathbb{F}'' \cup \{G''\} \succsim_{**} \mathbb{F}''$. □

Proof that Axioms S1–S3* and S8* imply Axiom S8**.* Choose $\mathbb{F}', \mathbb{G}' \in \mathcal{D}'$ with $\mathbb{F}' \supseteq \mathbb{G}'$ and $\mathbb{F}' \succ_* \mathbb{G}'$, as supplied by Axiom S8*. Because $\mathcal{D}' \subseteq \mathcal{D}''$ and Lemma S9 shows that the extension agrees with $\succsim_*$ on $\mathcal{D}'$, the same pair satisfies $\mathbb{F}' \succ_{**} \mathbb{G}'$. This proves Nontriviality. □

Together, the three propositions show that the relation $\succsim_{**}$ constructed from a preference satisfying Axioms 1–8 in the main text satisfies all eight adapted axioms in Step 3 there. This completes the proof of Lemma 15 in the main text.

References

Kopylov, Igor. Finite additive utility representations for preferences over menus. *Journal of Economic Theory*, 144(1):354–374, 2009.